Time Series Forecasting Project

Apr B Sunday Onkar 10:30 AM Batch

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1. Context

As an analyst at ABC Estate Wines, we are presented with historical data encompassing the sales of different types of wines throughout the 20th century. These datasets originate from the same company but represent sales figures for distinct wine varieties. Our objective is to delve into the data, analyze trends, patterns, and factors influencing wine sales over the course of the century. By leveraging data analytics and forecasting techniques, we aim to gain actionable insights that can inform strategic decision-making and optimize sales strategies for the future.

2. Objective

The primary objective of this project is to analyze and forecast wine sales trends for the 20th century based on historical data provided by ABC Estate Wines. We aim to equip ABC Estate Wines with the necessary insights and foresight to enhance sales performance, capitalize on emerging market opportunities, and maintain a competitive edge in the wine industry.

3. Data Dictionary

Variables	Description
YearMonth	Has the timestamp of the entire data.
Sparkling	Information about the sparkling wine sales.
Rose	Information about the rose wine sales.

4. Data Information

- There are two datasets.
- There are in total 187 rows and two columns in each dataset. One is timestamp and another is numeric.
- For Sparkling data, the description looks like: mean (2402.41), std (1295.11), min (1070.00), 25% (1605.00), 50% (1874.00), 75% (2549.00), max (7242.00).
- For Rose data, the description looks like: mean (187.00), mean (89.90), std (39.24), min (28.00), 25% (62.50), 50% (85.00), 75% (111.00), max (267.00).
- We have created Time Stamps and added to the data frame to make it a Time Series Data. Where the data starting from 1/1/1980 to 8/1/1995.
- There were two missing values present in the Rose data which we have done imputation with forward filling.

5. Sparkling Data

5.1. Understand the data by plotting

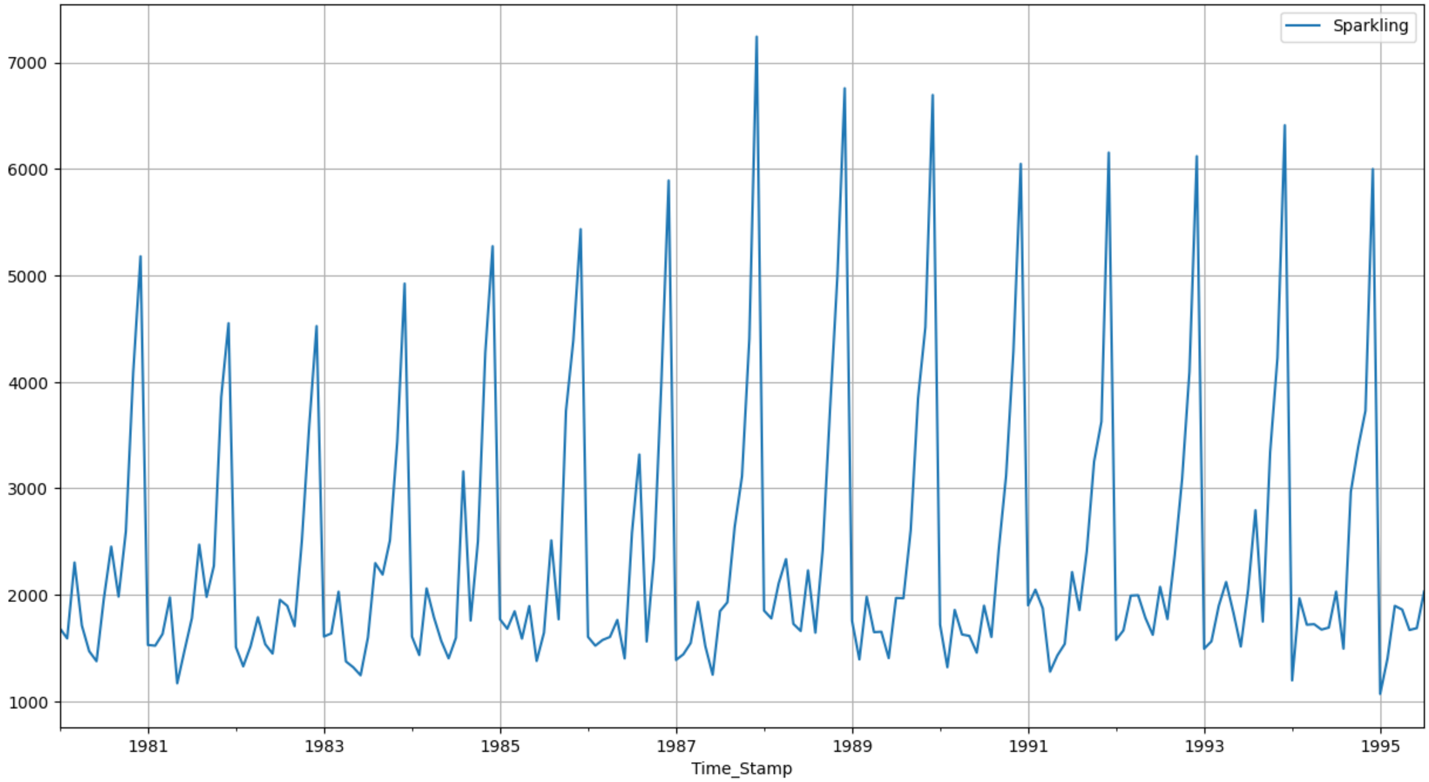

Figure 1: Trend of the sparkling wine sale over the past 8 years

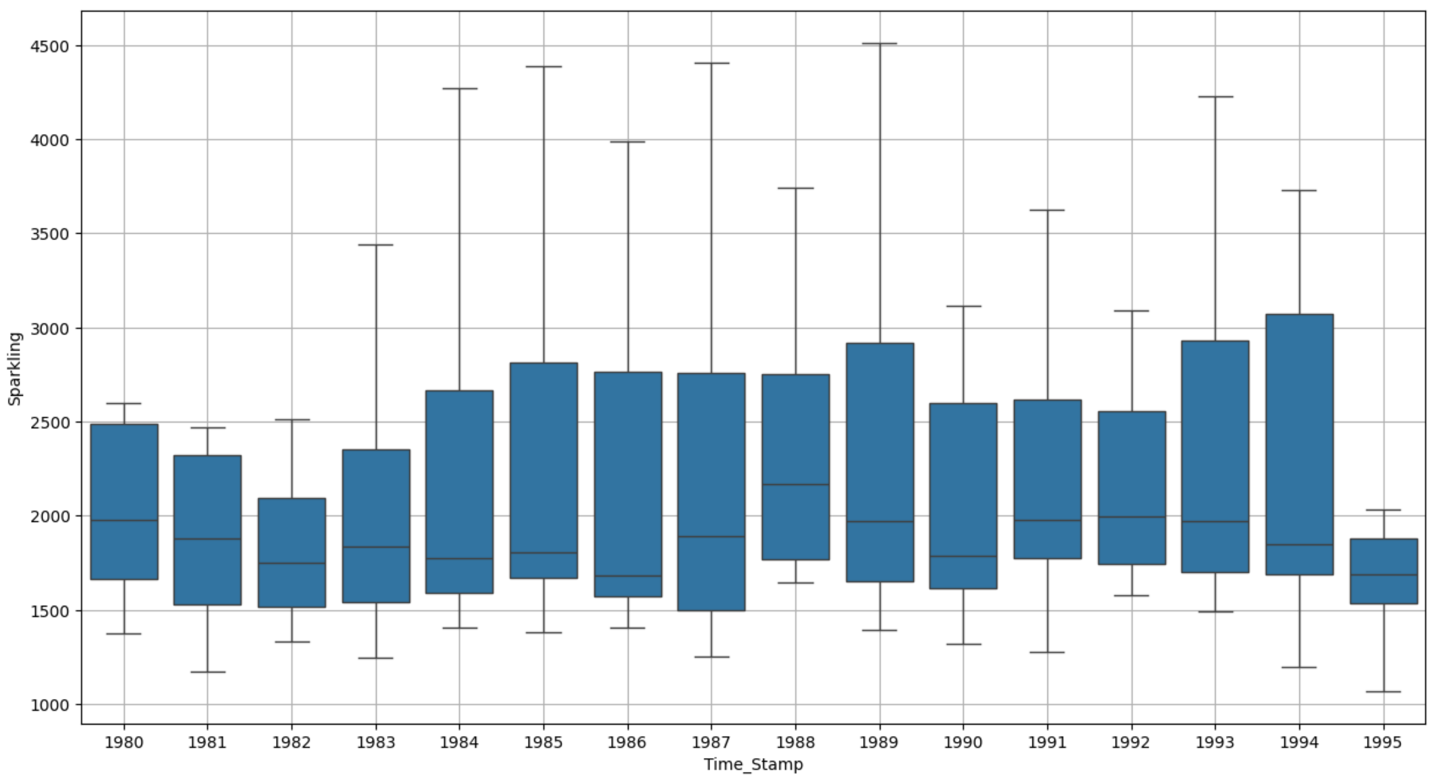

Figure 2: Box plot for sparkling wine sale over the years

- Sale of sparkling wine was at steady over the years.

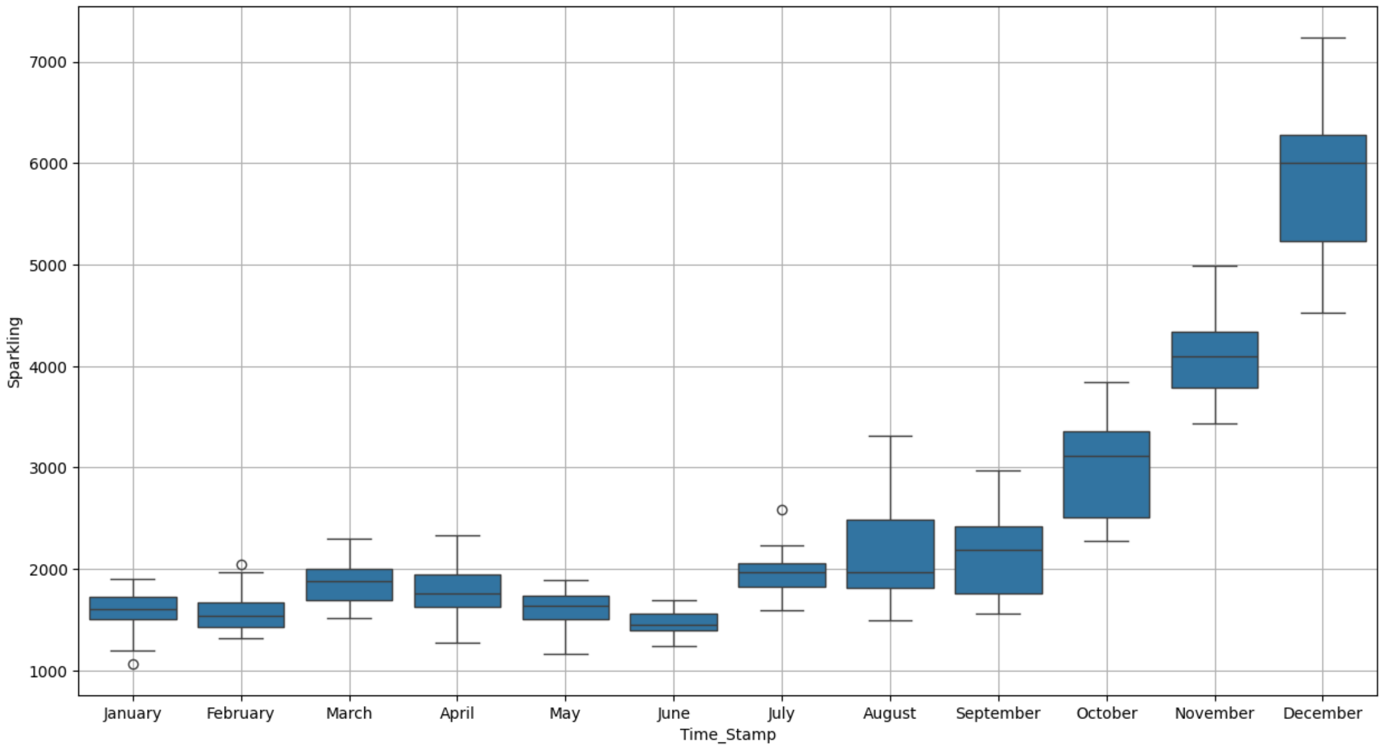

Figure 3: Box plot for sparkling wine sale trend in each month

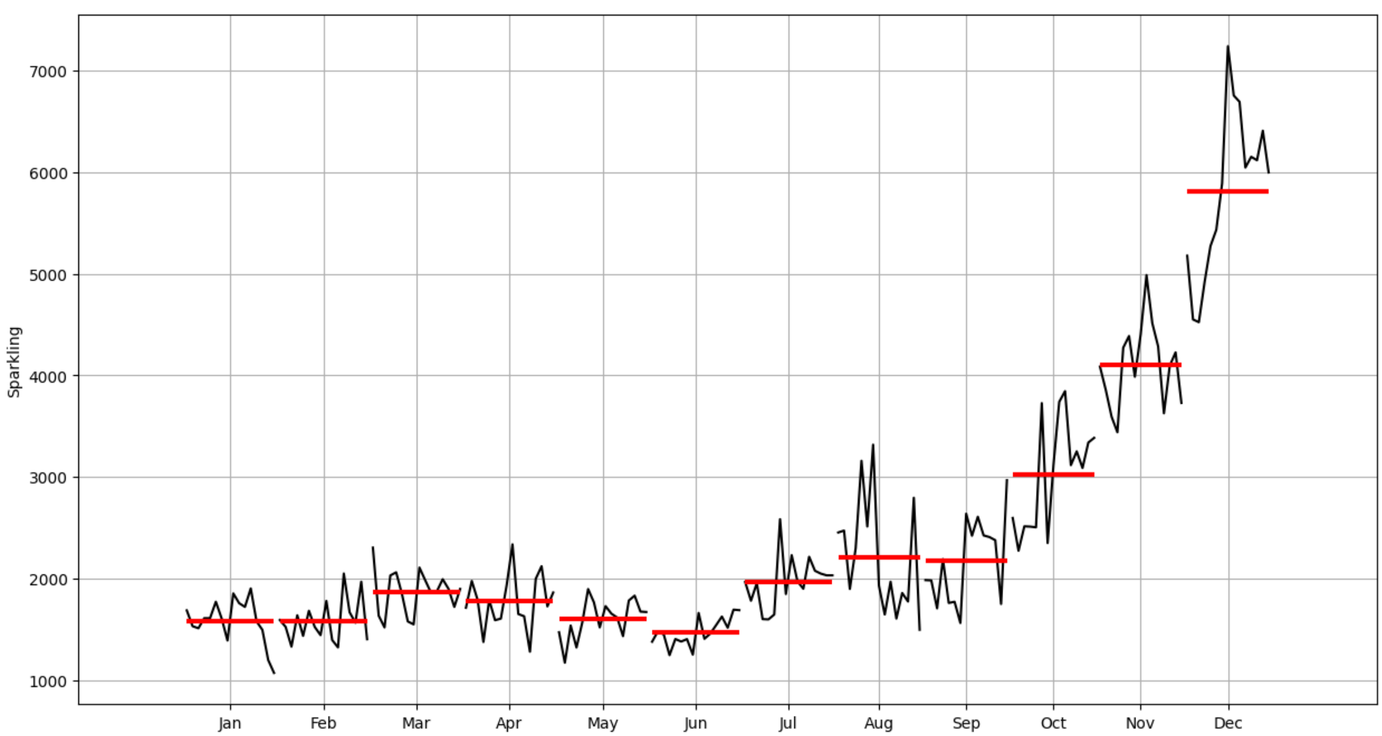

Figure 4: Trend for each month sparkling wine sale

- Form October till December the peak if sale increases and it is the season for the sale of sparkling wine.

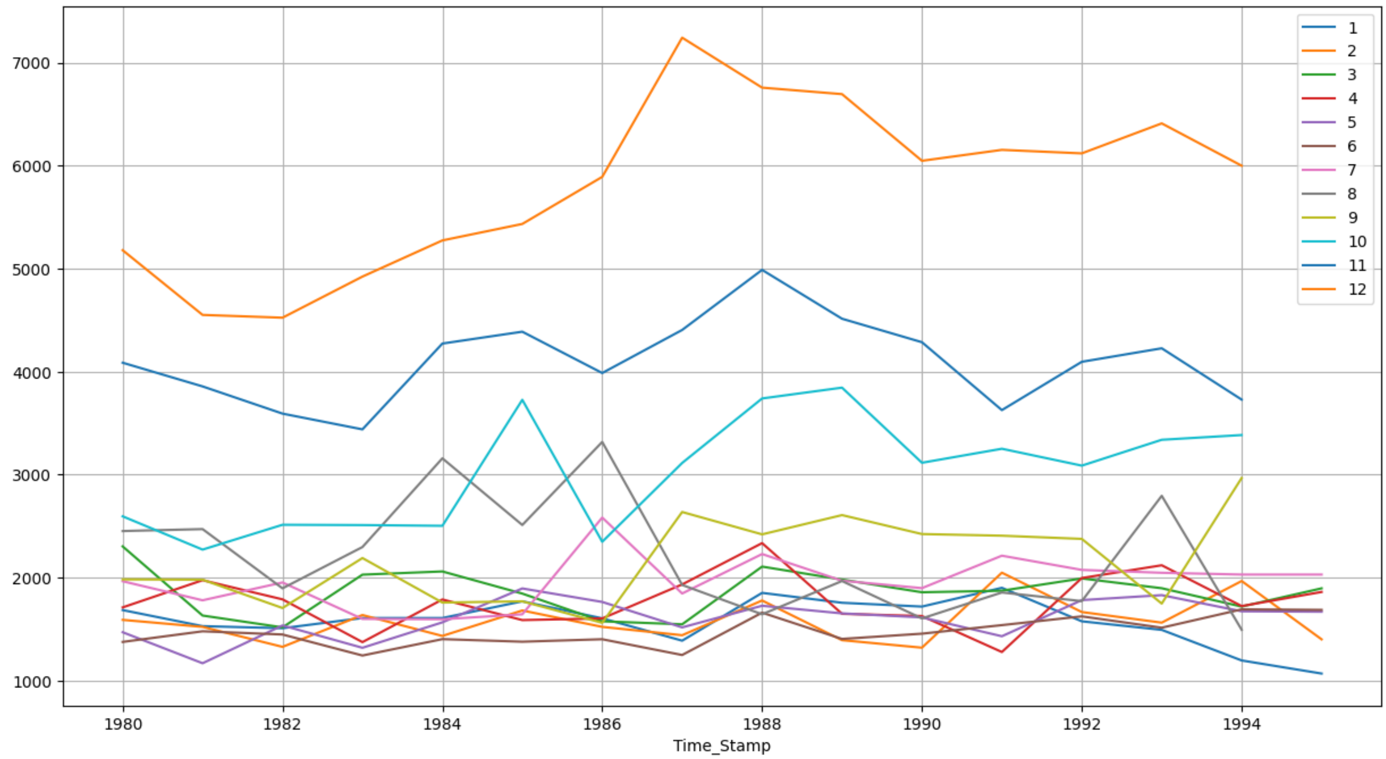

Figure 5: Monthly Sparkling wine sale across the years

- Trend shows that it is not additive.

5.2. Cumulative Distribution

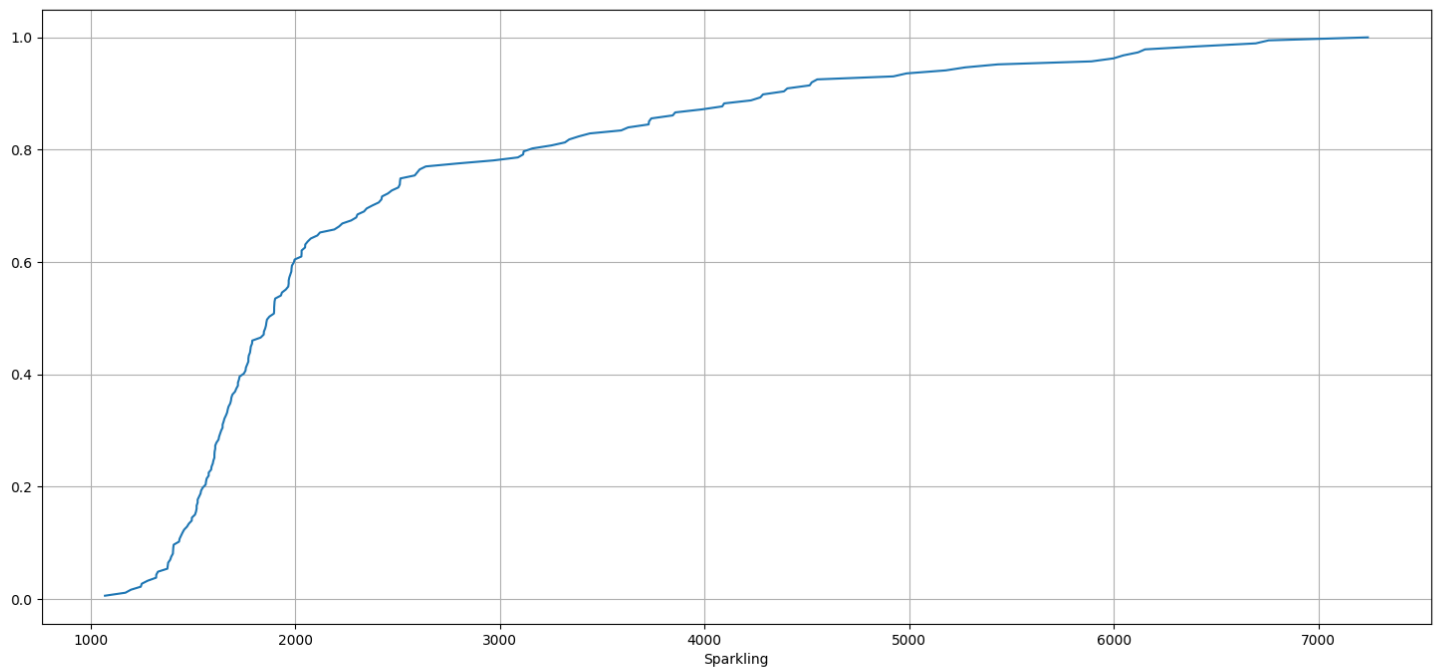

Figure 6: Distribution of sparkling data

- Over the years the sale has increased.

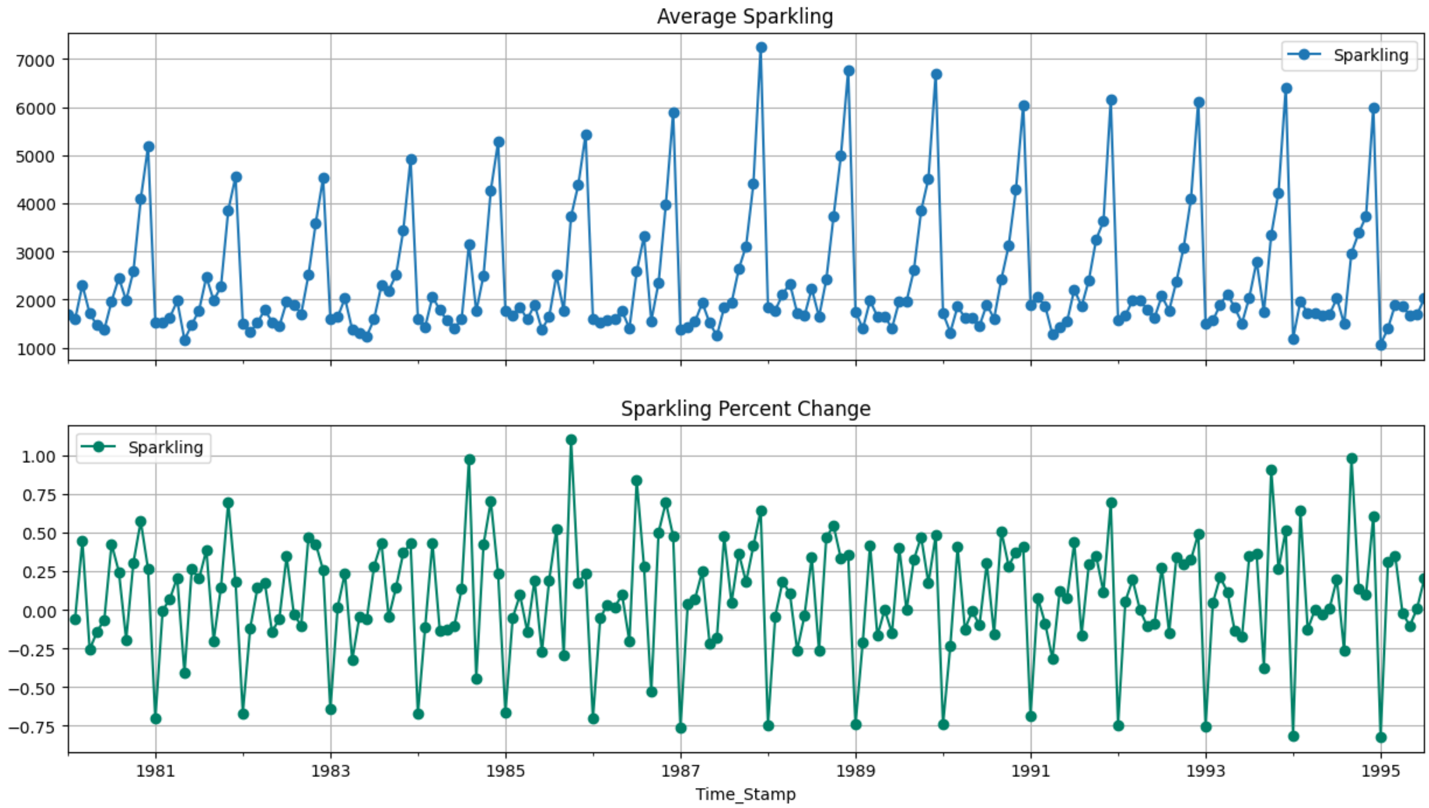

Figure 7: Average 'Sparkling' and the Percentage change of 'Sparkling' with respect to the time

- Stady trend can be observed over the years for sparkling wine sales.

5.3. Decompose the Time Series and plot different components

5.3.1. Additive

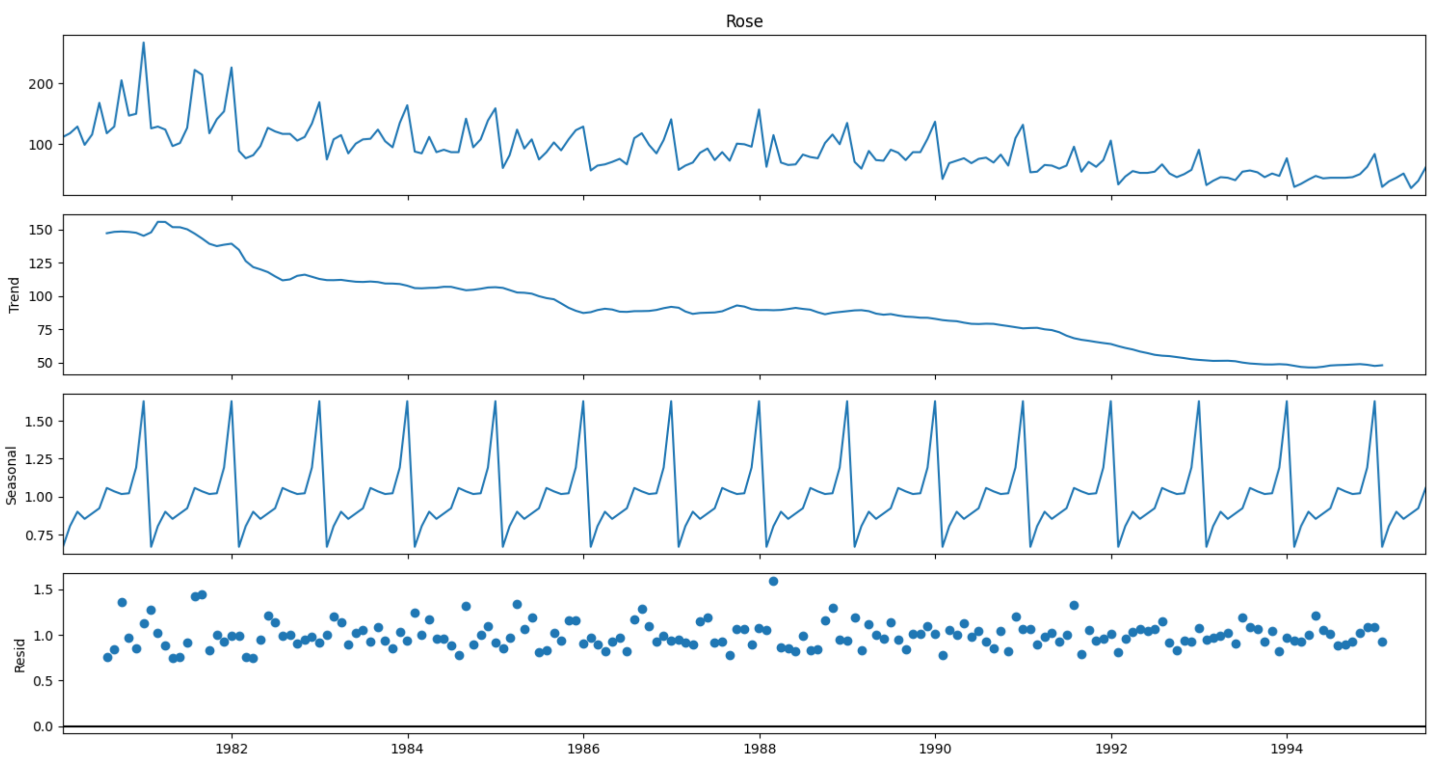

Figure 8: Additive decomposition of sparkling wine sale

- We see that the residuals are located around 0 from the plot of the residuals in the decomposition.

5.3.2. Multiplicative

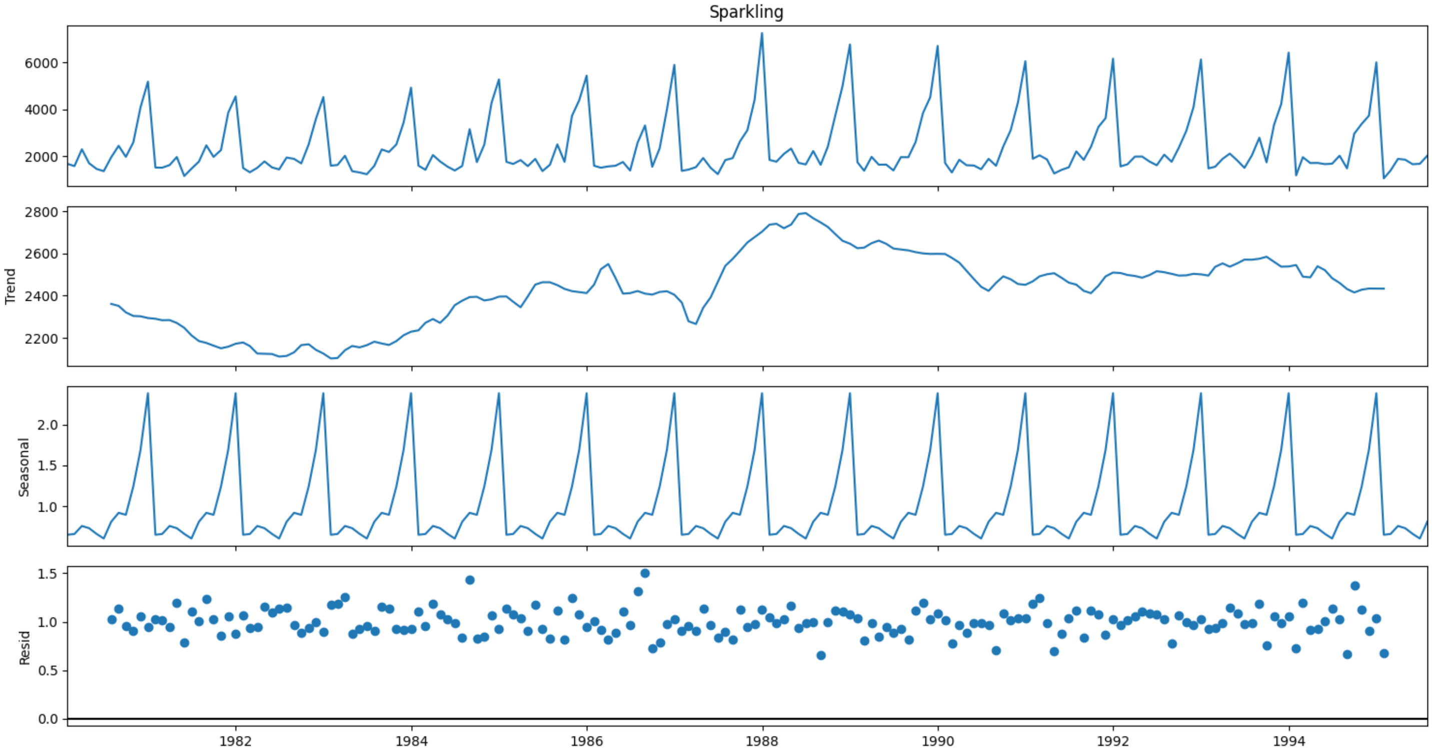

Figure 9: Multiplicative decomposition of sparkling wine sale

- For the multiplicative series, we see that a lot of residuals are located around 1. Thus, Multiplicative Decomposition is the right way to decompose the time series

5.4. Check for Stationarity of the Data

5.4.1. Whole data

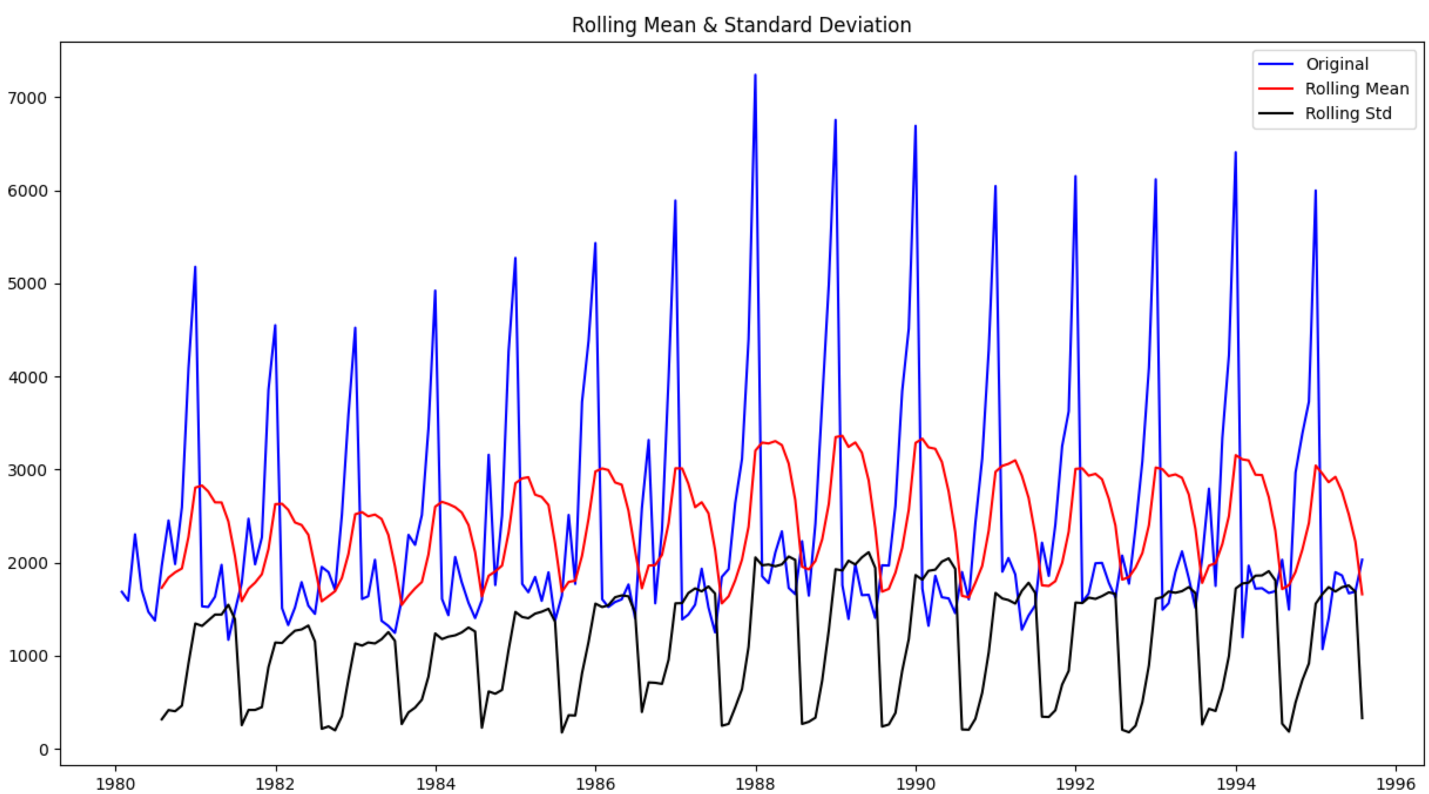

Figure 10: Stationarity of the sparkling data at α = 0.05

Results of Dickey-Fuller Test:

Test Statistic	-1.360497
p-value	0.601061
Lags Used	11.000000
Number of Observations Used	175.000000
Critical Value (1%)	-3.468280
Critical Value (5%)	-2.878202
Critical Value (10%)	-2.575653

- We see that at 5% significant level the Time Series is non-stationary.

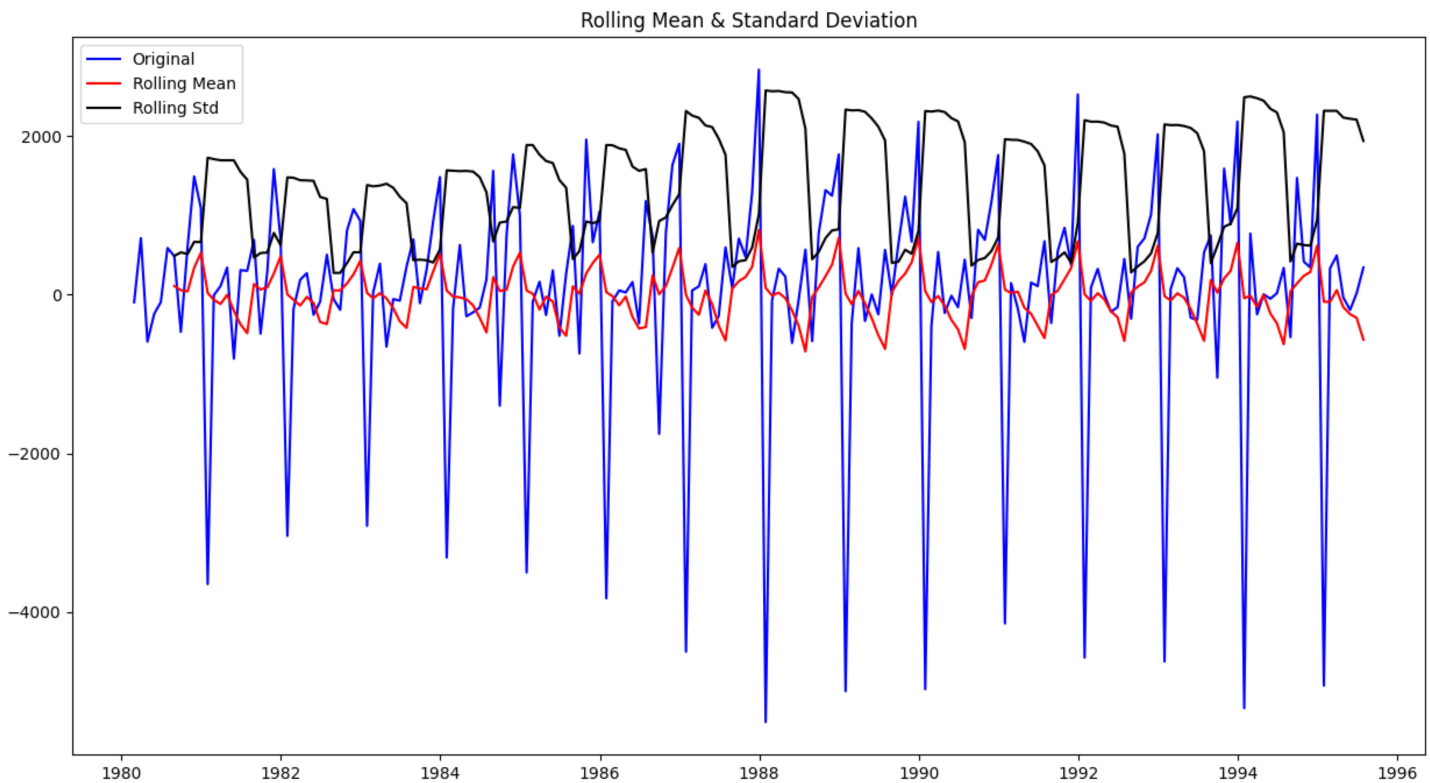

Figure 11: Stationarity of the sparkling data at order 1 and α = 0.05

Results of Dickey-Fuller Test:

Test Statistic	-45.050301
p-value	0.000000
Lags Used	10.000000
Number of Observations Used	175.000000
Critical Value (1%)	-3.468280
Critical Value (5%)	-2.878202
Critical Value (10%)	-2.575653

- We see that at $\alpha = 0.05$ the Time Series is indeed stationary.

5.4.2. Train Data

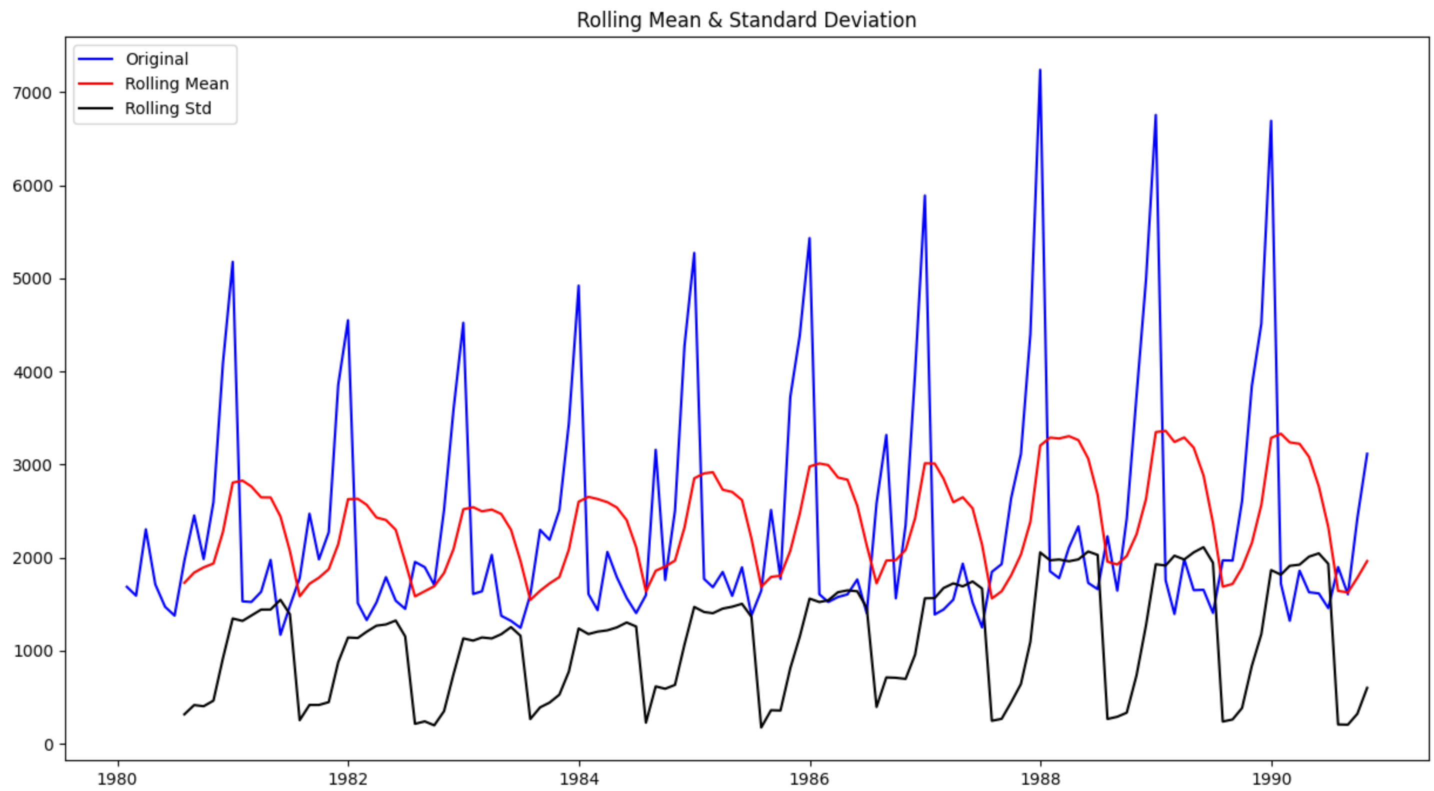

Figure 12: Stationarity of the training sparkling data at α = 0.05

Results of Dickey-Fuller Test:

Test Statistic	-1.155971
p-value	0.692243
#Lags Used	12.000000
Number of Observations Used	117.000000
Critical Value (1%)	-3.487517
Critical Value (5%)	-2.886578
Critical Value (10%)	-2.580124

- We see that the series is not stationary at $\alpha = 0.05$.

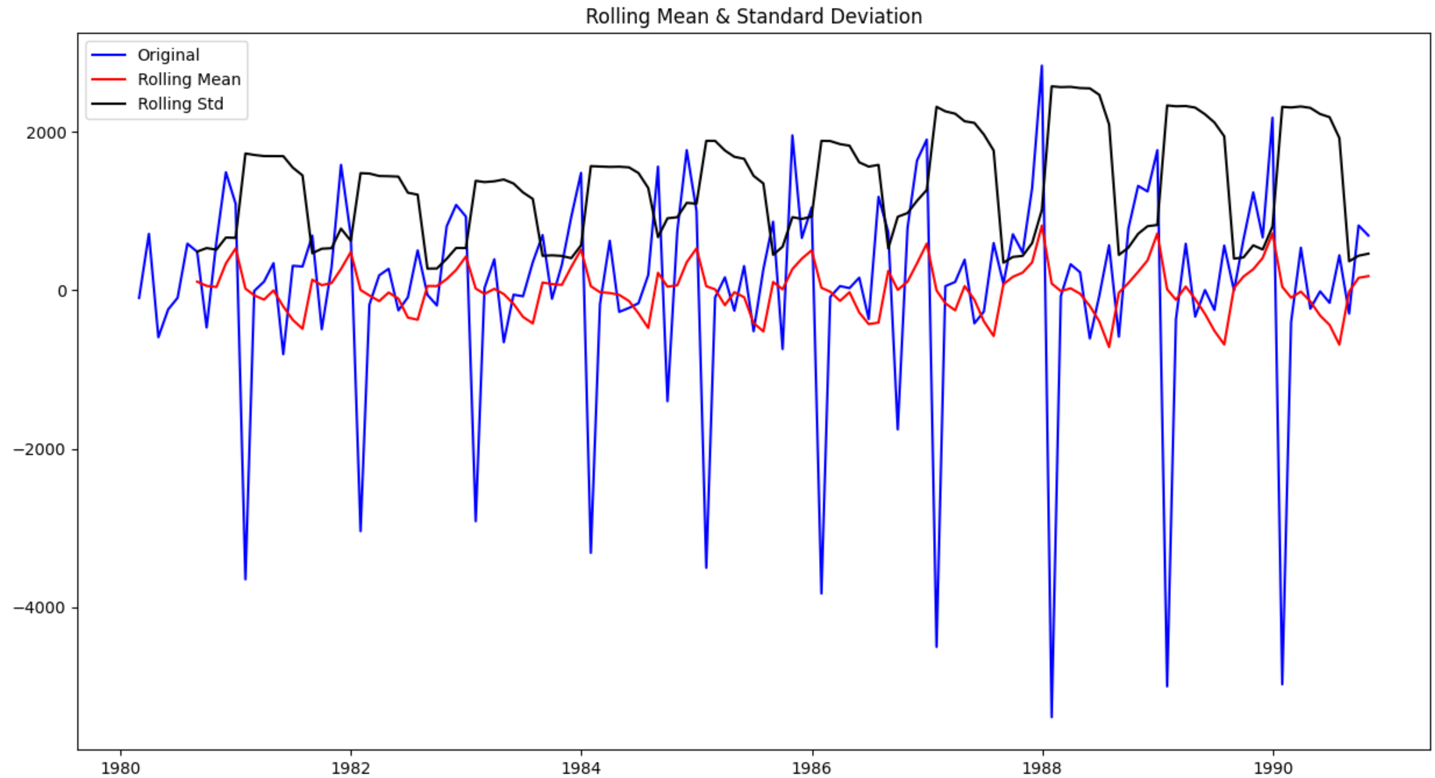

Figure 13: Stationarity of the training sparkling data at order 1 and α = 0.05

Results of Dickey-Fuller Test:

Test Statistic	-7.876510e+00
p-value	4.827963e-12
#Lags Used	1.100000e+01
Number of Observations Used	1.170000e+02
Critical Value (1%)	-3.487517e+00
Critical Value (5%)	-2.886578e+00
Critical Value (10%)	-2.580124e+00

- We see that after taking a difference of order 1 the series have become stationary at $\alpha = 0.05$

5.5. Autocorrelation Function (ACF)

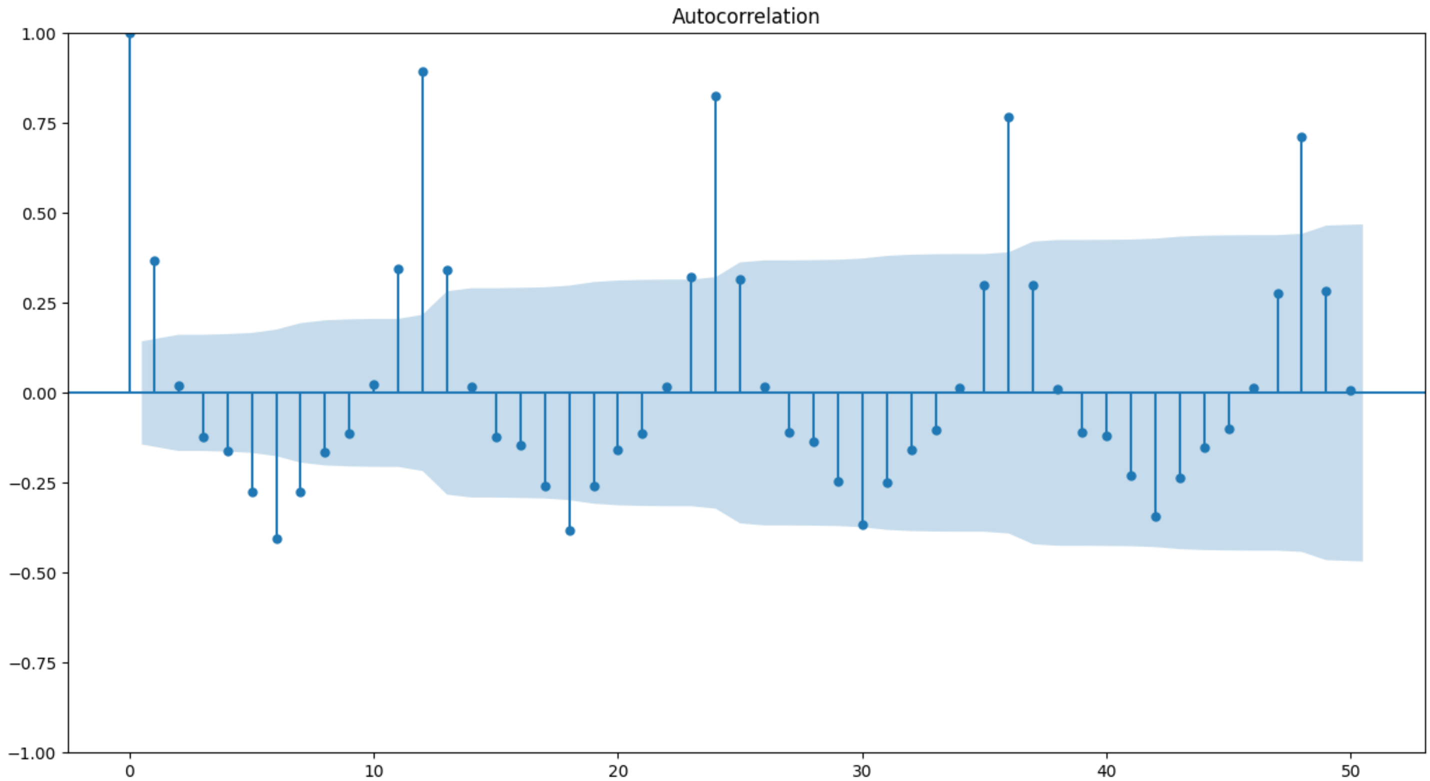

Figure 14: Autocorrelation Function Plot of whole sparkling data

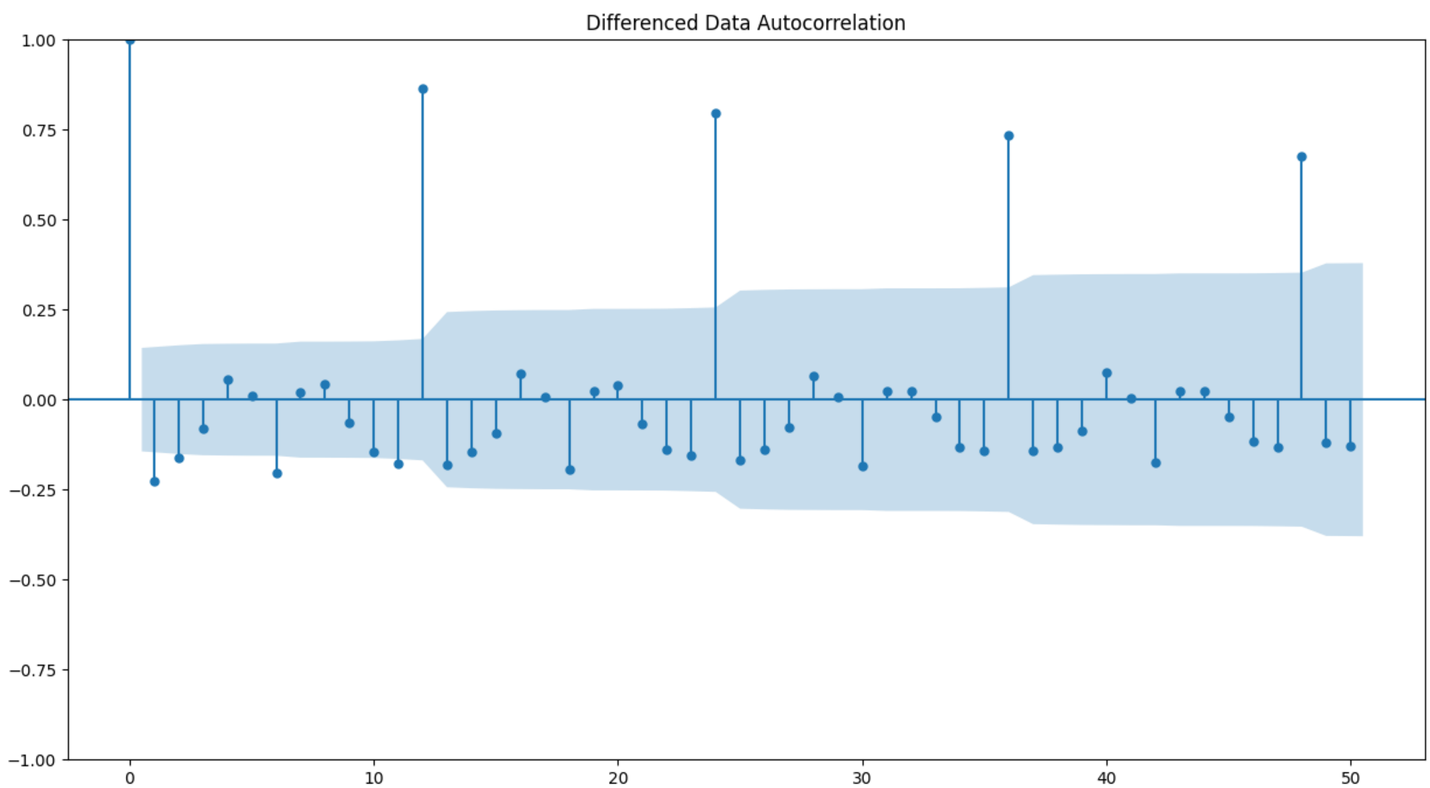

Figure 15: Autocorrelation Function Plot of whole sparkling data at difference 1

From the above plots, we can say that there seems to be a seasonality in the data.

We see that there can be a seasonality of 12. But from the decomposition at the start we ascertained that visually it looks like the seasonality =12 and thus using the same

5.6. Data Split

We have did train and test split at a ratio of 70 : 30 respectively.

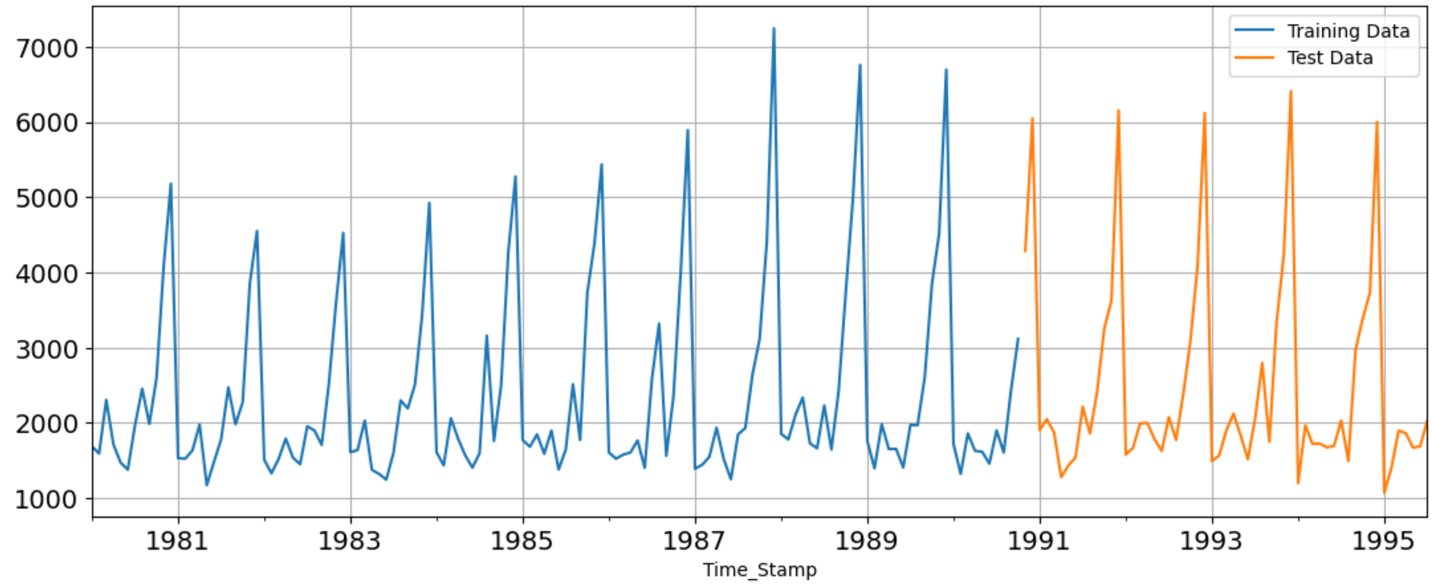

Figure 16: Plot of train and test data

5.7. Model Building

5.7.1. Linear Regression Model

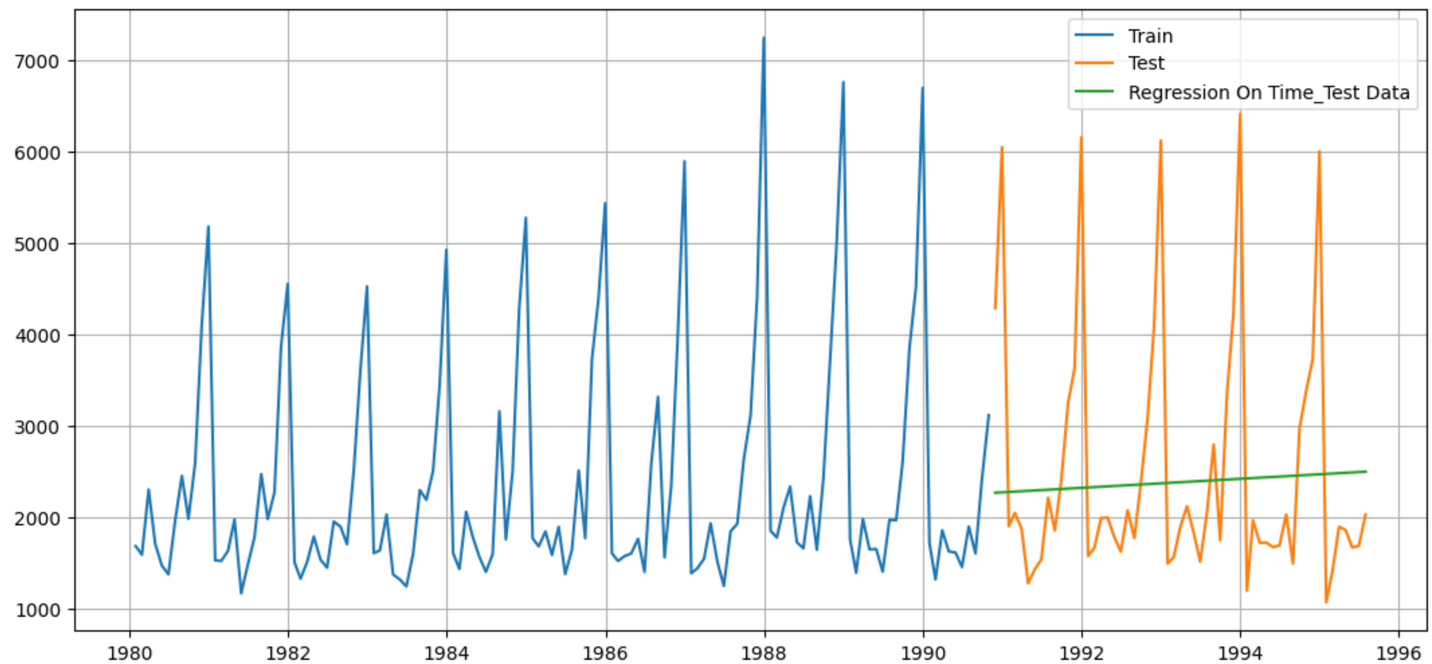

Figure 17: Linear regression performance on Sparkling data

For RegressionOnTime forecast on the Test Data, RMSE is 1374.55

5.7.2. Simple Average

We have calculated the mean of all the scores to plot the simple average.

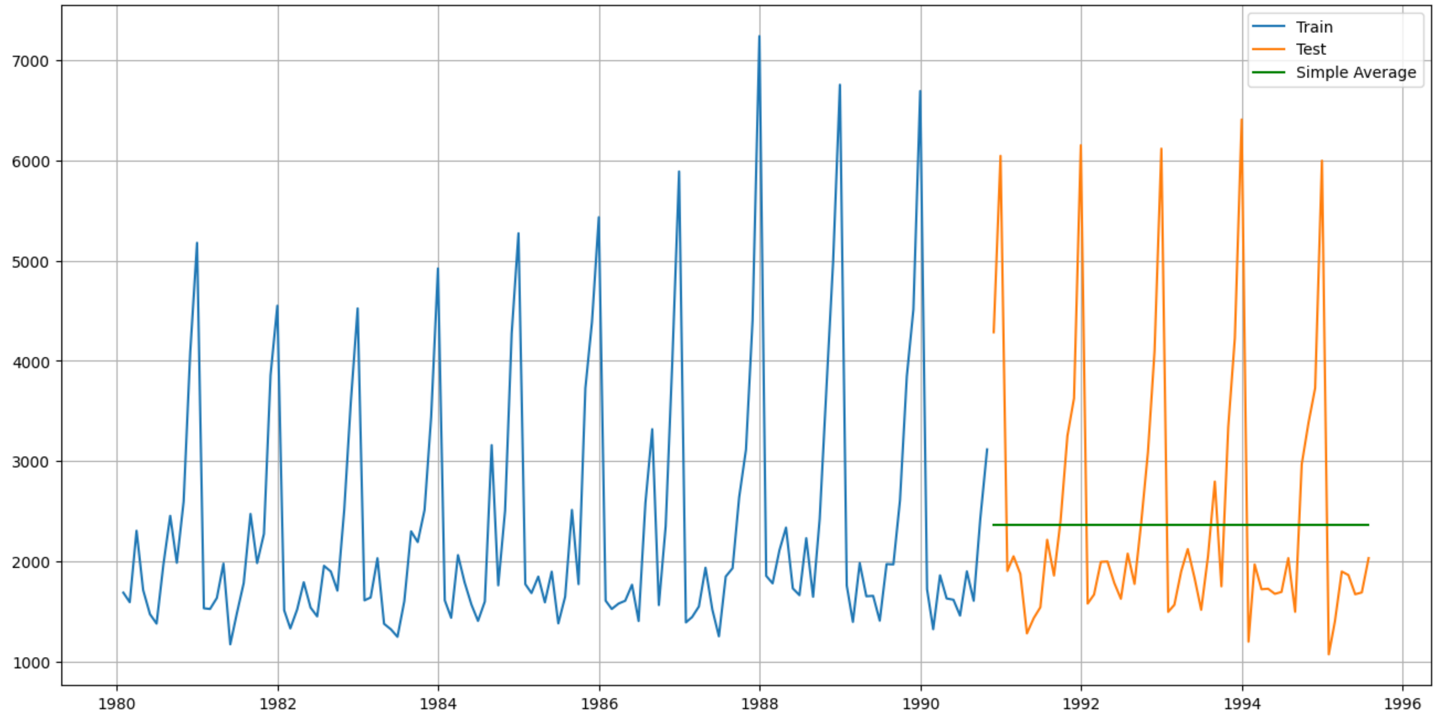

Figure 18: Simple Average for Sparkling Data

- For Simple Average forecast on the Test Data, RMSE is 1368.75

5.7.3. Moving Average (MA)

For the moving average model, we are going to calculate rolling means (or moving averages) for different intervals. The best interval can be determined by the maximum accuracy (or the minimum error) over here.

- For Moving Average, we are going to average over the entire data

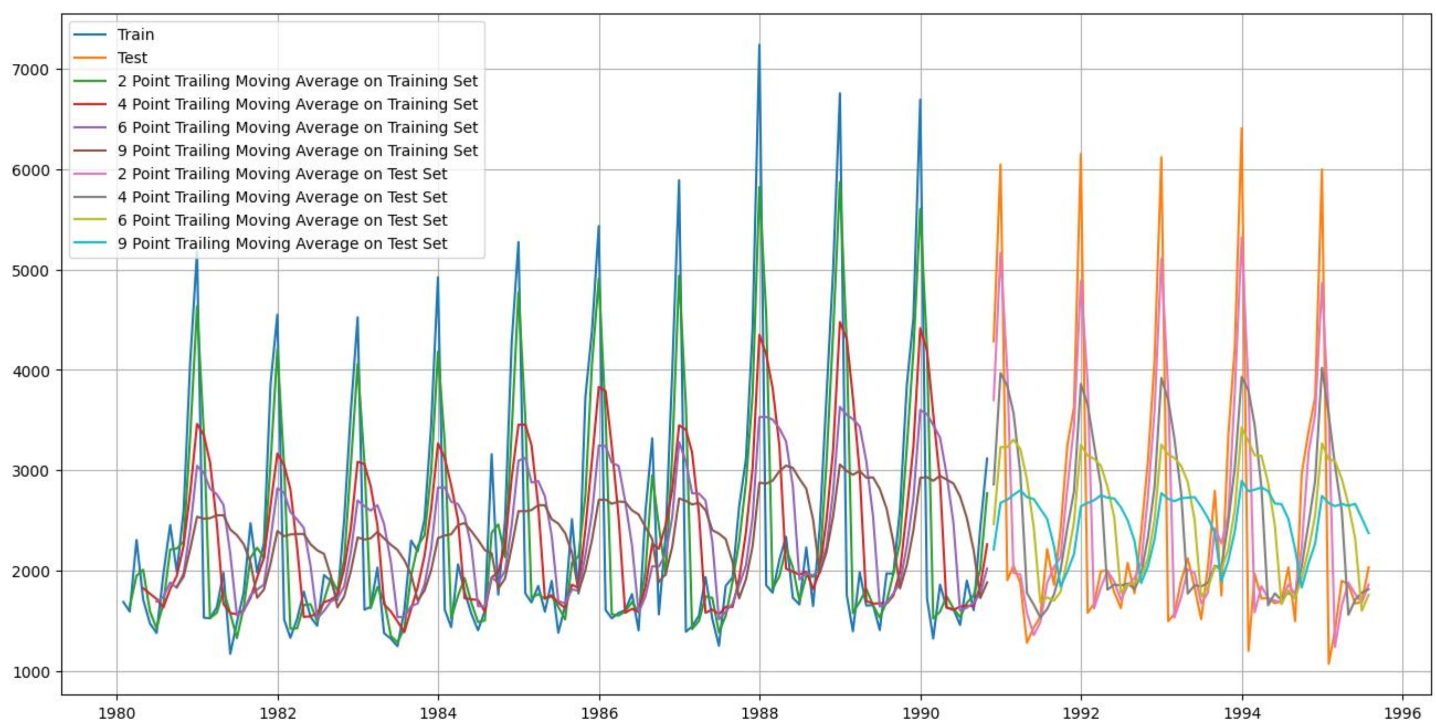

Figure 19: Moving Average Model performance on Sparkling data

For 2 point Moving Average Model forecast on the Training Data, RMSE is 811.179

For 4 point Moving Average Model forecast on the Training Data, RMSE is 1184.213

For 6 point Moving Average Model forecast on the Training Data, RMSE is 1337.201

For 9 point Moving Average Model forecast on the Training Data, RMSE is 1422.653

5.7.4. Model comparison

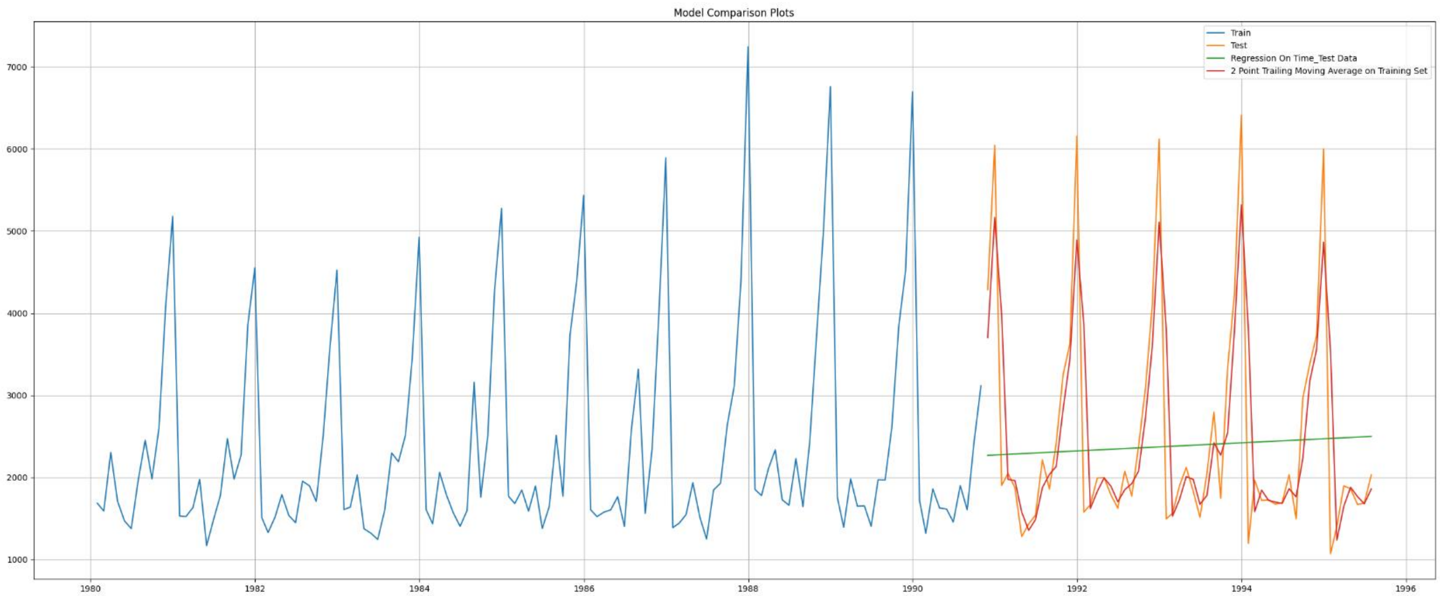

Figure 20: Model performance comparison 1 on sparkling data

- 2-point trailing is performing well than linear regression.

5.7.5. Simple Exponential Smoothing (SES)

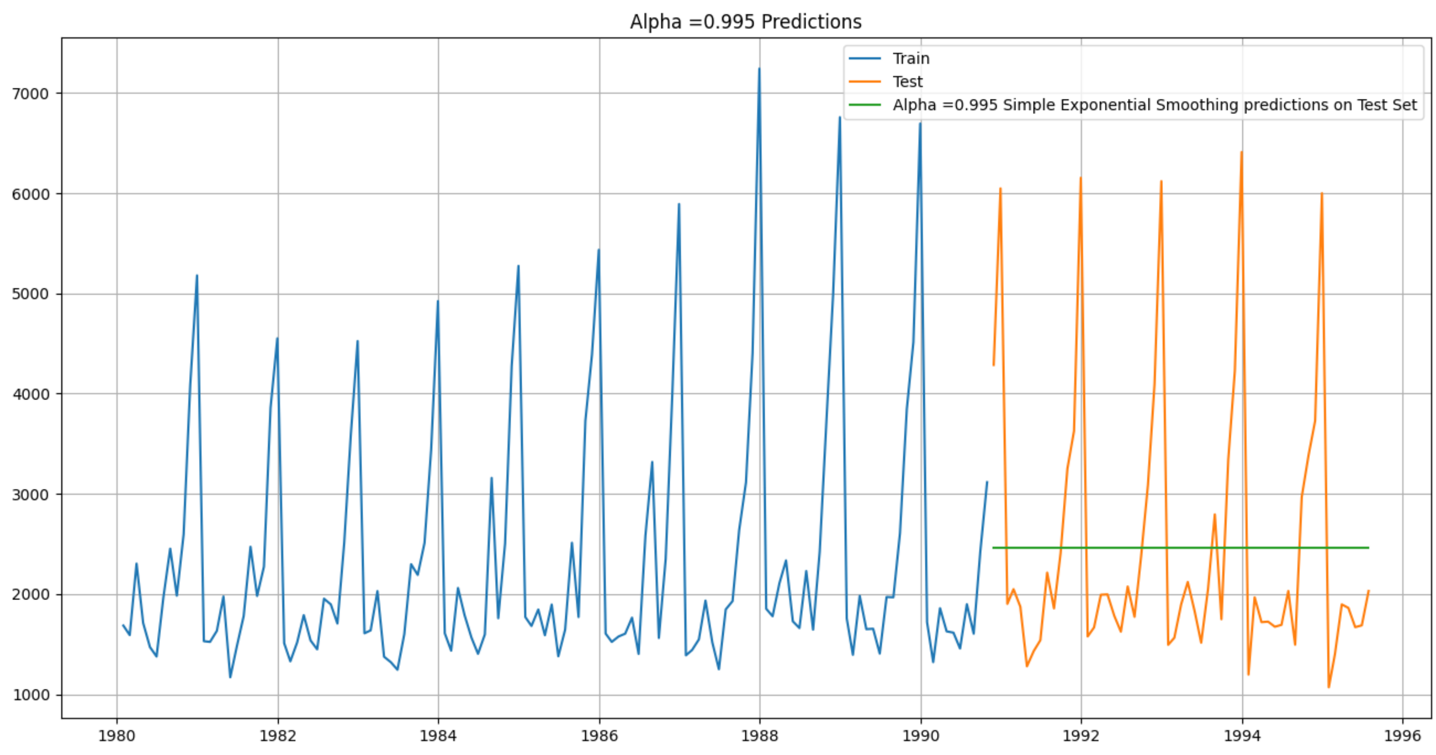

Figure 21: SES model performance (α=0.995) on Sparkling Data

- For Alpha =0.995 Simple Exponential Smoothing Model forecast on the Test Data, RMSE is 1362.429

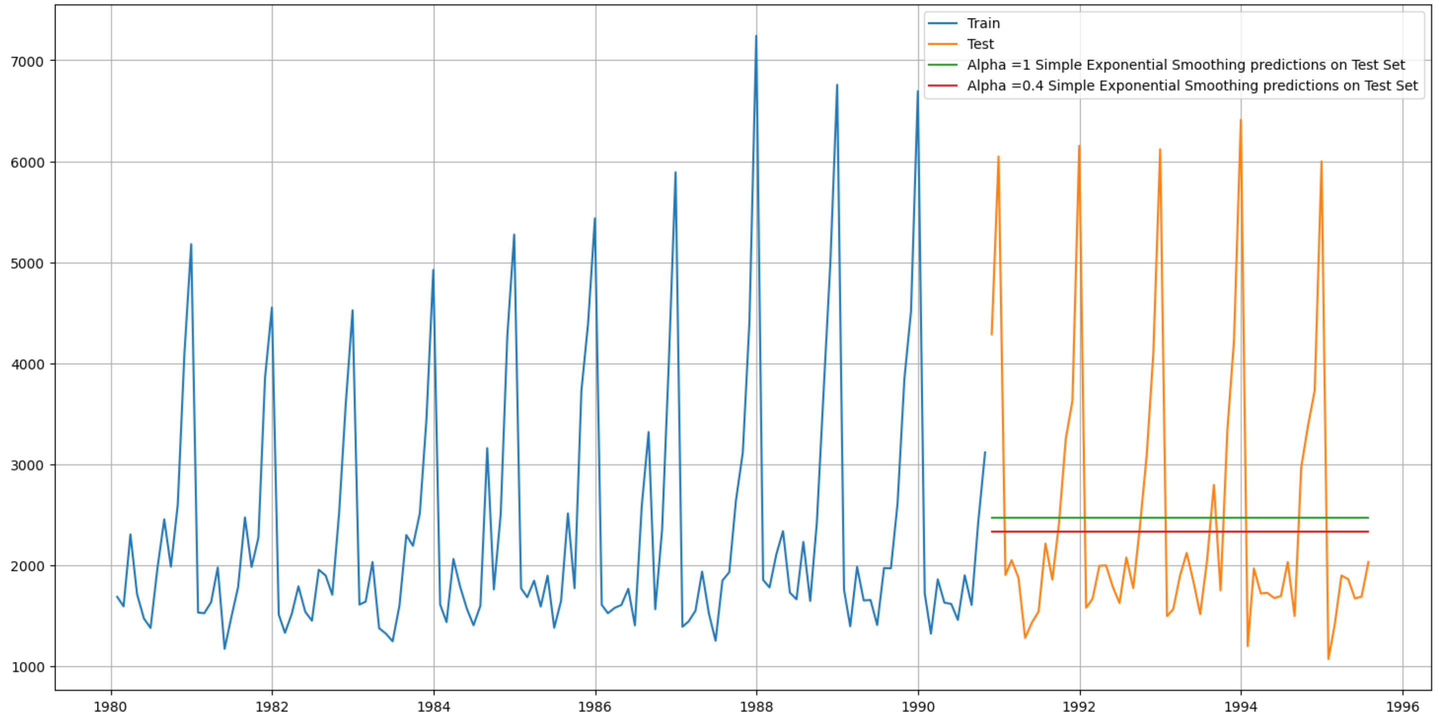

Figure 22: SES model performance (α=0.4) on Sparkling Data

RMSE scores:

Alpha=0.995, Simple Exponential Smoothing 1362.428949

Alpha=0.4, Simple Exponential Smoothing 1363.037803

- At Alpha = 0.995 the SES model is giving better RMSE score than the Alpha=0.4.

5.7.6. Double Exponential Smoothing (Holt's Model)

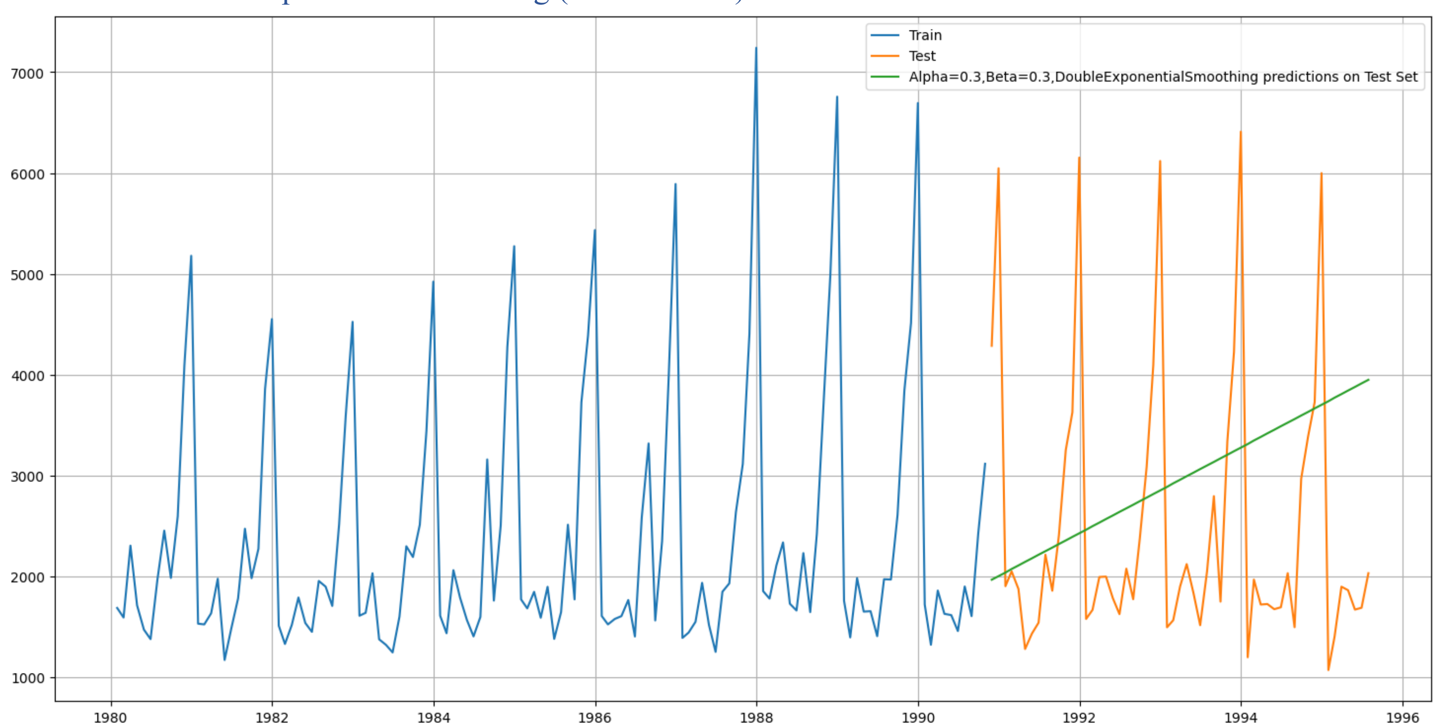

Figure 23: DES model performance in Sparkling data

Alpha=0.3, Beta=0.3, Double Exponential Smoothing 1597.853999

5.7.7. Triple Exponential Smoothing (Holt - Winter's Model)

TES autofit model gave us below best values for alpha, beta, gamma:
smoothing_level: 0.07571445210103464, smoothing_trend: 0.06489808813237438, smoothing_seasonal: 0.3765608370780376

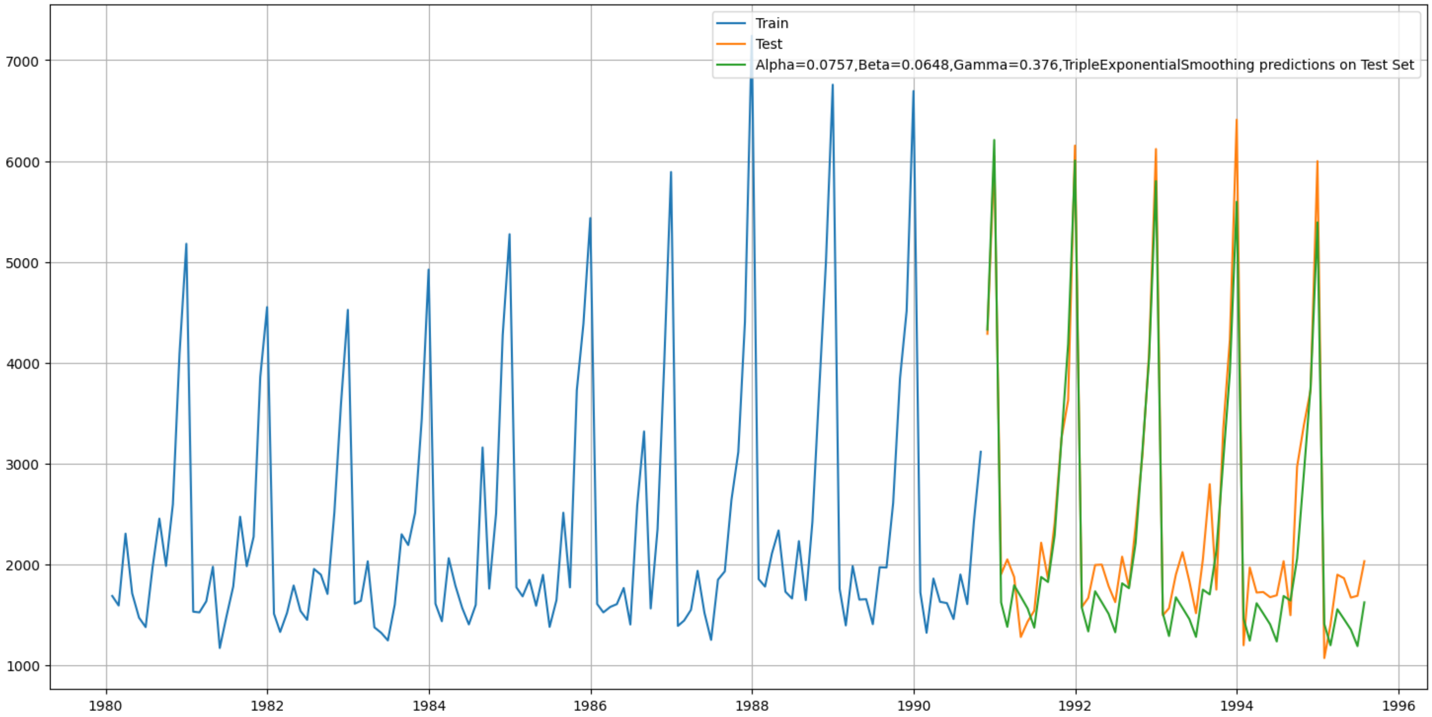

Figure 24: TES Performance on sparkling data

- For Alpha=0.0757, Beta=0.0648, Gamma=0.376, Triple Exponential Smoothing Model forecast on the Test Data, RMSE is 381.657
- For Alpha=0.075, Beta=0.064, Gamma=0.376, Triple Exponential Smoothing Model forecast on the Test Data, RMSE is 381.657232

5.7.8. Exponential Smoothing Models Comparison

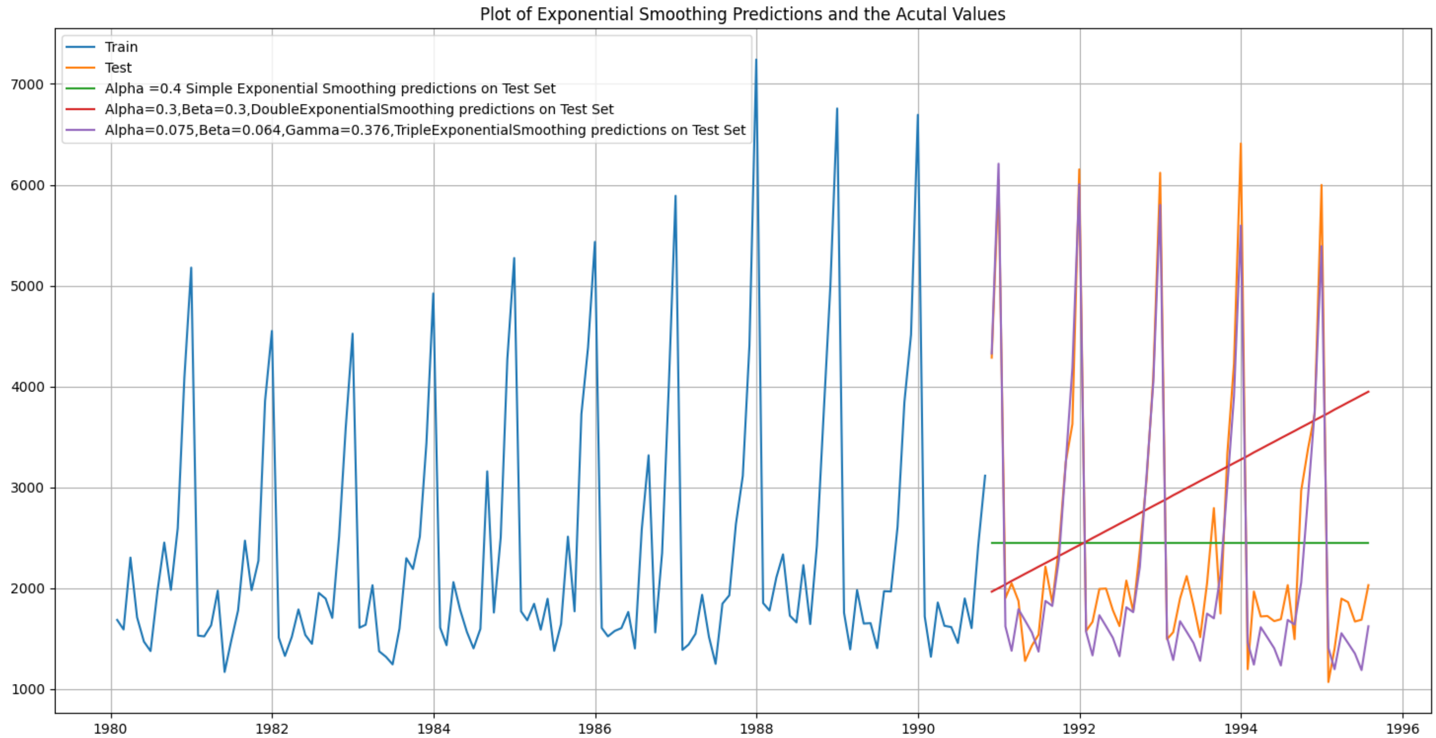

Figure 25: Exponential Smoothing Models Comparison on Sparkling Data

In this particular we have built several models and went through a model building exercise. This particular exercise has given us an idea as to which particular model gives us the least error on our test set for this data. But in Time Series Forecasting, we need to be very vigil about the fact that after we have done this exercise we need to build the model on the whole data. Remember, the training data that we have used to build the model stops much before the data ends. In order to forecast using any of the models built, we need to build the models again (this time on the complete data) with the same parameters. For this particular mentored learning session, we will go ahead and build only the top 2 models which gave us the best accuracy (least RMSE).

The best model to be built on the whole data is:

Alpha=0.075,Beta=0.064,Gamma=0.376,Triple Exponential Smoothing

5.7.9. ARMA Model

We are trying to build an Automated version of an ARMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

param	AIC
(2, 0, 1)	2197.084
(0, 0, 1)	2204.87
(2, 0, 0)	2204.881
(0, 0, 2)	2206.111
(1, 0, 1)	2206.142
(1, 0, 2)	2207.163
(1, 0, 0)	2207.502
(2, 0, 2)	2208.121
(0, 0, 0)	2228.484

Table 1: AIC values of Automatic search for best ARMA models for Rose Data

- The lowest value found to be at (2, 0, 1).

```
                               SARIMAX Results
==============================================================================
Dep. Variable:              Sparkling   No. Observations:                  130
Model:                 ARIMA(2, 0, 1)   Log Likelihood               -1093.542
Date:                Thu, 05 Dec 2024   AIC                           2197.084
Time:                        16:16:56   BIC                           2211.422
Sample:                    01-31-1980   HQIC                          2202.910
                         - 10-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       2379.9376    112.866     21.086      0.000    2158.724    2601.151
ar.L1          1.2114      0.135      8.991      0.000       0.947       1.475
ar.L2         -0.4998      0.124     -4.046      0.000      -0.742      -0.258
ma.L1         -0.8128      0.152     -5.349      0.000      -1.111      -0.515
sigma2      1.182e+06   1.28e+05      9.214      0.000    9.31e+05    1.43e+06
===================================================================================
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):                39.43
Prob(Q):                              0.90   Prob(JB):                         0.00
Heteroskedasticity (H):               2.19   Skew:                             0.96
Prob(H) (two-sided):                  0.01   Kurtosis:                         4.90
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 26: ARMA Model Summery on sparkling train data

- ARMA model RMSE score on test data is 1338.139903948421

5.7.10. ARIMA Model

We will build an Automated version of an ARIMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

param	AIC
(2, 1, 2)	2178.11
(2, 1, 1)	2193.975
(0, 1, 2)	2194.034
(1, 1, 2)	2194.96
(1, 1, 1)	2196.05
(0, 1, 1)	2217.939
(2, 1, 0)	2223.899
(1, 1, 0)	2231.138
(0, 1, 0)	2232.719

Table 2:AIC values of Automatic search for best ARIMA models for Sparkling Data

- The lowest value found to be at (2, 1, 2).

```
                               SARIMAX Results
==============================================================================
Dep. Variable:              Sparkling   No. Observations:                  130
Model:                 ARIMA(2, 1, 2)   Log Likelihood               -1084.055
Date:                Thu, 05 Dec 2024   AIC                           2178.110
Time:                        16:17:00   BIC                           2192.409
Sample:                    01-31-1980   HQIC                          2183.920
                         - 10-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          1.3020      0.046     28.556      0.000       1.213       1.391
ar.L2         -0.5360      0.079     -6.744      0.000      -0.692      -0.380
ma.L1         -1.9914      0.110    -18.144      0.000      -2.207      -1.776
ma.L2          0.9997      0.110      9.069      0.000       0.784       1.216
sigma2      1.085e+06   2.04e-07   5.31e+12      0.000    1.08e+06    1.08e+06
===================================================================================
Ljung-Box (L1) (Q):                   0.10   Jarque-Bera (JB):                19.53
Prob(Q):                              0.75   Prob(JB):                         0.00
Heteroskedasticity (H):               2.30   Skew:                             0.71
Prob(H) (two-sided):                  0.01   Kurtosis:                         4.27
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 27: ARIMA Model Summery on sparkling train data

- ARIMA model RMSE score on test data is 1325.1667432185322

5.7.11. SARIMA Model

We will build an Automated version of a SARIMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

Earlier we have seen the seasonality is at 12.

param	seasonal	AIC
(1, 1, 2)	(2, 0, 2, 12)	1521.738
(1, 1, 2)	(1, 0, 2, 12)	1521.949
(2, 1, 2)	(2, 0, 2, 12)	1523.218
(2, 1, 2)	(1, 0, 2, 12)	1523.525
(0, 1, 2)	(2, 0, 2, 12)	1523.707

Figure 28:AIC values of Automatic search for best SARIMA models for Sparkling Data

- The lowest value found to be at param (1,1,2) and seasonal (2, 0, 2, 12)

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                    y   No. Observations:                  130
Model:             SARIMAX(1, 1, 2)x(1, 0, 2, 12)   Log Likelihood                -753.975
Date:                            Thu, 05 Dec 2024   AIC                           1521.949
Time:                                    16:18:30   BIC                           1540.324
Sample:                                         0   HQIC                          1529.390
                                             - 130
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.6325      0.242     -2.619      0.009      -1.106      -0.159
ma.L1         -0.1412      0.213     -0.663      0.507      -0.558       0.276
ma.L2         -0.7662      0.164     -4.661      0.000      -1.088      -0.444
ar.S.L12       1.0588      0.015     70.109      0.000       1.029       1.088
ma.S.L12      -0.5816      0.090     -6.437      0.000      -0.759      -0.405
ma.S.L24      -0.1154      0.113     -1.021      0.307      -0.337       0.106
sigma2      1.432e+05   2.06e+04      6.960      0.000    1.03e+05    1.83e+05
===================================================================================
Ljung-Box (L1) (Q):                   0.11   Jarque-Bera (JB):                 7.79
Prob(Q):                              0.74   Prob(JB):                         0.02
Heteroskedasticity (H):               1.44   Skew:                             0.23
Prob(H) (two-sided):                  0.29   Kurtosis:                         4.27
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 29: SARIMA Model Summery on sparkling train data

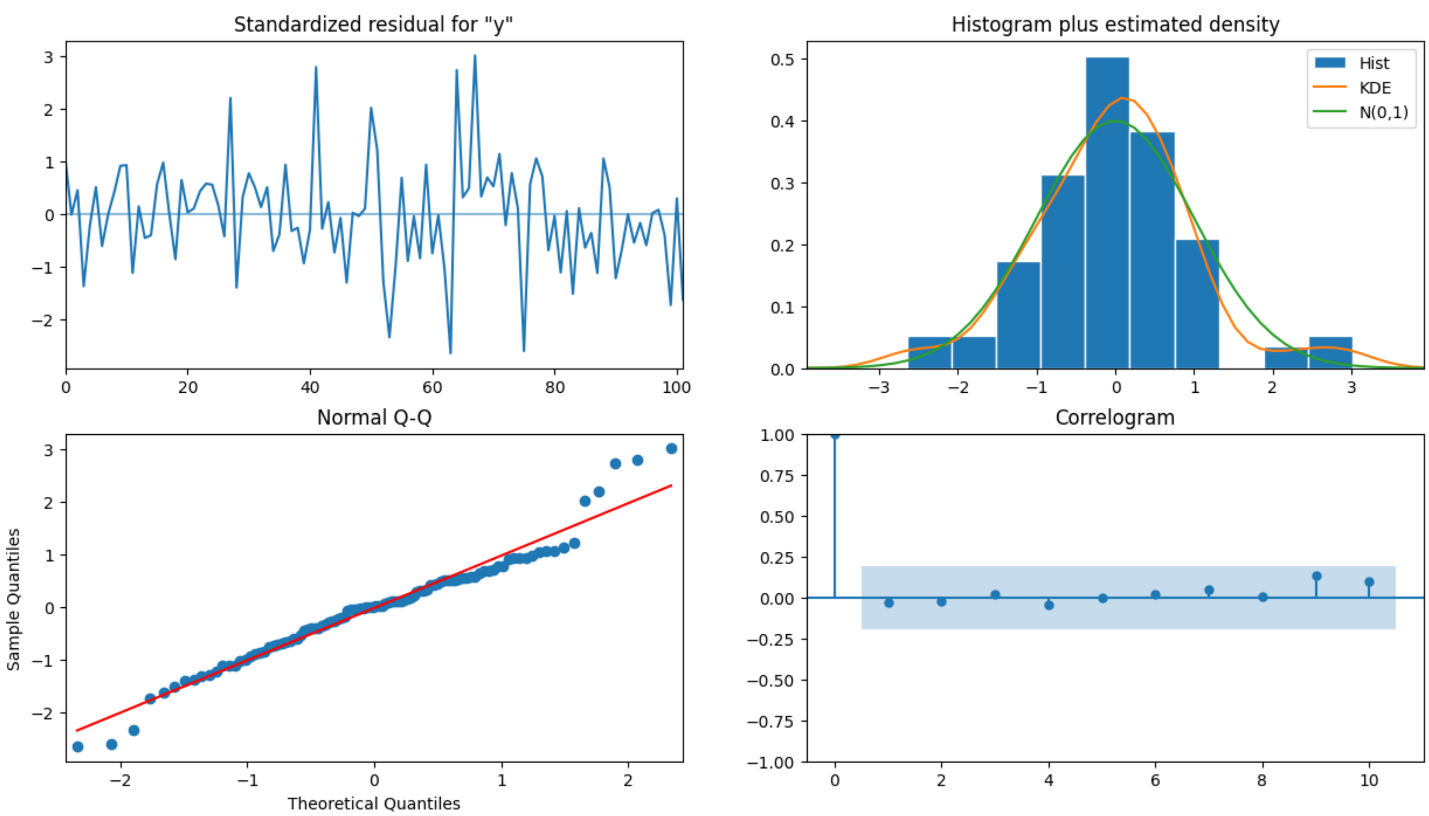

Figure 30: SARIMA Diagnostic Plot on Sparkling Train Data

- From the model diagnostics plot, we can see that all the individual diagnostics plots almost follow the theoretical numbers and thus we cannot develop any pattern from these plots.
- SARIMA model RMSE score on test data is 697.0976063790248

5.7.11.1. Optimum SARIMA model on the Full Data

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                            Sparkling   No. Observations:                  187
Model:             SARIMAX(1, 1, 2)x(1, 0, 2, 12)   Log Likelihood               -1173.413
Date:                              Thu, 05 Dec 2024   AIC                           2360.827
Time:                                      16:18:34   BIC                           2382.309
Sample:                                  01-31-1980   HQIC                          2369.551
                                       - 07-31-1995
Covariance Type:                                opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.6610      0.242     -2.734      0.006      -1.135      -0.187
ma.L1         -0.2739      0.200     -1.369      0.171      -0.666       0.118
ma.L2         -0.8113      0.227     -3.578      0.000      -1.256      -0.367
ar.S.L12       1.0157      0.012     84.456      0.000       0.992       1.039
ma.S.L12      -1.3873      0.338     -4.102      0.000      -2.050      -0.724
ma.S.L24      -0.1461      0.146     -1.001      0.317      -0.432       0.140
sigma2      5.948e+04   1.84e+04      3.231      0.001    2.34e+04    9.56e+04
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                27.47
Prob(Q):                              0.96   Prob(JB):                         0.00
Heteroskedasticity (H):               1.03   Skew:                             0.52
Prob(H) (two-sided):                  0.93   Kurtosis:                         4.76
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 31: Optimum SARIMA Model Summery on whole sparkling data

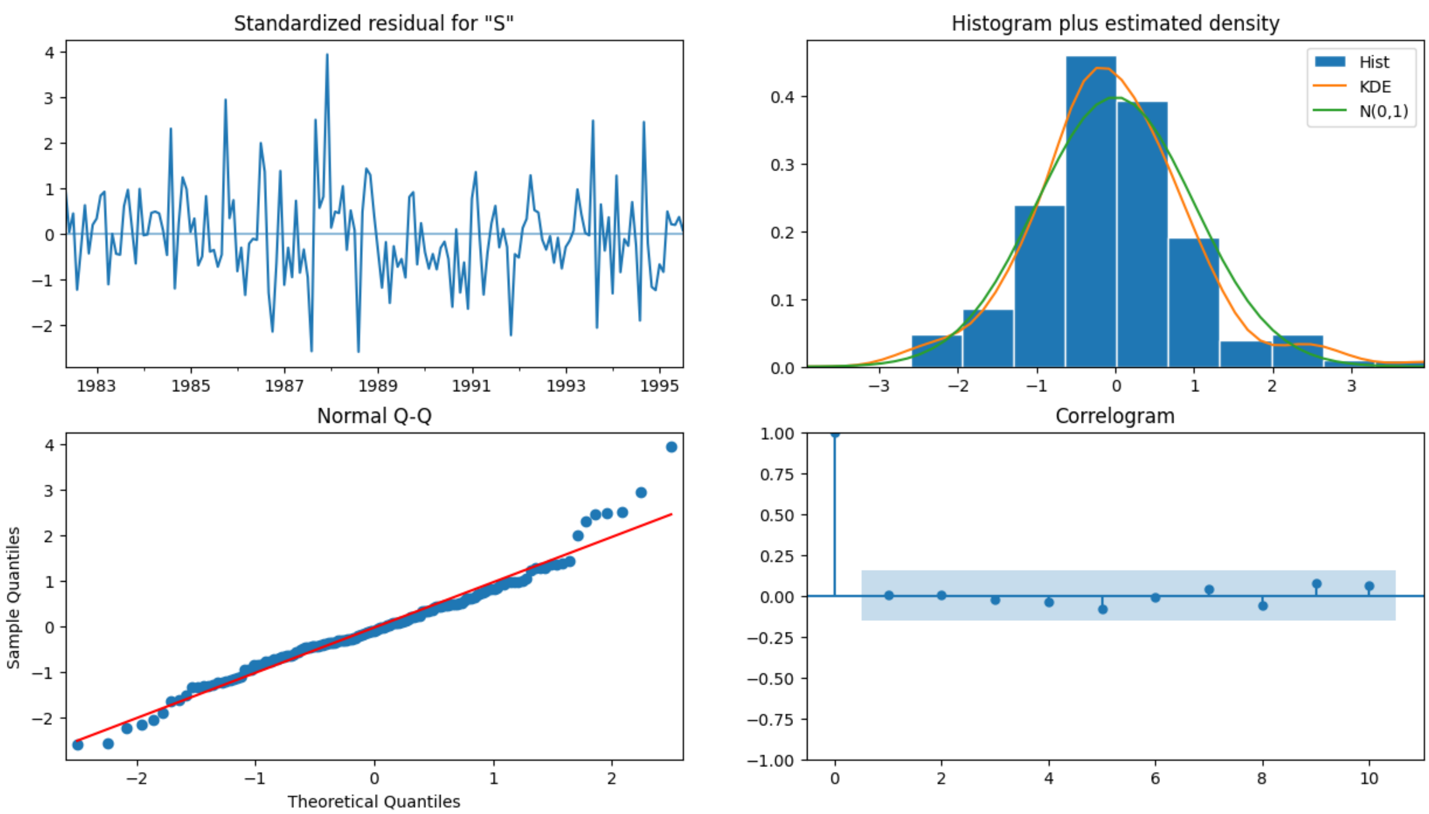

Figure 32: SARIMA Diagnostic Plot on Whole Sparkling Data

5.7.11.2. Forecast of future 57 months performance using SARIMA Model

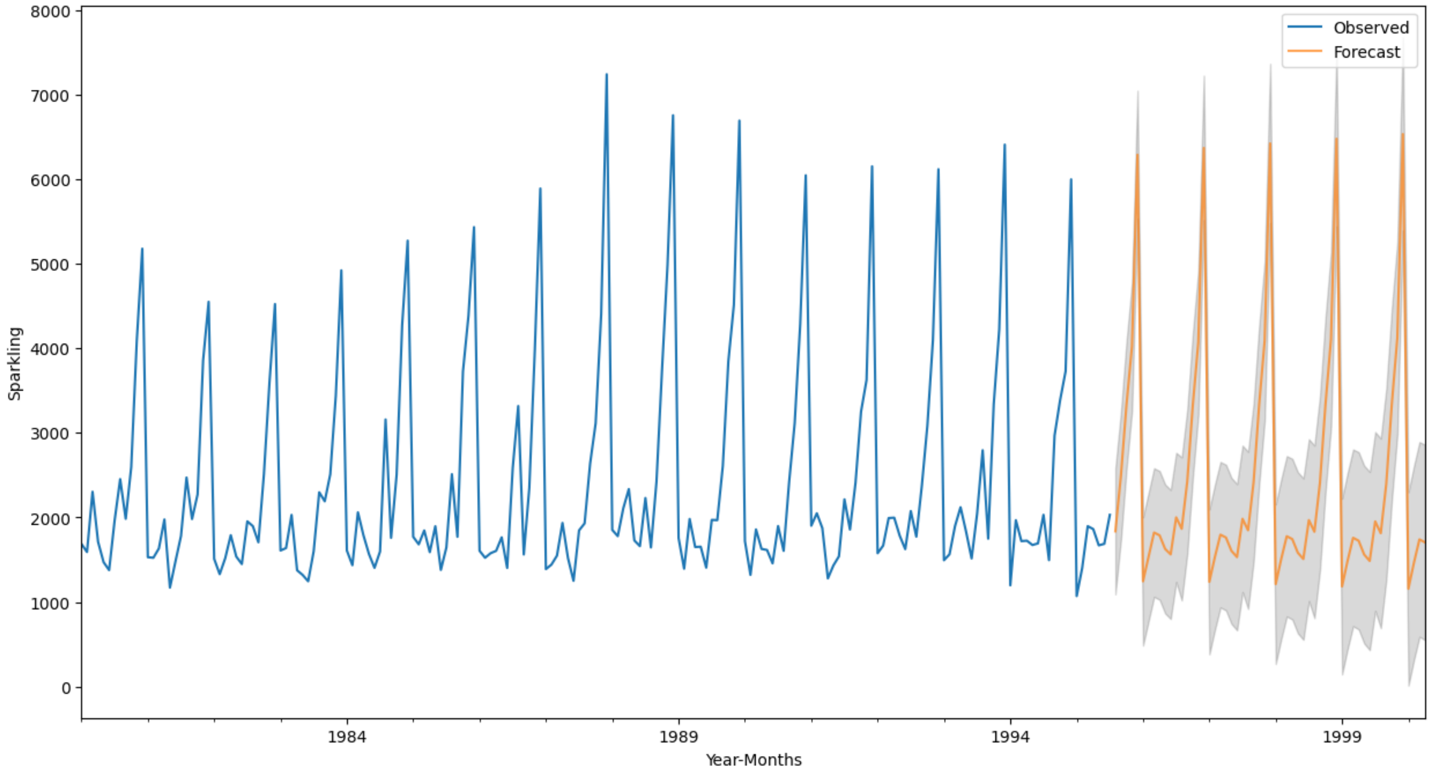

Figure 33: Predict 57 months Sparkling wine sale into the future using SARIMA Model

5.8. Model Comparison

Model	RMSE Score
Alpha=0.075, Beta=0.064, Gamma=0.376, Triple Exponential Smoothing	381.657232
SARIMA(1, 1, 2)(1, 0, 2, 12)	697.097606
2point Trailing Moving Average	811.178937
4point Trailing Moving Average	1184.2133
ARIMA(2,1,2)	1325.16674
6point Trailing Moving Average	1337.20052
ARIMA(2, 0, 1)	1338.1399
Alpha=0.995, Simple Exponential Smoothing	1362.42895
Alpha=0.4, Simple Exponential Smoothing	1363.0378
Simple Average	1368.746717
RegressionOnTime	1374.5502
9point Trailing Moving Average	1422.65328
Alpha=0.3,Beta=0.3,Double Exponential Smoothing	1597.854
Alpha=0.3, Beta=0.5, Gamma=1.0, Triple Exponential Smoothing	1656.7291

Table 3: Model Comparison based on RMSE Score

- The best/lowest RMSE Score obtained in TES with Alpha=0.075, Beta=0.064, Gamma=0.376
- The next best model is Sarima and the highest RMSE score obtained in TES with Alpha=0.3, Beta=0.5, Gamma=1.0

5.9. Forecast with best model

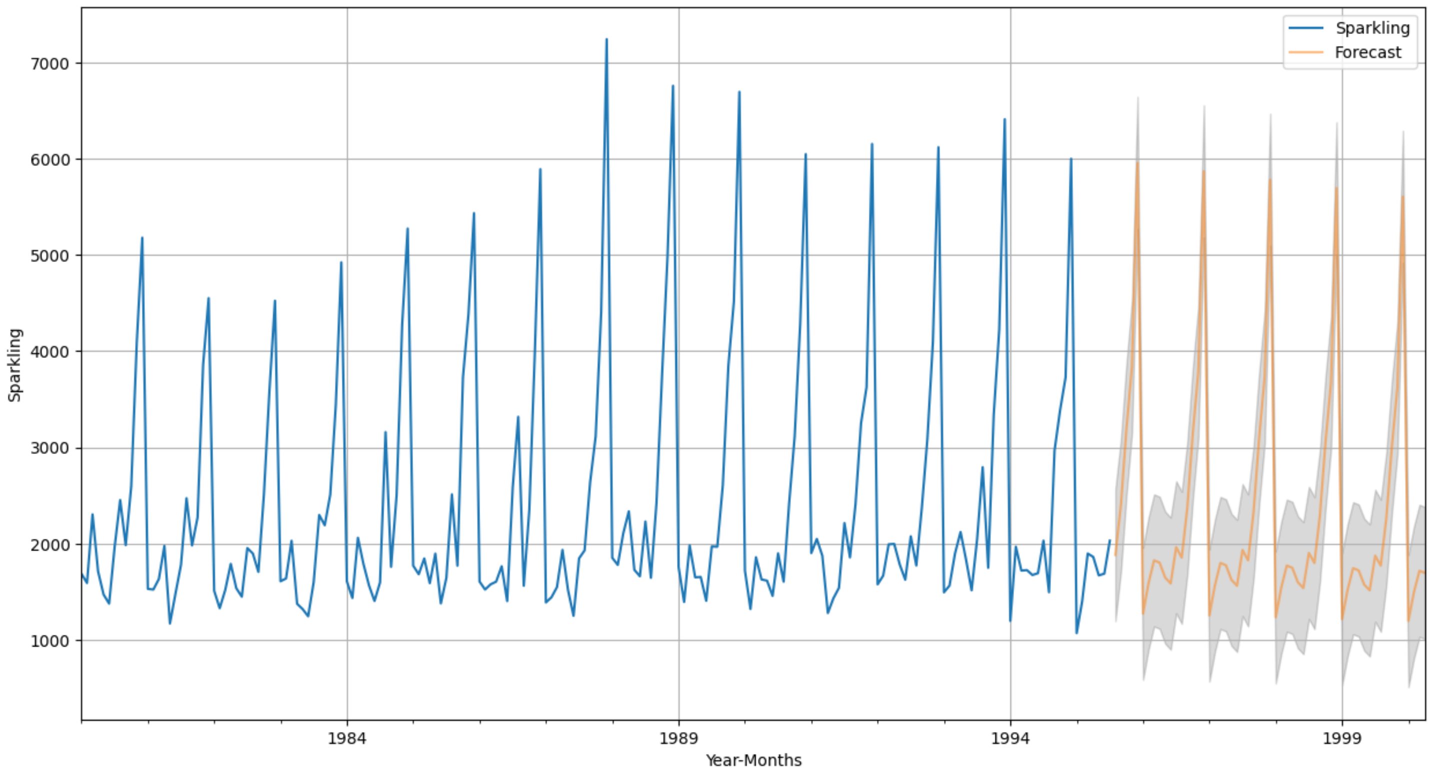

Figure 34: TES Model Forecast on whole Sparkling data

- The above trend represents the prediction on Rose wine sale over next 57 months (Model: TES with Alpha=0.075, Beta=0.064, Gamma=0.376)
- Sparkling wine sale will be at steady state.

5.10. Actionable Insights & Recommendations

Sales Trends:

- Steady growth in sparkling wine sales over the years.
- Peak sales observed during October to December, aligning with seasonal demand.

Best Forecasting Model:

- Triple Exponential Smoothing (TES) with parameters (Alpha=0.075, Beta=0.064, Gamma=0.376) achieved the lowest RMSE of 381.66, making it the most reliable for forecasting.

Seasonality Insights:

- Clear seasonality with a periodicity of 12 months.
- Multiplicative decomposition effectively captures the seasonal component.

Recommendations:

- Target Seasonal Peaks: Focus marketing campaigns and inventory buildup from October to December to maximize sales.
- Expand Production Capacity: Plan production in advance to meet increased seasonal demand while maintaining stock during off-seasons.
- Leverage Forecasts: Use TES forecasts to make data-driven decisions for production, inventory, and sales strategies over the next 57 months.

6. Rose Data

6.1. Understand the data by plotting

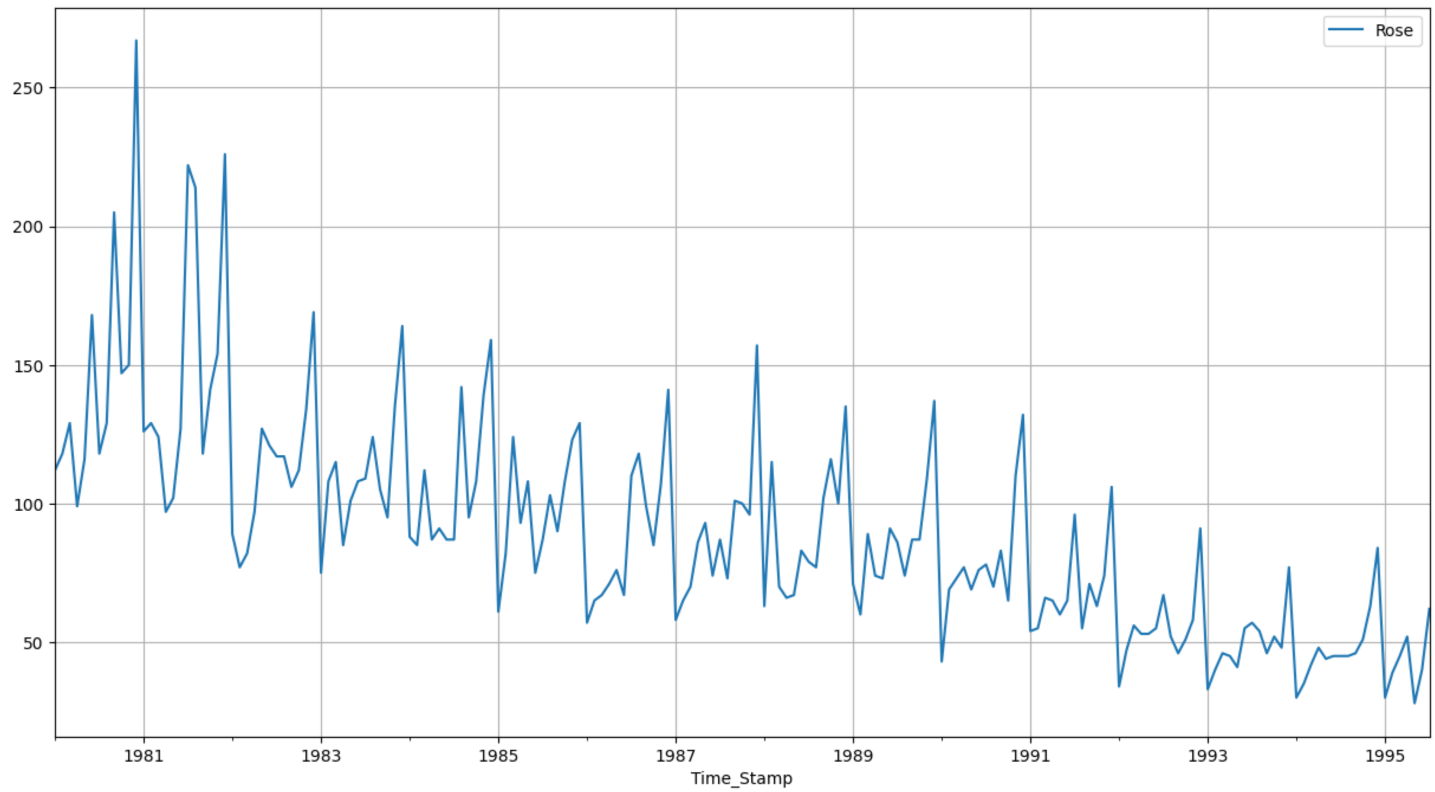

Figure 35: Trend of the rose wine sale over the past 8 years

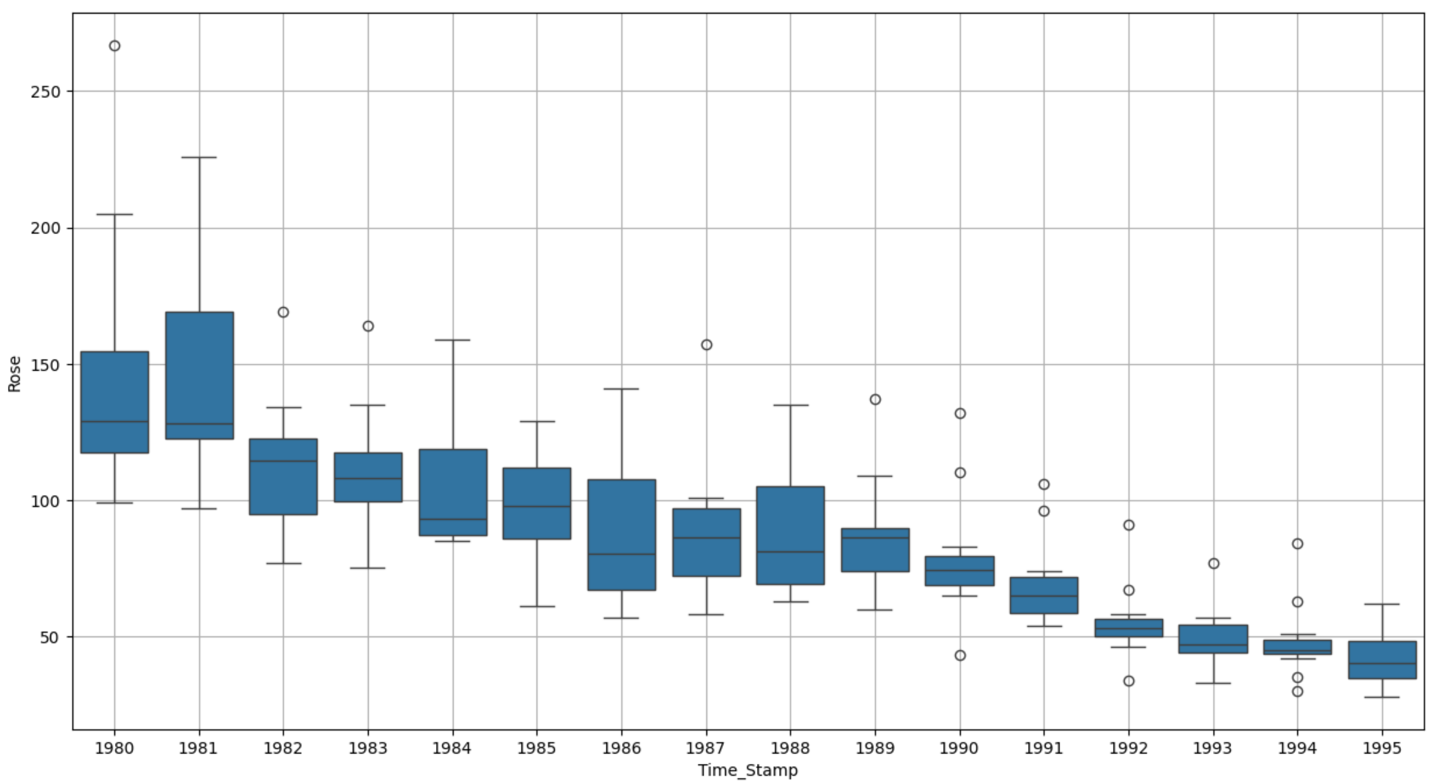

Figure 36: Box plot for rose wine sale over the years

- Sale of rose wine reduced over the years.

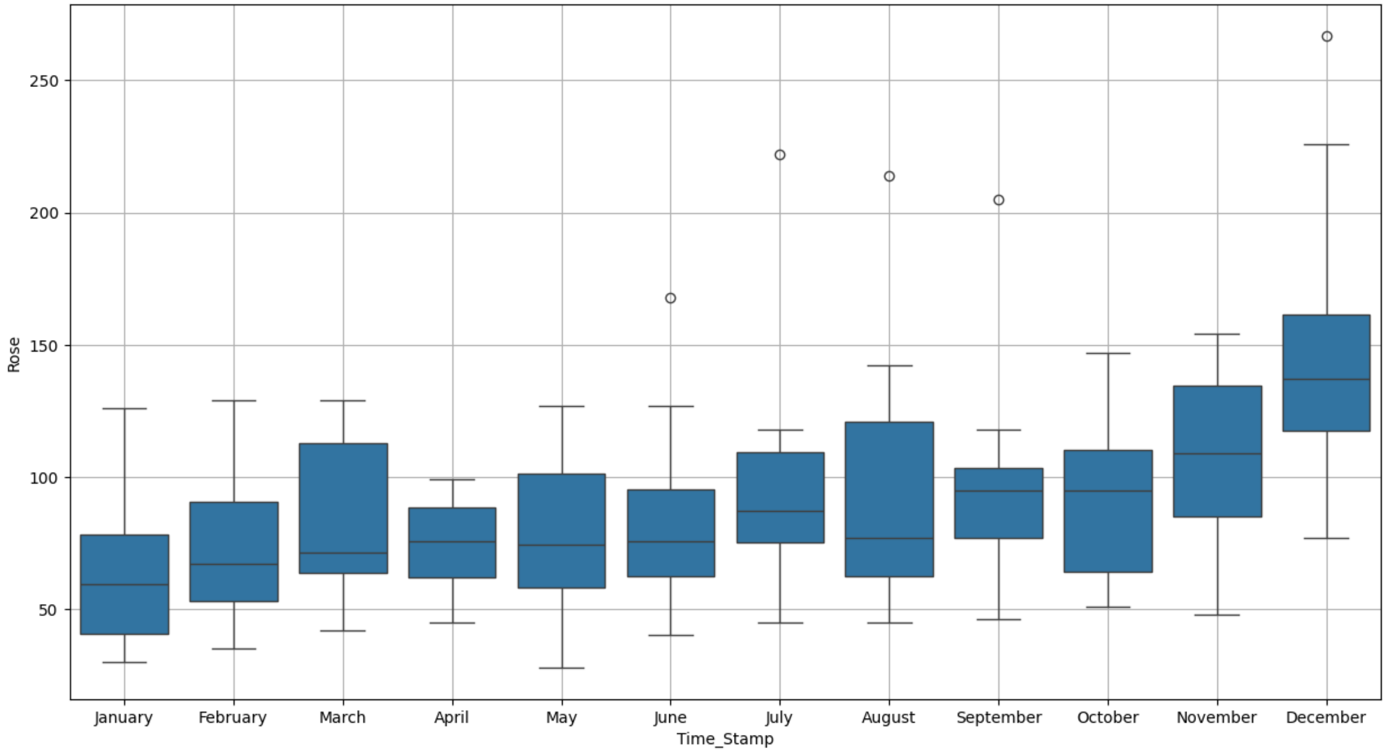

Figure 37: Box plot for rose wine sale trend in each month

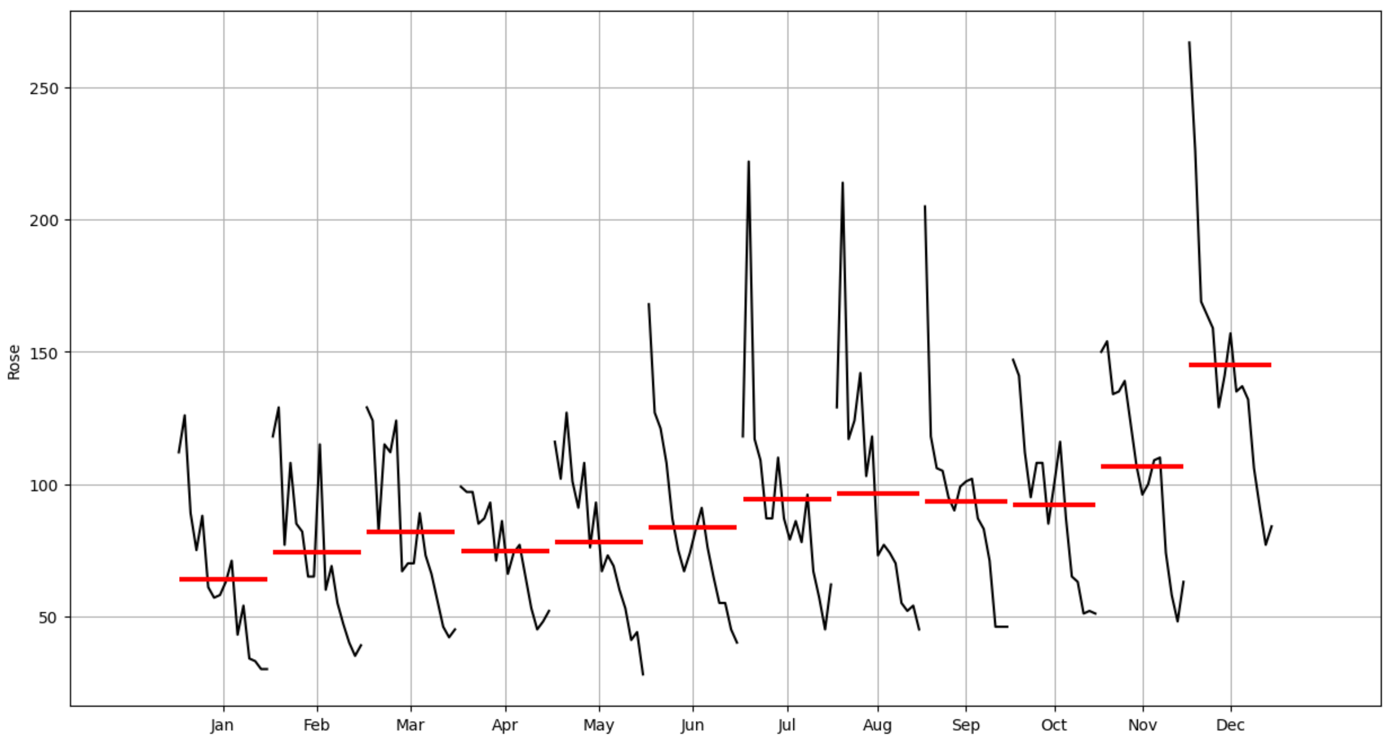

Figure 38: Trend for each month rose wine sale

- In December the peak if sale increases and it is the season for the sale of rose wine.

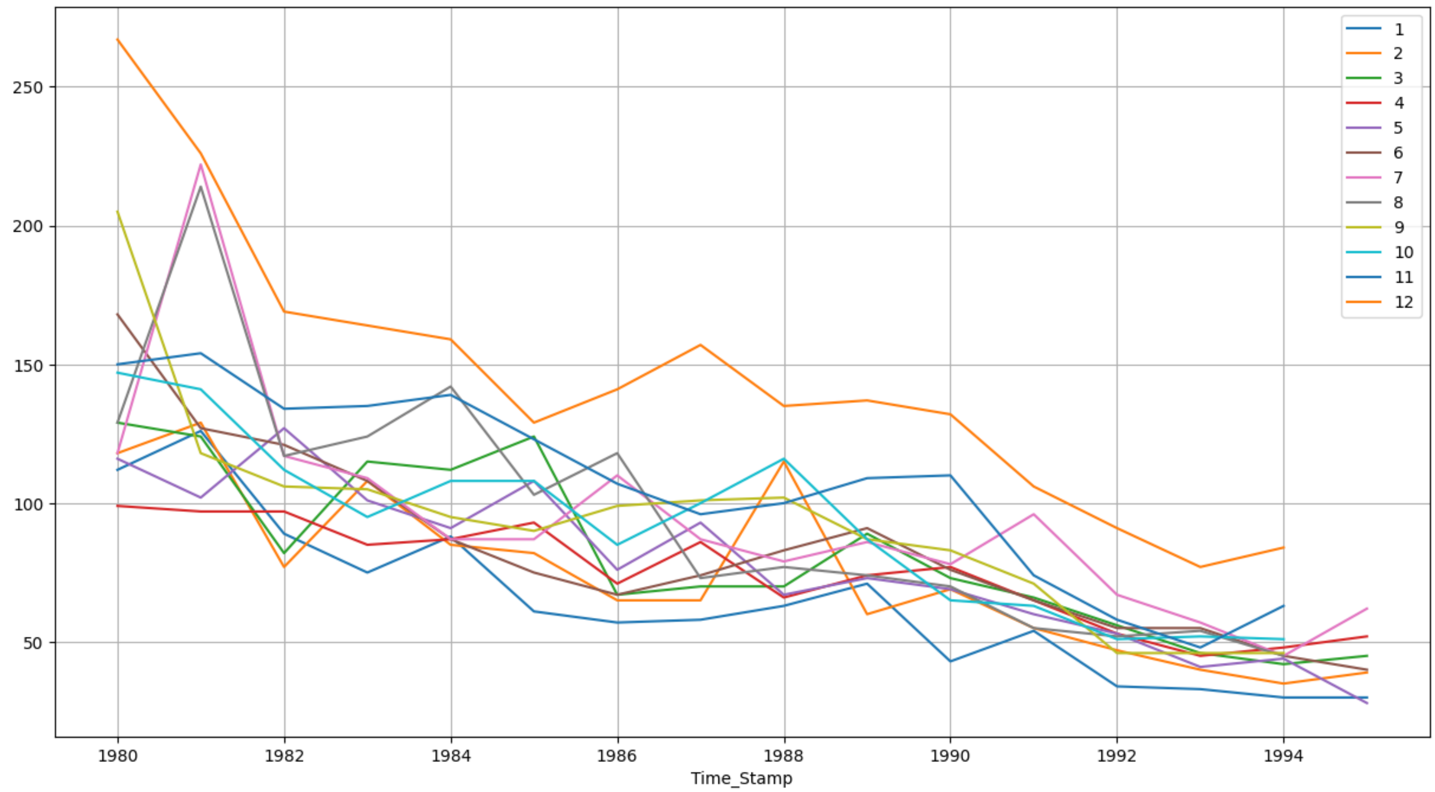

Figure 39: Monthly Rose wine sale across the years

- Trend shows that it is not additive.

6.2. Cumulative Distribution

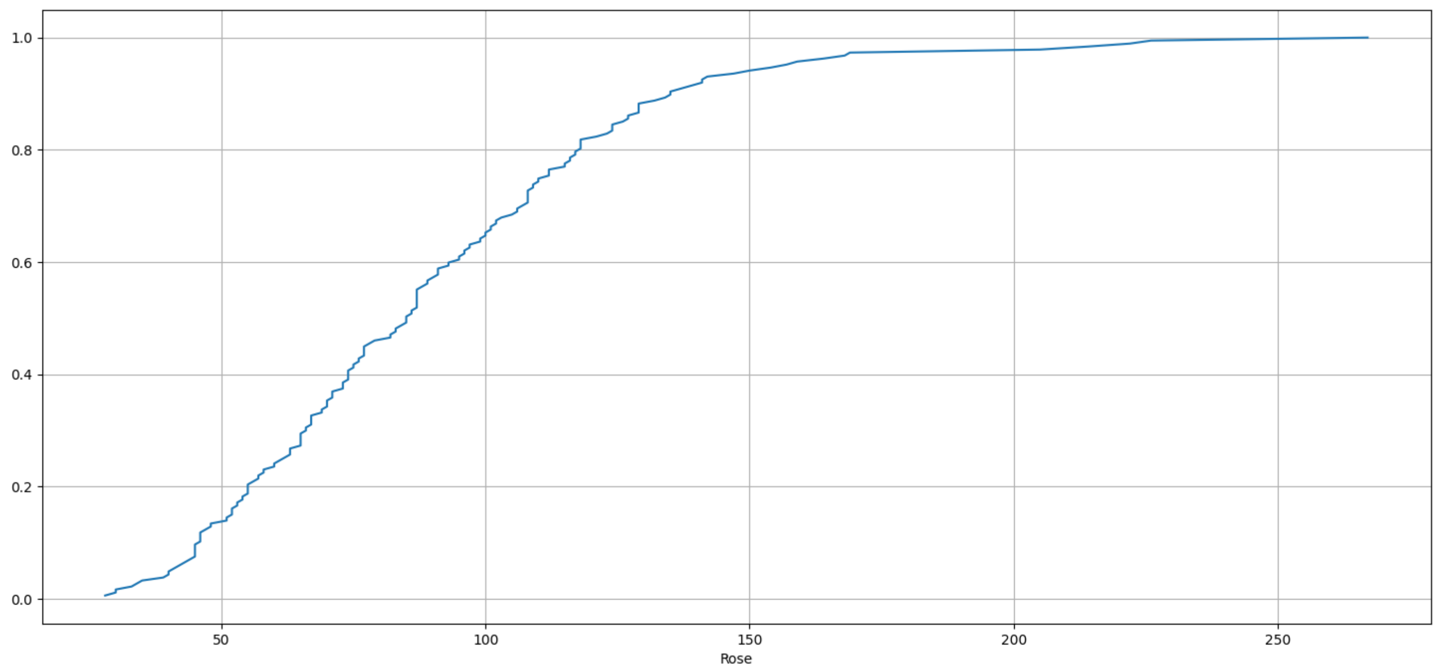

Figure 40: Distribution of rose data

- Over the years the sale has increased.

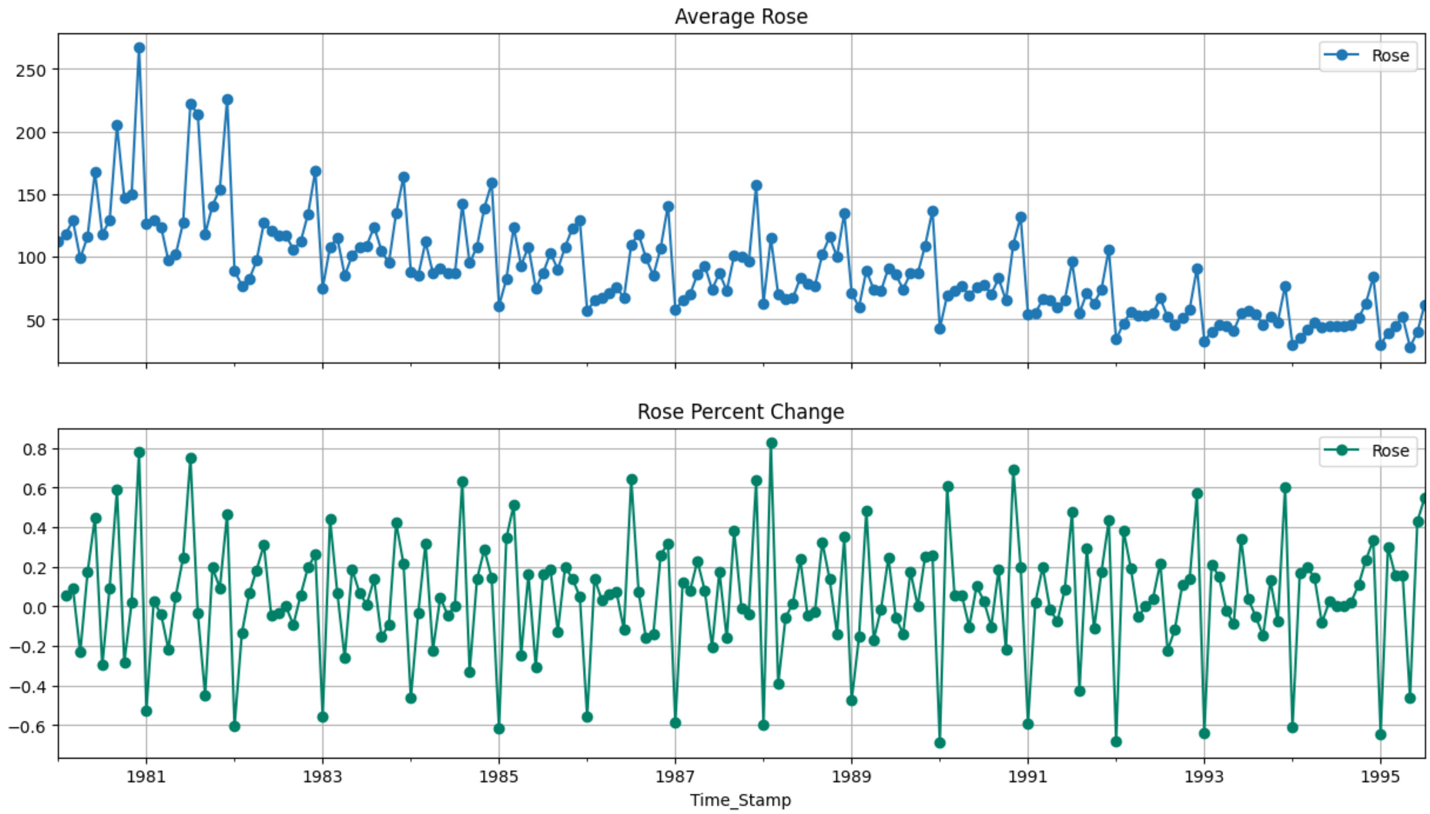

Figure 41: Average 'Rose' and the Percentage change of 'Rose' with respect to the time

- Stady trend can be observed over the years for rose wine sales.

6.3. Decompose the Time Series and plot different components

6.3.1. Additive

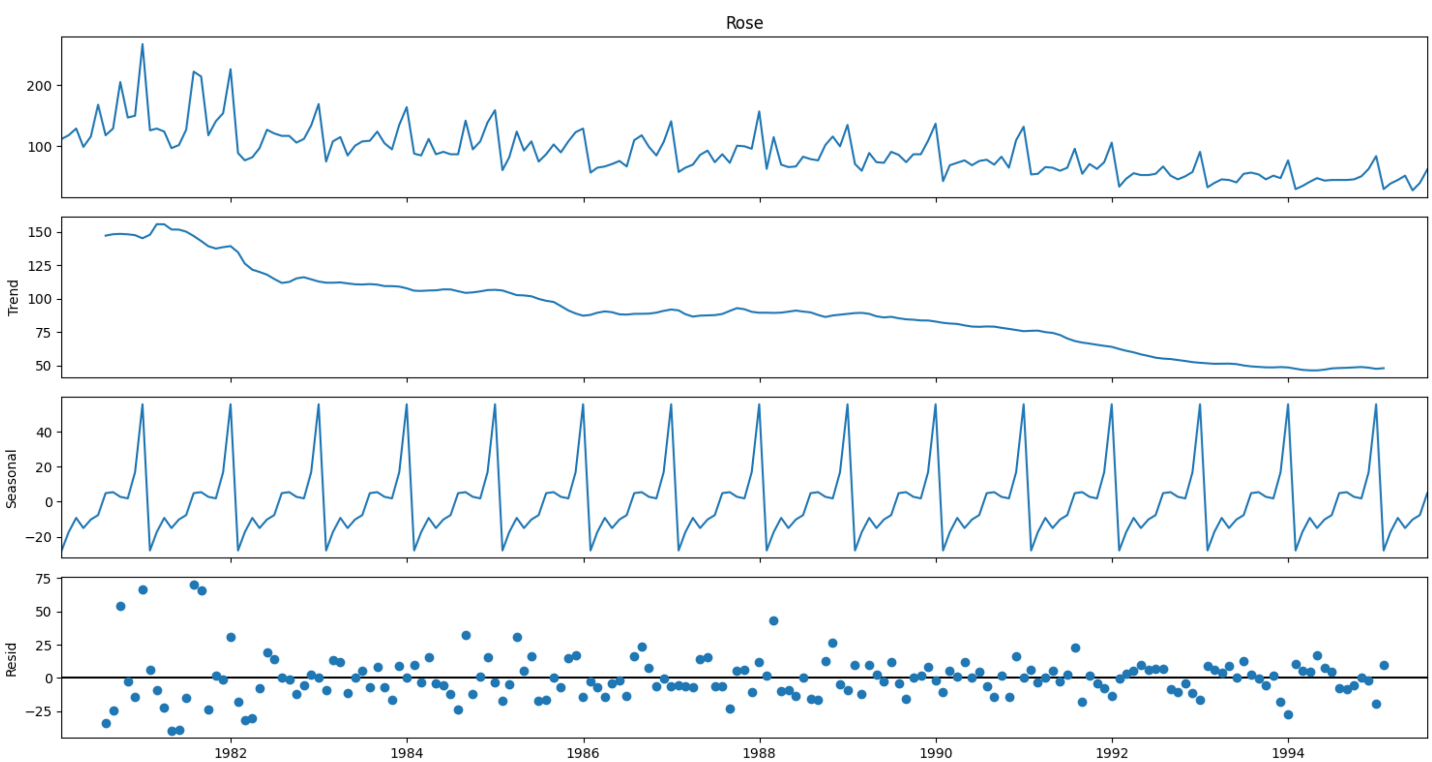

Figure 42: Additive decomposition of rose wine sale

- We see that the residuals are located around 0 from the plot of the residuals in the decomposition.

6.3.2. Multiplicative

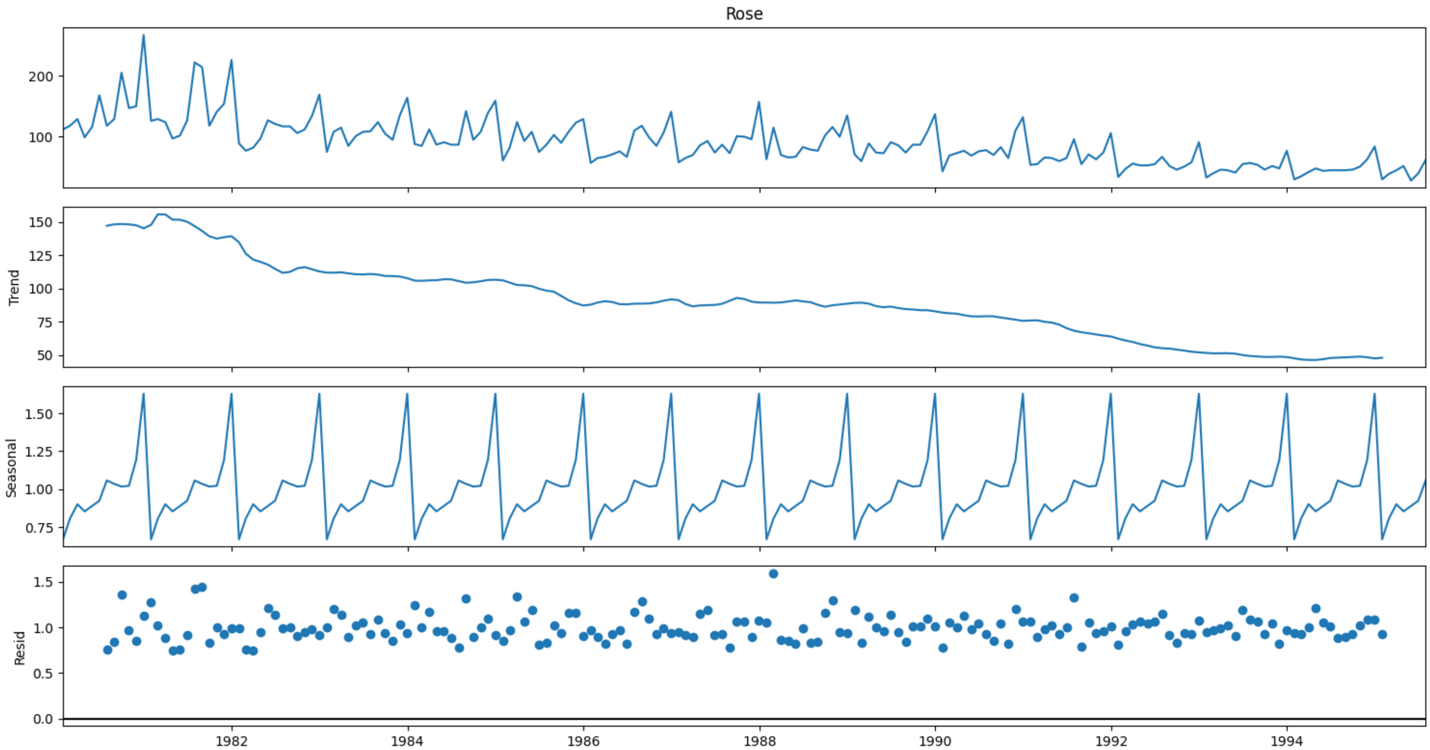

Figure 43: Multiplicative decomposition of rose wine sale

- For the multiplicative series, we see that a lot of residuals are located around 1. Thus, Multiplicative Decomposition is the right way to decompose the time series

6.4. Check for Stationarity of the Data

6.4.1. Whole Data

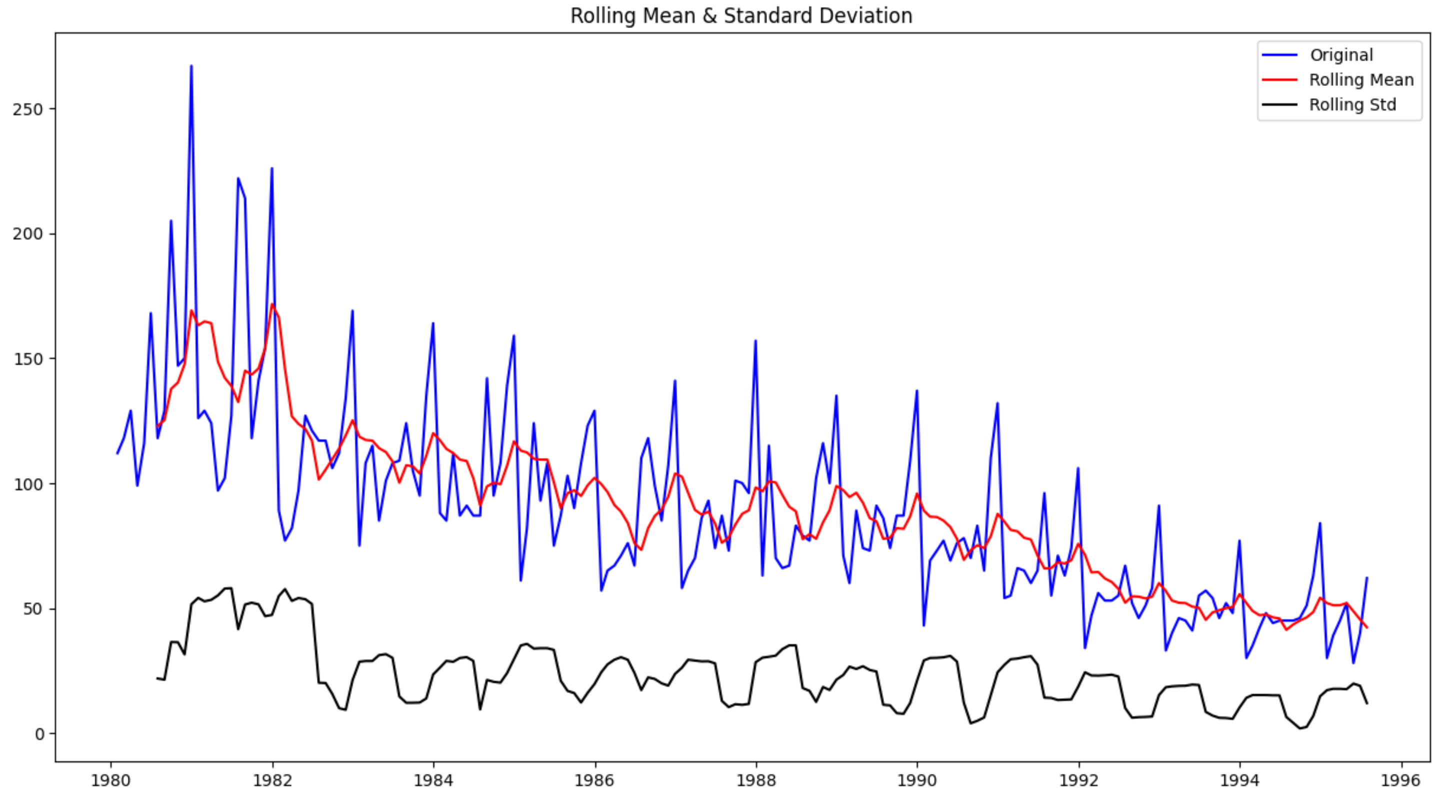

Figure 44: Stationarity of the Rose data at α = 0.05

Results of Dickey-Fuller Test:

Test Statistic -1.874856

p-value 0.343981

Lags Used 13.000000

Number of Observations Used 173.000000

Critical Value (1%) -3.468726

Critical Value (5%) -2.878396

Critical Value (10%) -2.575756

- We see that at 5% significant level the Time Series is non-stationary.

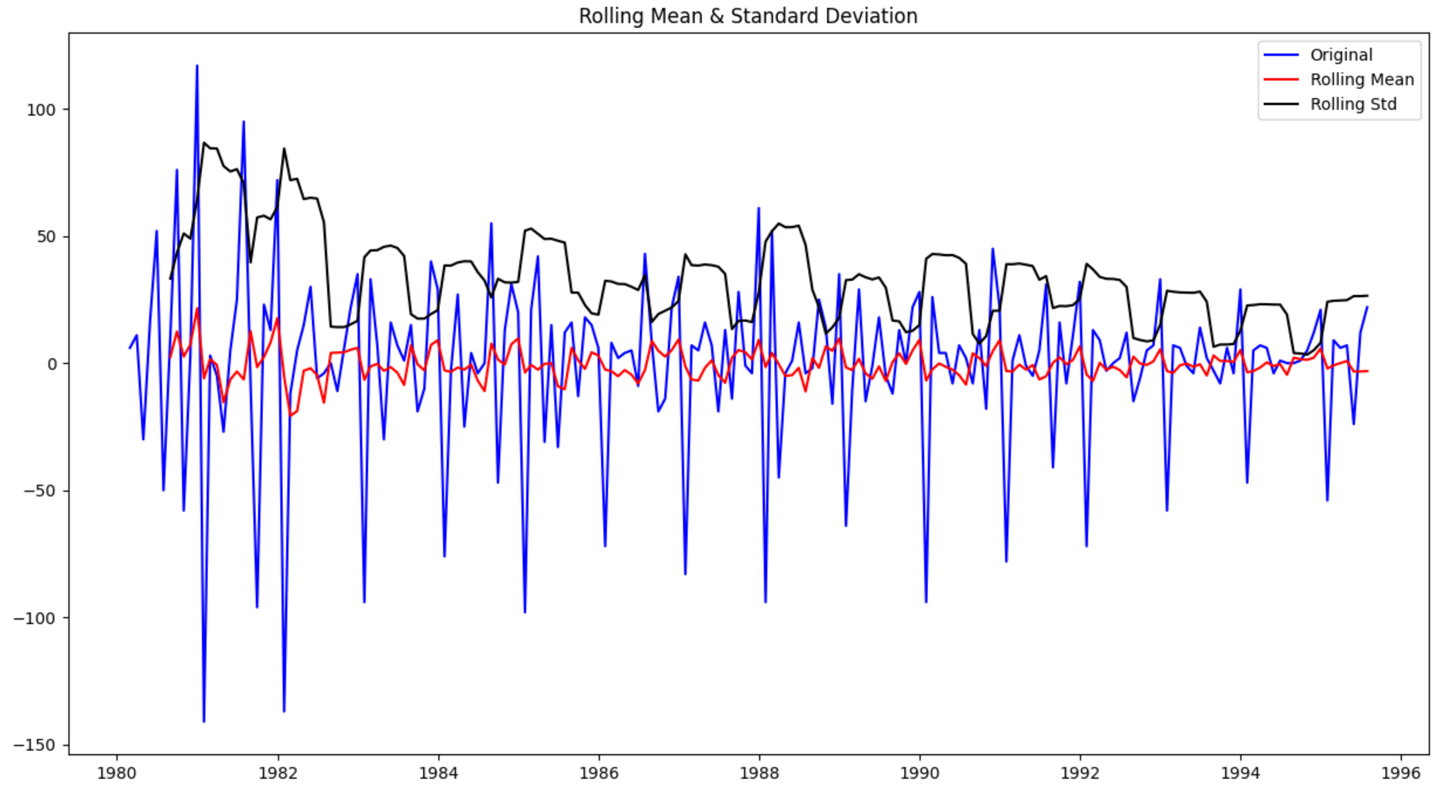

Figure 45: Stationarity of the Rose data at order 1 and α = 0.05

Results of Dickey-Fuller Test:

Test Statistic -8.044139e+00

p-value 1.813580e-12

Lags Used 1.200000e+01

Number of Observations Used 1.730000e+02

Critical Value (1%) -3.468726e+00

Critical Value (5%) -2.878396e+00

Critical Value (10%) -2.575756e+00

- We see that at α = 0.05 and rolling order of one the Time Series is indeed stationary.

6.4.2. Test Data

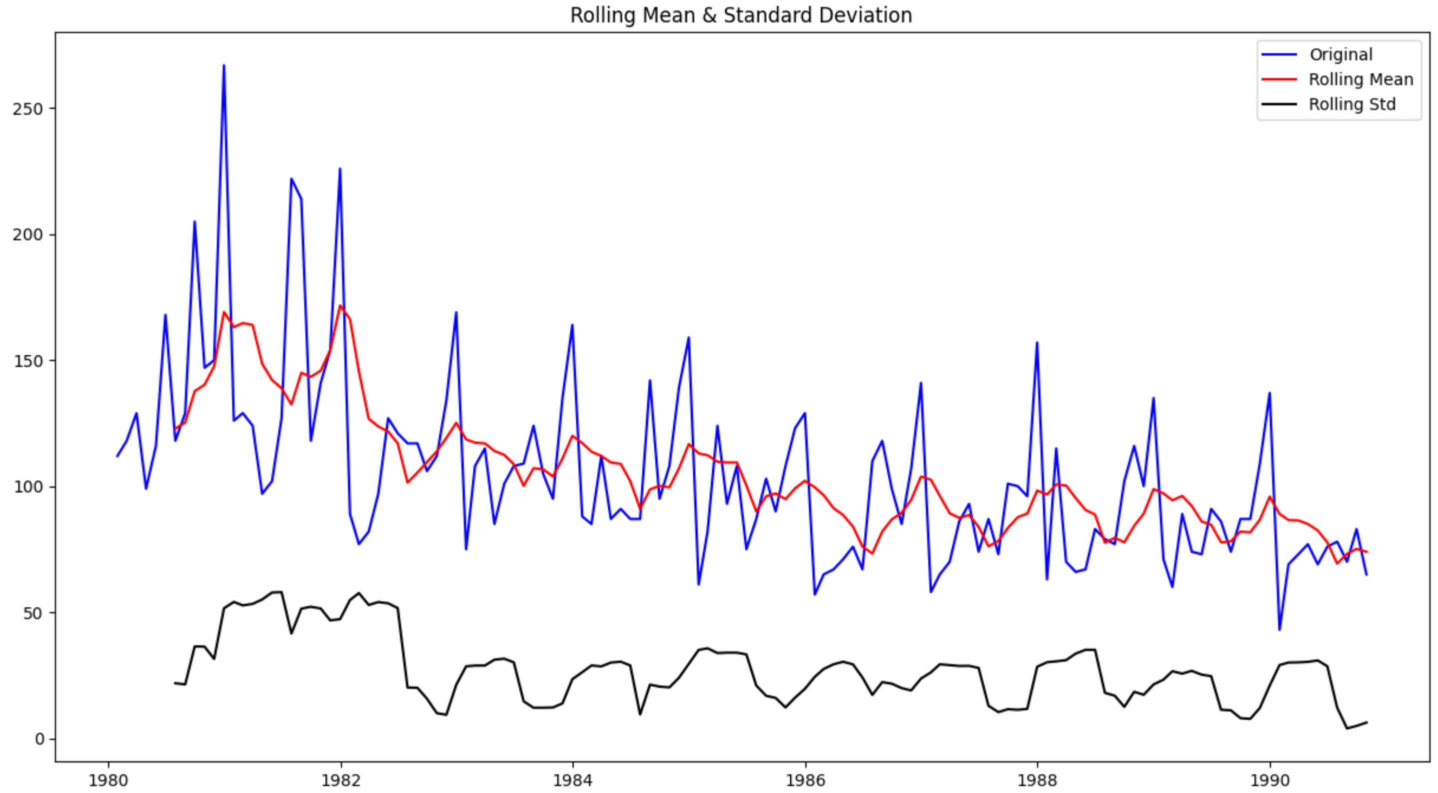

Figure 46: Stationarity of the training rose data at α = 0.05

Results of Dickey-Fuller Test:

Test Statistic -1.939916

p-value 0.313523

Lags Used 13.000000

Number of Observations Used 116.000000

Critical Value (1%) -3.488022

Critical Value (5%) -2.886797

Critical Value (10%) -2.580241

- We see that the series is not stationary at α = 0.05

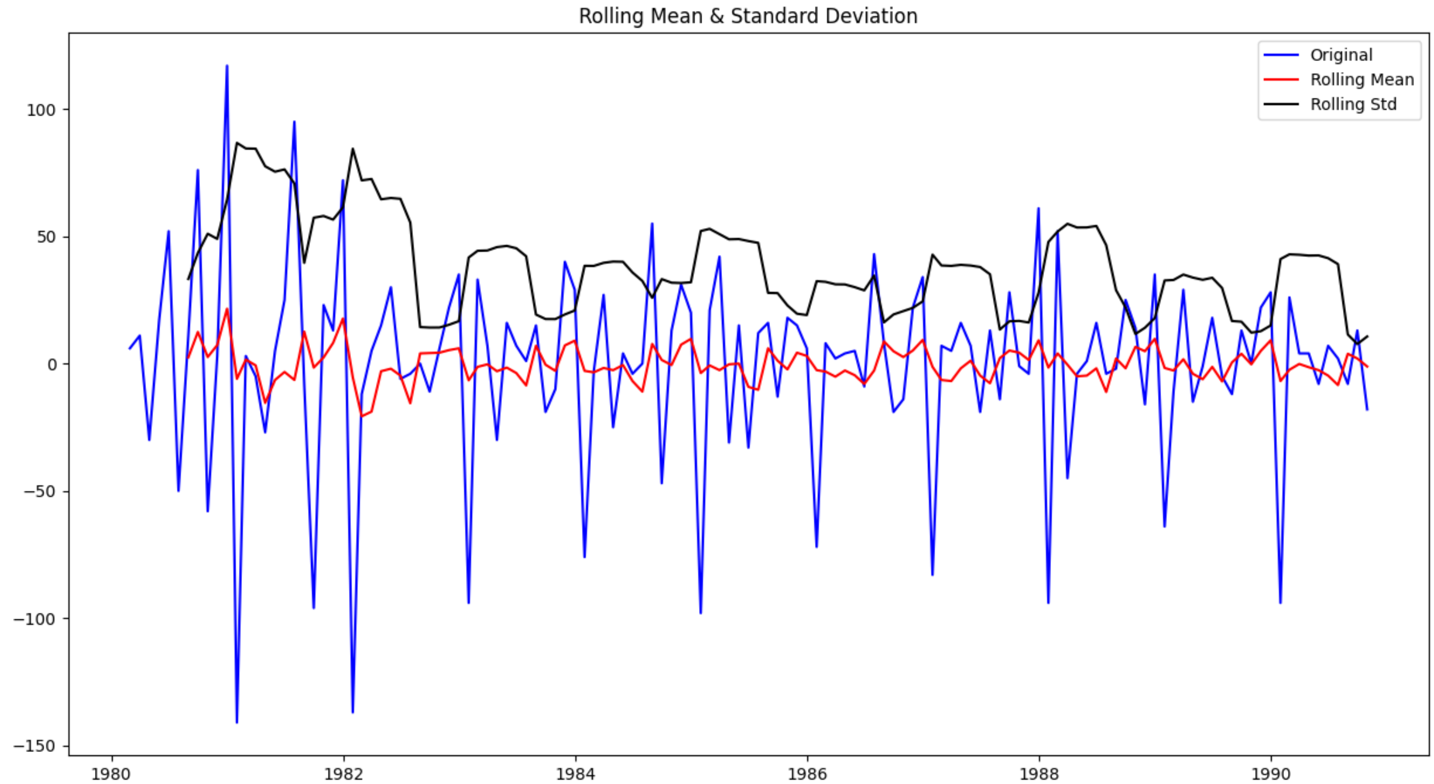

Figure 47: Stationarity of the training rose data at order1 and α = 0.05

Results of Dickey-Fuller Test:

Test Statistic	-6.486071e+00
p-value	1.260009e-08
Lags Used	1.200000e+01
Number of Observations Used	1.160000e+02
Critical Value (1%)	-3.488022e+00
Critical Value (5%)	-2.886797e+00
Critical Value (10%)	-2.580241e+00

- We see that after taking a difference of order 1 the series have become stationary at $\alpha = 0.05$

6.5. Autocorrelation Function (ACF)

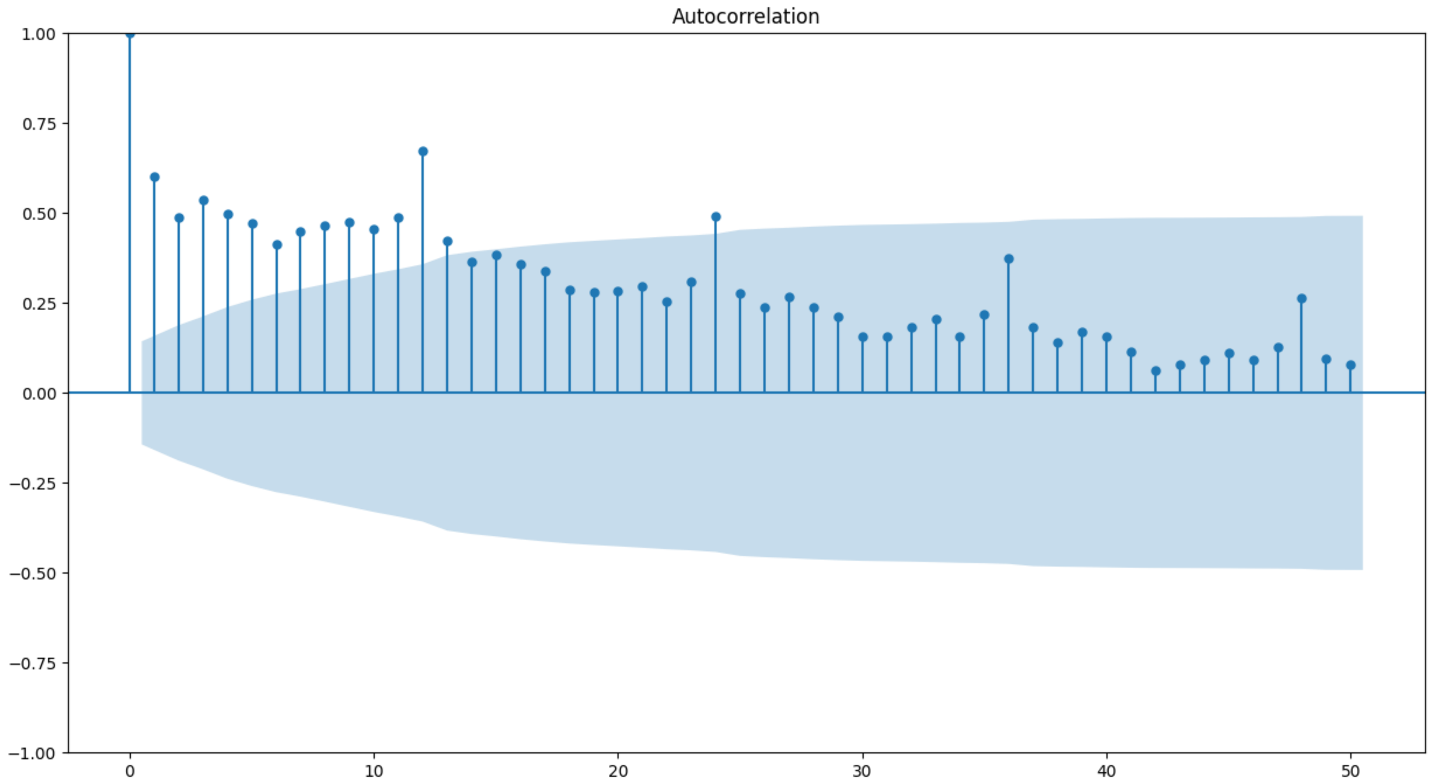

Figure 48: Autocorrelation Function Plot of whole rose data

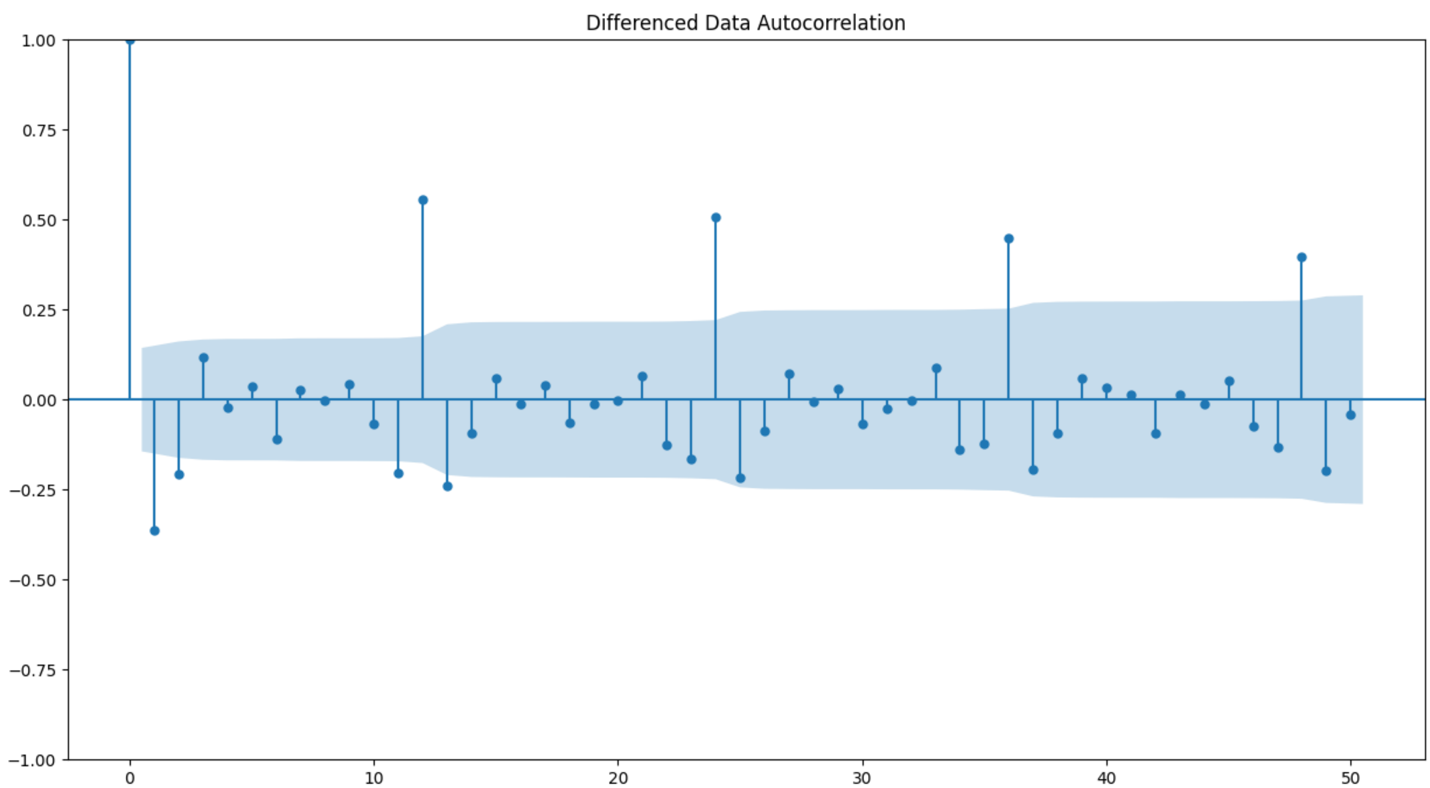

Figure 49: Autocorrelation Function Plot of whole rose data at difference 1

From the above plots, we can say that there seems to be a seasonality in the data.

We see that there can be a seasonality of 12. But from the decomposition at the start we ascertained that visually it looks like the seasonality =12 and thus using the same.

6.6. Data Split

We have did train and test split at a ratio of 70 : 30 respectively.

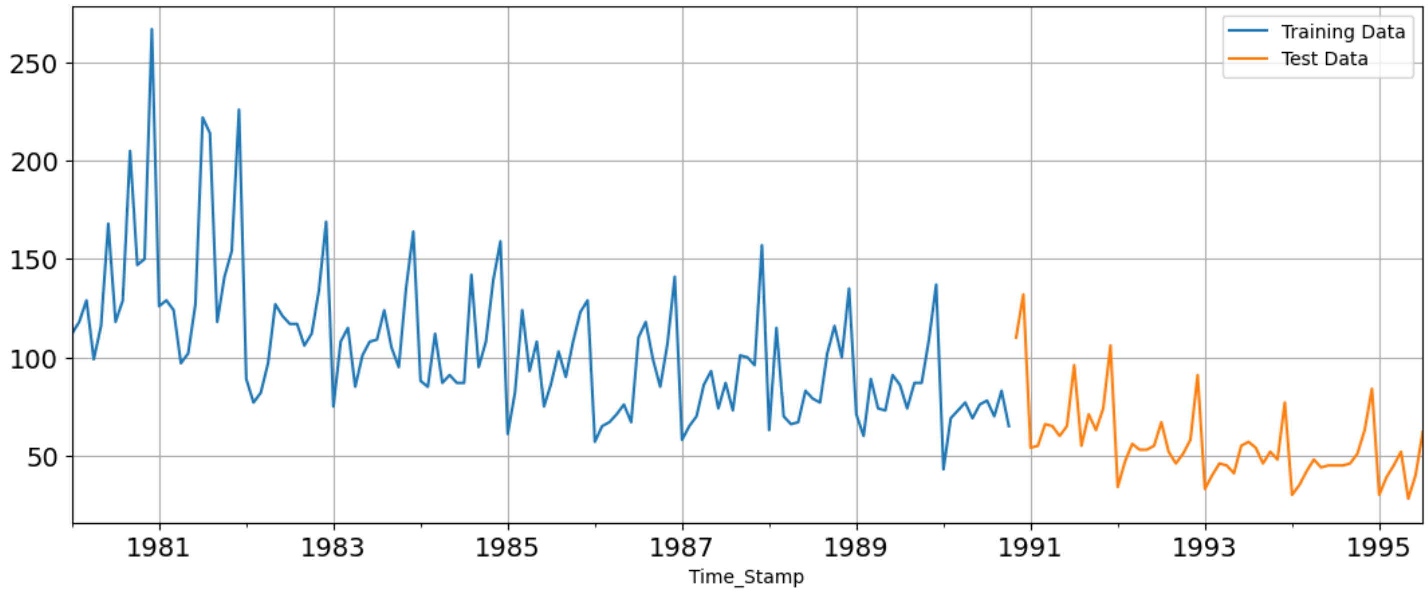

Figure 50: Plot of train and test data

6.7. Model Building

6.7.1. Linear Regression Model

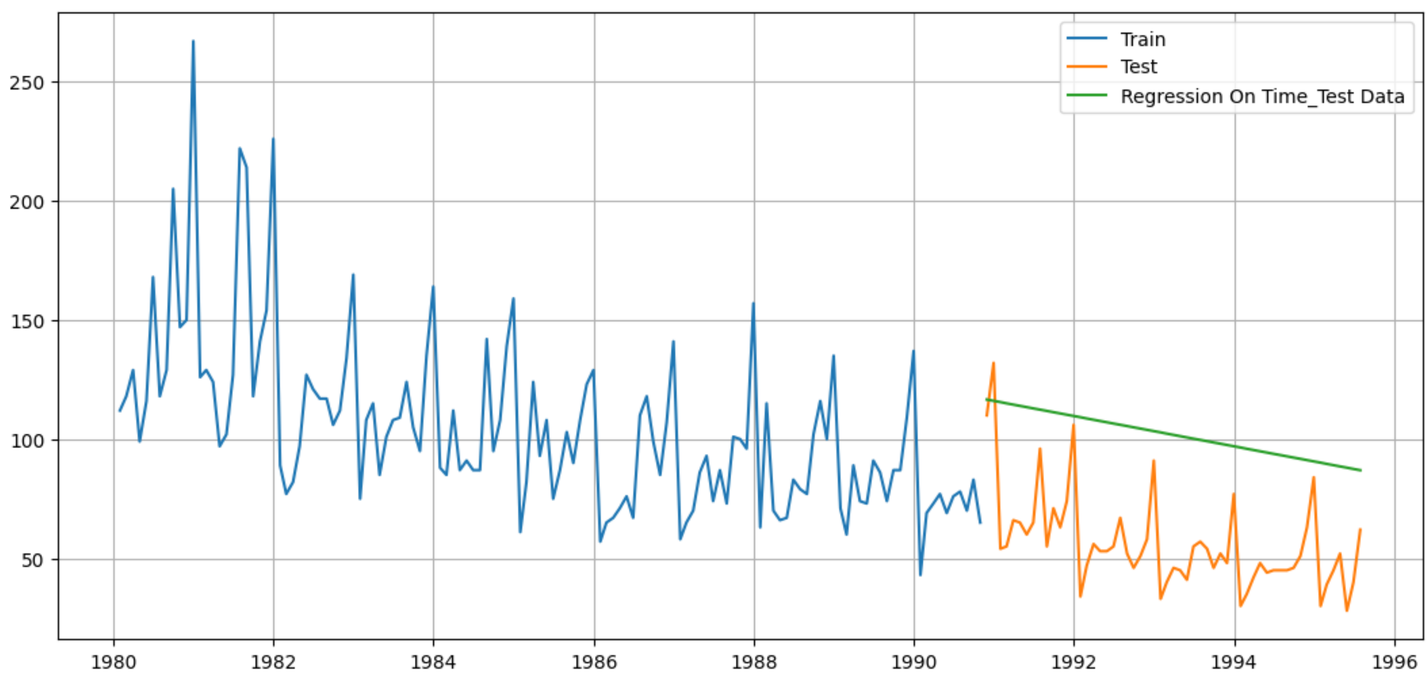

Figure 51: Linear regression performance on Rose data

For RegressionOnTime forecast on the Test Data, RMSE is 48.77

6.7.2. Simple Average

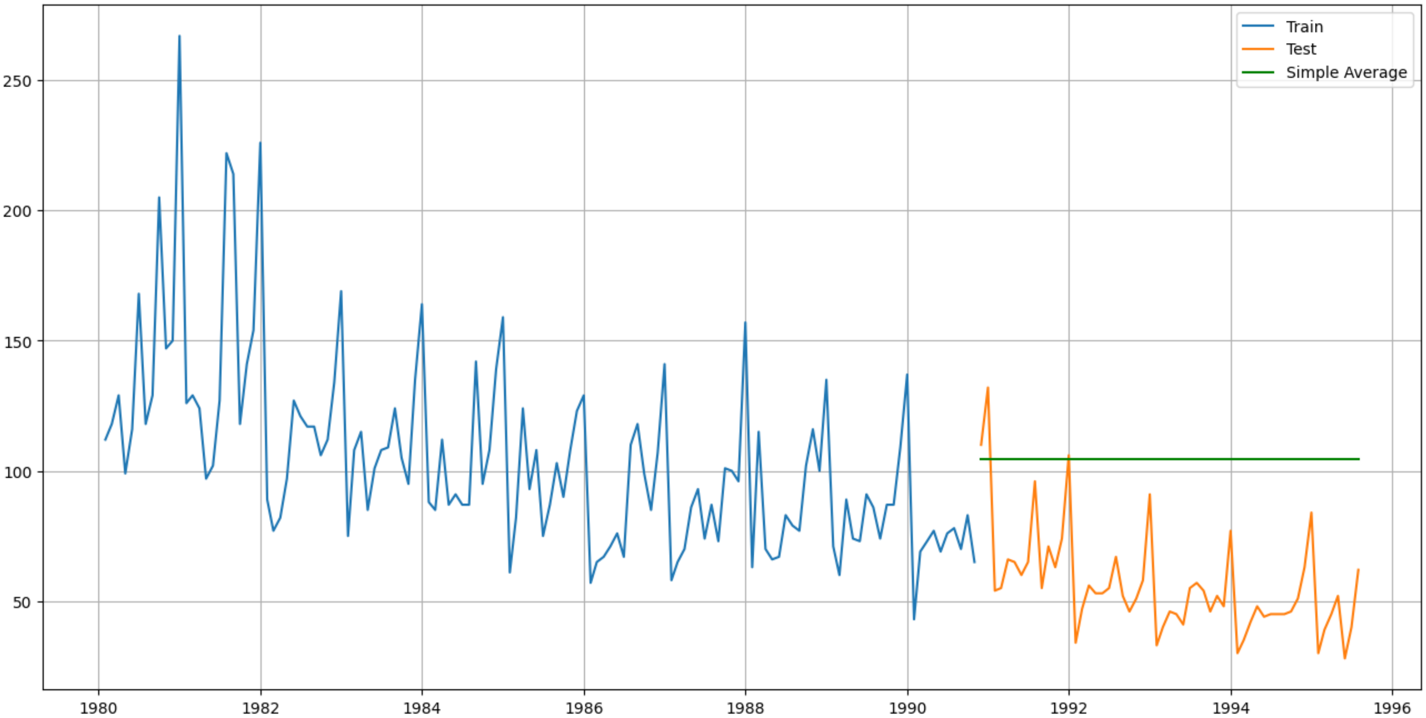

Figure 52: Simple Average Rose Data

For Simple Average forecast on the Test Data, RMSE is 52.43

6.7.3. Moving Average (MA)

For the moving average model, we are going to calculate rolling means (or moving averages) for different intervals. The best interval can be determined by the maximum accuracy (or the minimum error) over here.

- For Moving Average, we are going to average over the entire data

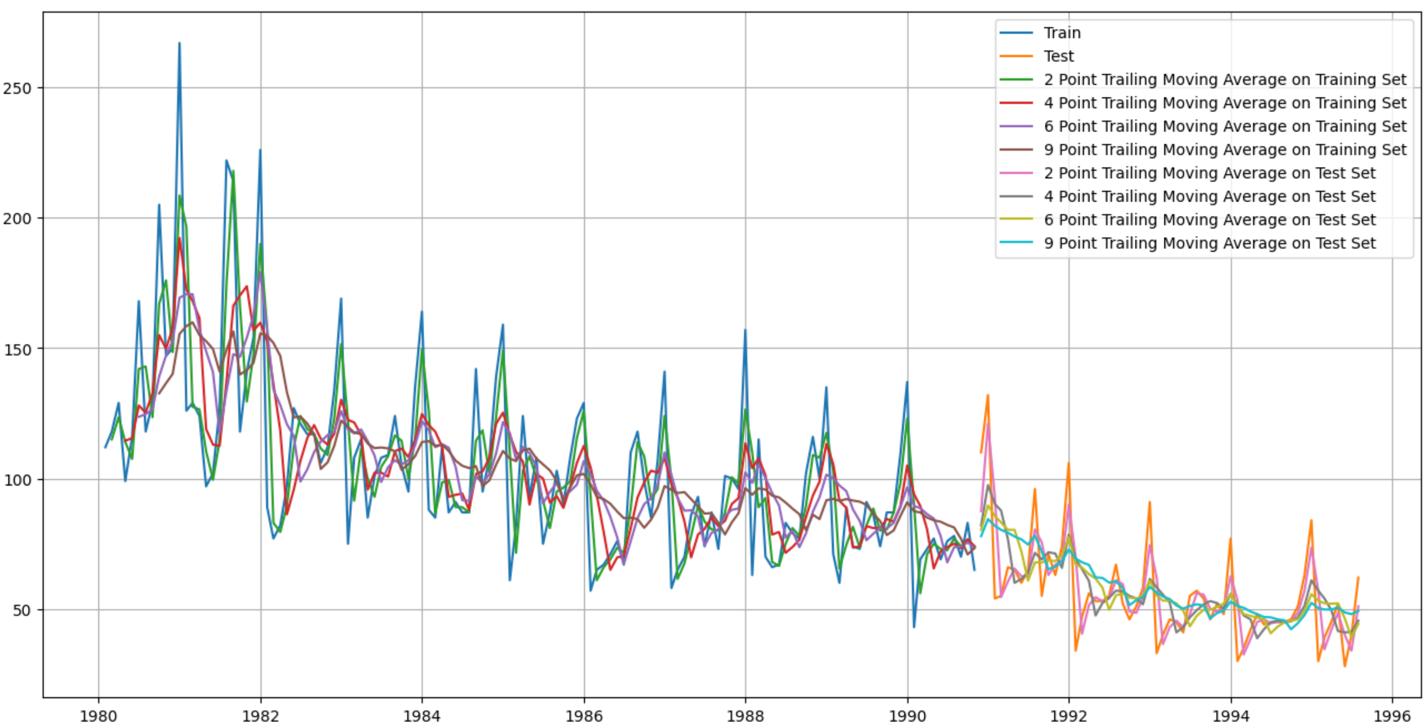

Figure 53:Moving Average Model performance on Rose data

For 2 point Moving Average Model forecast on the Training Data, RMSE is 11.801

For 4 point Moving Average Model forecast on the Training Data, RMSE is 15.371

For 6 point Moving Average Model forecast on the Training Data, RMSE is 15.867

For 9 point Moving Average Model forecast on the Training Data, RMSE is 16.345

6.7.4. Model comparison

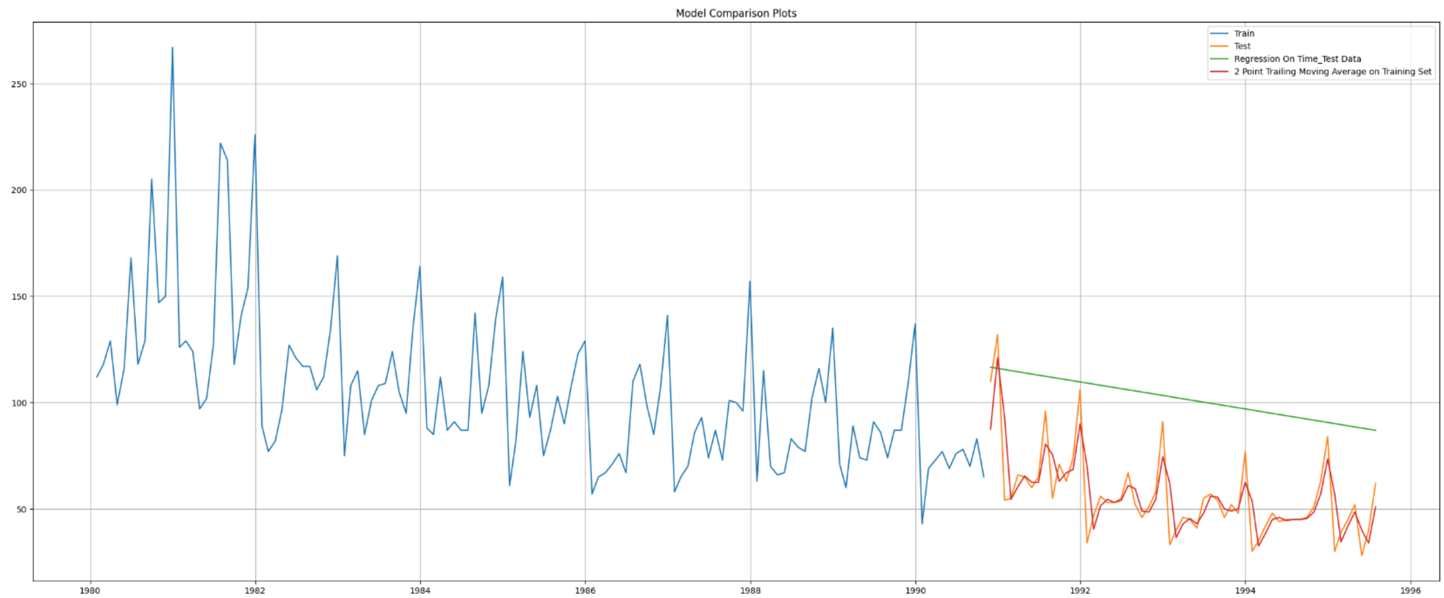

Figure 54: Model performance comparison 1 on rose data

- 2-point trailing is performing well than linear regression.

6.7.5. Simple Exponential Smoothing (SES)

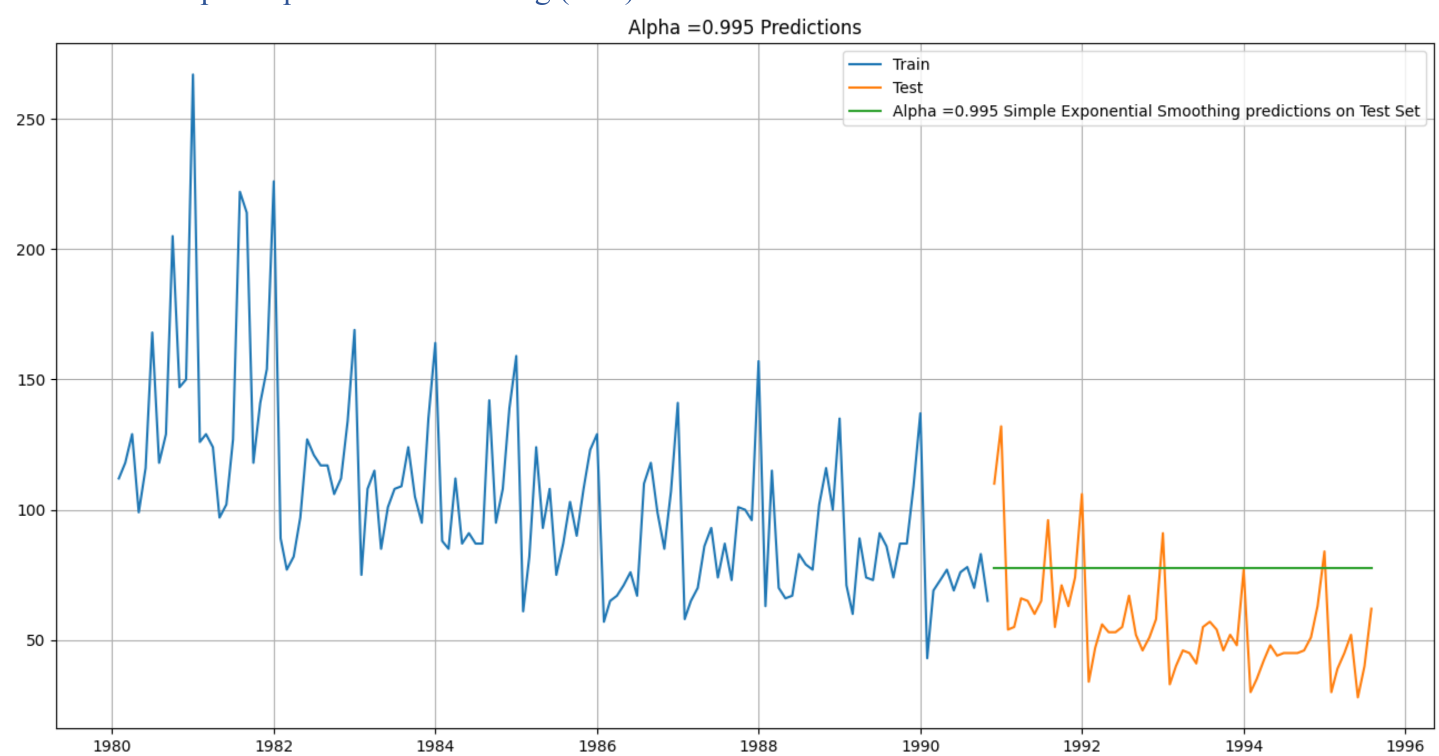

Figure 55: SES model performance (α=0.995) on Rose Data

- For Alpha =0.995 Simple Exponential Smoothing Model forecast on the Test Data, RMSE is 29.243

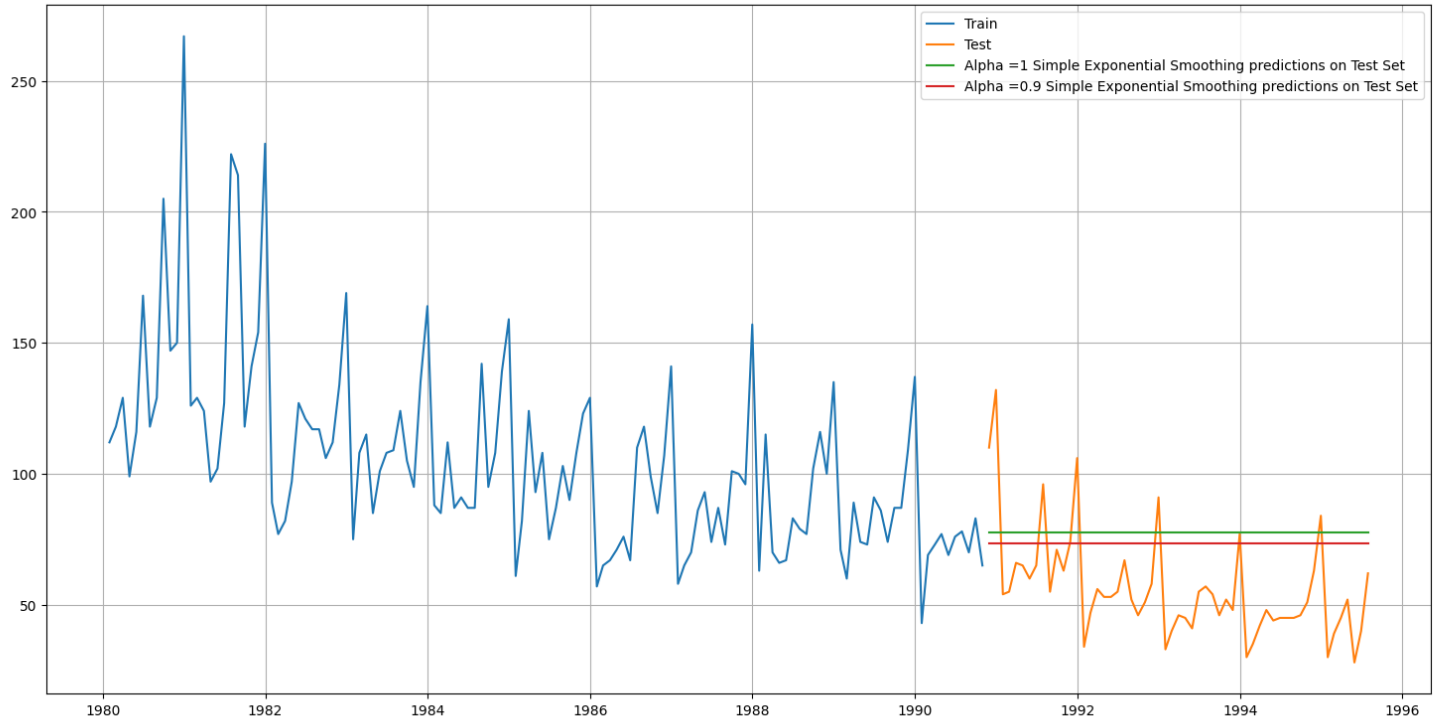

Figure 56: SES model performance (α=0.9) on Rose Data

RMSE scores:

Alpha=0.995, Simple Exponential Smoothing 29.243074

Alpha=0.9, Simple Exponential Smoothing 22.513502

- At Alpha = 0.9 the SES model is giving better RMSE score than the Alpha=0.995.

6.7.6. Double Exponential Smoothing (Holt's Model)

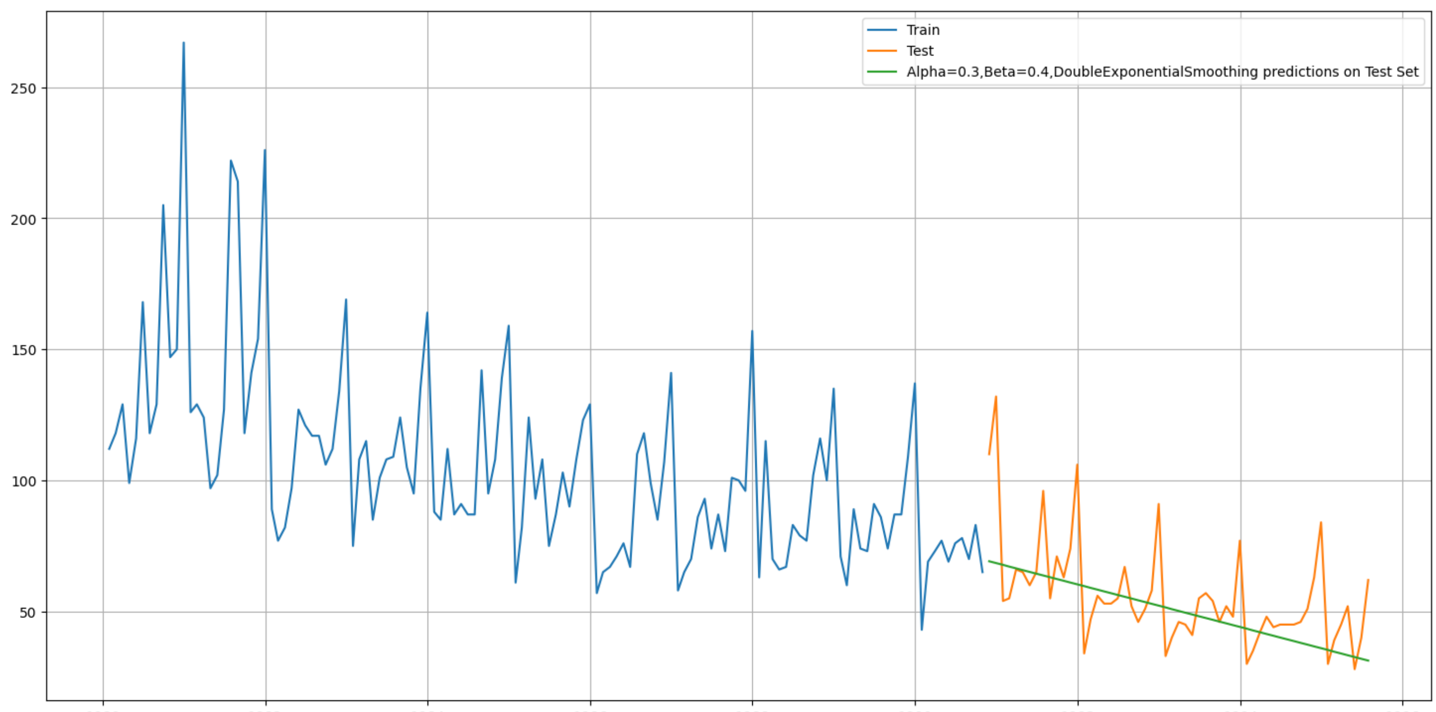

Figure 57:DES model performance in Rose data

Alpha=0.3, Beta=0.4, Double Exponential Smoothing 18.337178

6.7.7. Triple Exponential Smoothing (Holt - Winter's Model)

TES autofit model gave us below best values for alpha, beta, gamma:

smoothing_level: 0.0999080139189177, smoothing_trend: 1.9932826568022853e-06, smoothing_seasonal: 0.00017683239767298466

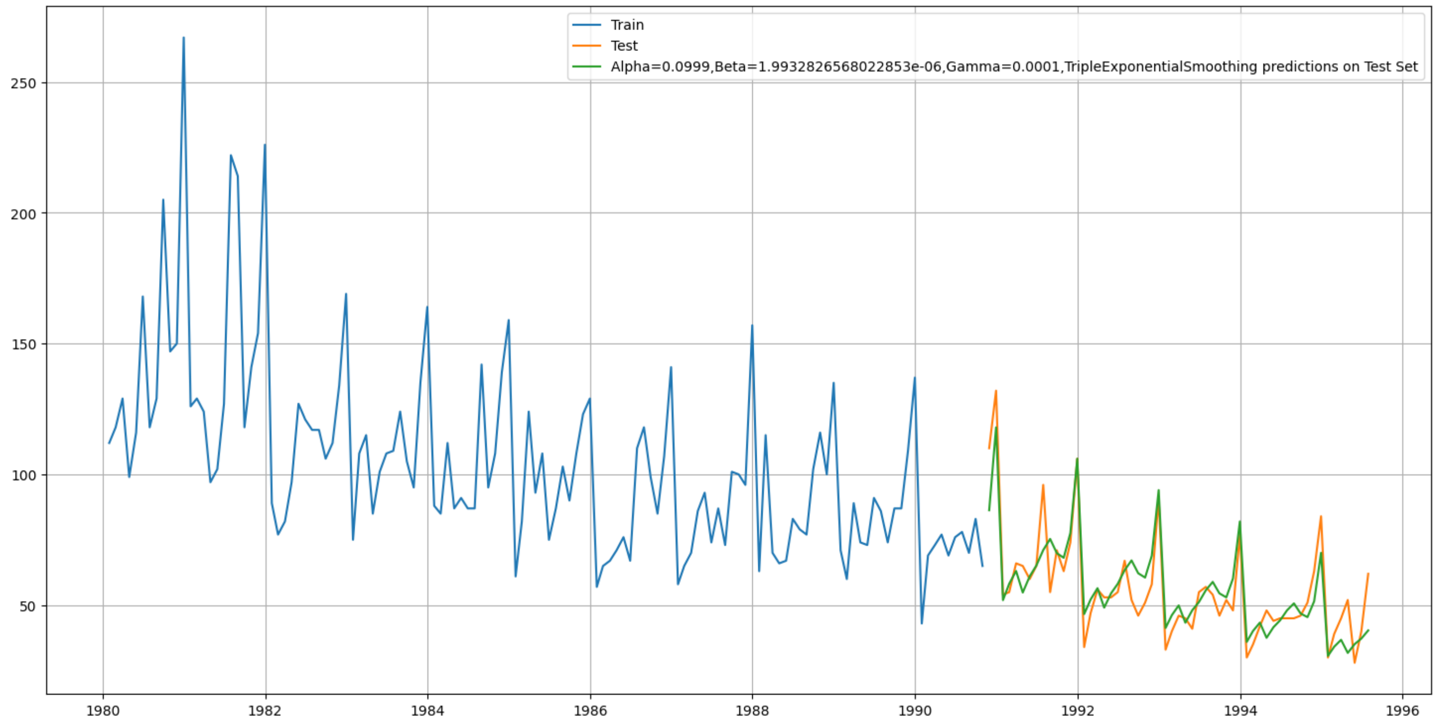

Figure 58:TES Performance on rose data

- For Alpha=0.0999, Beta=1.9932, Gamma=0.0001, Triple Exponential Smoothing Model forecast on the Test Data, RMSE is 9.337
- For Alpha=0.5, Beta=0.3, Gamma=1.0, Triple Exponential Smoothing Model forecast on the Test Data, RMSE is 62.140816

6.7.8. Exponential Smoothing Models Comparison

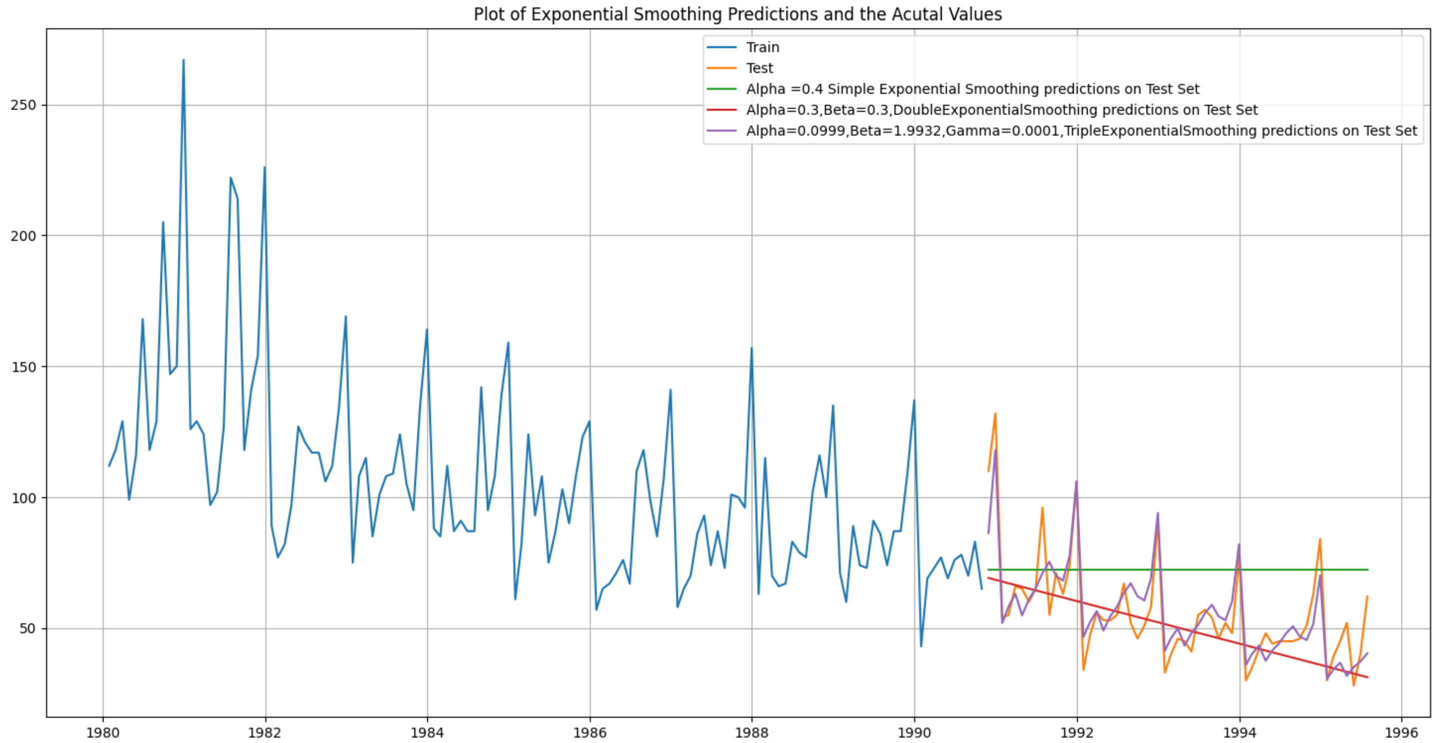

Figure 59: Exponential Smoothing Models Comparison on Rose Data

In this particular we have built several models and went through a model building exercise. This particular exercise has given us an idea as to which particular model gives us the least error on our test set for this data. But in Time Series Forecasting, we need to be very vigil about the fact that after we have done this exercise we need to build the model on the whole data. Remember, the training data that we have used to build the model stops much before the data ends. In order to forecast using any of the models built, we need to build the models again (this time on the complete data) with the same parameters. For this particular mentored learning session, we will go ahead and build only the top 2 models which gave us the best accuracy (least RMSE).

The best model to be built on the whole data is:

- Alpha=0.0999,Beta=1.9932,Gamma=0.0001,Triple Exponential Smoothing

6.7.9. ARMA Model

We are trying to build an Automated version of an ARIMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

param	AIC
(1, 0, 2)	1272.009
(2, 0, 2)	1272.231
(2, 0, 1)	1272.785
(1, 0, 1)	1273.97
(1, 0, 0)	1282.884
(2, 0, 0)	1283.462
(0, 0, 1)	1287.093
(0, 0, 2)	1288.117
(0, 0, 0)	1306.289

Table 4: AIC values of Automatic search for best ARMA models for Rose Data

- The lowest value found to be at (1, 0, 2).

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  130
Model:                 ARIMA(1, 1, 2)   Log Likelihood                -625.737
Date:                Thu, 05 Dec 2024   AIC                           1259.473
Time:                        16:17:30   BIC                           1270.912
Sample:                    01-31-1980   HQIC                          1264.121
                         - 10-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4649      0.274     -1.698      0.090      -1.002       0.072
ma.L1         -0.2485      0.253     -0.983      0.326      -0.744       0.247
ma.L2         -0.5971      0.208     -2.874      0.004      -1.004      -0.190
sigma2       945.0250     87.810     10.762      0.000     772.921    1117.129
===================================================================================
Ljung-Box (L1) (Q):                   0.03   Jarque-Bera (JB):                40.04
Prob(Q):                              0.86   Prob(JB):                         0.00
Heteroskedasticity (H):               0.33   Skew:                             0.84
Prob(H) (two-sided):                  0.00   Kurtosis:                         5.14
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 60:ARMA Model Summery on rose train data

- ARMA model RMSE score on test data is 38.667284935069056

6.7.10. ARIMA Model

We will build an Automated version of an ARIMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

param	AIC
(0, 1, 2)	1259.248
(1, 1, 2)	1259.473
(1, 1, 1)	1260.037
(2, 1, 1)	1261.014
(0, 1, 1)	1261.327
(2, 1, 2)	1261.472
(2, 1, 0)	1278.135
(1, 1, 0)	1297.077
(0, 1, 0)	1313.176

Table 5: AIC values of Automatic search for best ARIMA models for Rose Data

- The lowest value found to be at (0, 1, 2).

```
                                SARIMAX Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  130
Model:                 ARIMA(1, 1, 2)   Log Likelihood                -625.737
Date:                Thu, 05 Dec 2024   AIC                           1259.473
Time:                        16:17:30   BIC                           1270.912
Sample:                    01-31-1980   HQIC                          1264.121
                         - 10-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4649      0.274     -1.698      0.090      -1.002       0.072
ma.L1         -0.2485      0.253     -0.983      0.326      -0.744       0.247
ma.L2         -0.5971      0.208     -2.874      0.004      -1.004      -0.190
sigma2       945.0250     87.810     10.762      0.000     772.921    1117.129
===================================================================================
Ljung-Box (L1) (Q):                   0.03   Jarque-Bera (JB):                40.04
Prob(Q):                              0.86   Prob(JB):                         0.00
Heteroskedasticity (H):               0.33   Skew:                             0.84
Prob(H) (two-sided):                  0.00   Kurtosis:                         5.14
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 61: ARIMA Model Summery on rose train data

- ARIMA model RMSE score on test data is 30.487641263961414

6.7.11. SARIMA Model

We will build an Automated version of a SARIMA model for which the best parameters are selected in accordance with the lowest Akaike Information Criteria (AIC).

Earlier we have seen the seasonality is at 12.

param	seasonal	AIC
(0, 1, 2)	(2, 0, 2, 12)	871.075238
(1, 1, 2)	(2, 0, 2, 12)	873.003875
(2, 1, 2)	(2, 0, 2, 12)	874.213961
(2, 1, 1)	(2, 0, 0, 12)	879.792363
(2, 1, 2)	(2, 0, 0, 12)	880.763857

Table 6:AIC values of Automatic search for best SARIMA models for Rose Data

- The lowest value found to be at param (0,1,2) and seasonal (2, 0, 2, 12)

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                  y   No. Observations:                  130
Model:             SARIMAX(0, 1, 2)x(2, 0, 2, 12)   Log Likelihood                -428.538
Date:                            Thu, 05 Dec 2024   AIC                            871.075
Time:                                    16:18:12   BIC                            889.450
Sample:                                         0   HQIC                           878.516
                                            - 130
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1         -0.8367    239.070     -0.003      0.997    -469.404     467.731
ma.L2         -0.1633     39.021     -0.004      0.997     -76.642      76.315
ar.S.L12       0.3494      0.079      4.408      0.000       0.194       0.505
ar.S.L24       0.3067      0.075      4.103      0.000       0.160       0.453
ma.S.L12       0.0454      0.134      0.338      0.735      -0.218       0.309
ma.S.L24      -0.0912      0.145     -0.628      0.530      -0.376       0.193
sigma2       250.7786      6e+04      0.004      0.997   -1.17e+05    1.18e+05
===================================================================================
Ljung-Box (L1) (Q):                   0.09   Jarque-Bera (JB):                 3.10
Prob(Q):                              0.76   Prob(JB):                         0.21
Heteroskedasticity (H):               0.88   Skew:                             0.43
Prob(H) (two-sided):                  0.71   Kurtosis:                         3.05
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 62: SARIMA Model Summery on Rose train data

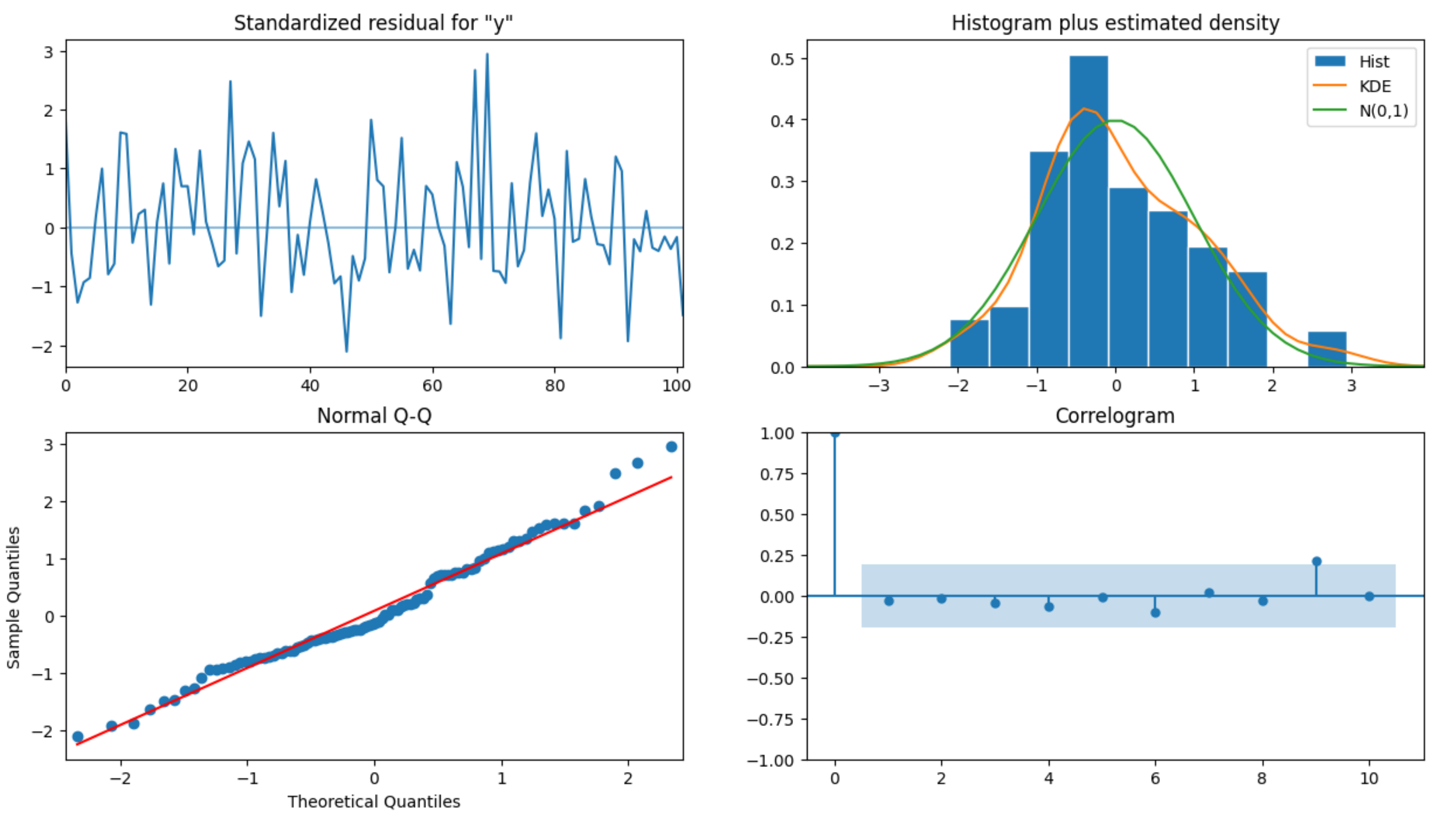

Figure 63: SARIMA Diagnostic Plot on Rose Train Data

- From the model diagnostics plot, we can see that all the individual diagnostics plots almost follow the theoretical numbers and thus we cannot develop any pattern from these plots.
- SARIMA model RMSE score on test data is 25.363820478510938

6.7.11.1. Optimum SARIMA model on the Full Data

```
                                      SARIMAX Results                                       
==========================================================================================
Dep. Variable:                                Rose   No. Observations:                  187
Model:             SARIMAX(1, 1, 2)x(2, 0, 2, 12)   Log Likelihood                -647.370
Date:                            Thu, 05 Dec 2024   AIC                           1310.740
Time:                                    16:18:14   BIC                           1335.291
Sample:                                01-31-1980   HQIC                          1320.710
                                     - 07-31-1995                                         
Covariance Type:                              opg                                         
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.0244      0.300     -0.081      0.935      -0.612       0.564
ma.L1         -0.7409      0.321     -2.306      0.021      -1.371      -0.111
ma.L2         -0.1606      0.281     -0.571      0.568      -0.712       0.391
ar.S.L12       0.3987      0.054      7.345      0.000       0.292       0.505
ar.S.L24       0.3373      0.050      6.727      0.000       0.239       0.436
ma.S.L12       0.0091      0.092      0.099      0.921      -0.171       0.189
ma.S.L24      -0.1494      0.099     -1.504      0.133      -0.344       0.045
sigma2       199.2740     21.793      9.144      0.000     156.560     241.988
===================================================================================
Ljung-Box (L1) (Q):                   0.10   Jarque-Bera (JB):                 8.83
Prob(Q):                              0.75   Prob(JB):                         0.01
Heteroskedasticity (H):               0.26   Skew:                             0.50
Prob(H) (two-sided):                  0.00   Kurtosis:                         3.58
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Figure 64: Optimum SARIMA Model Summery on whole rose data

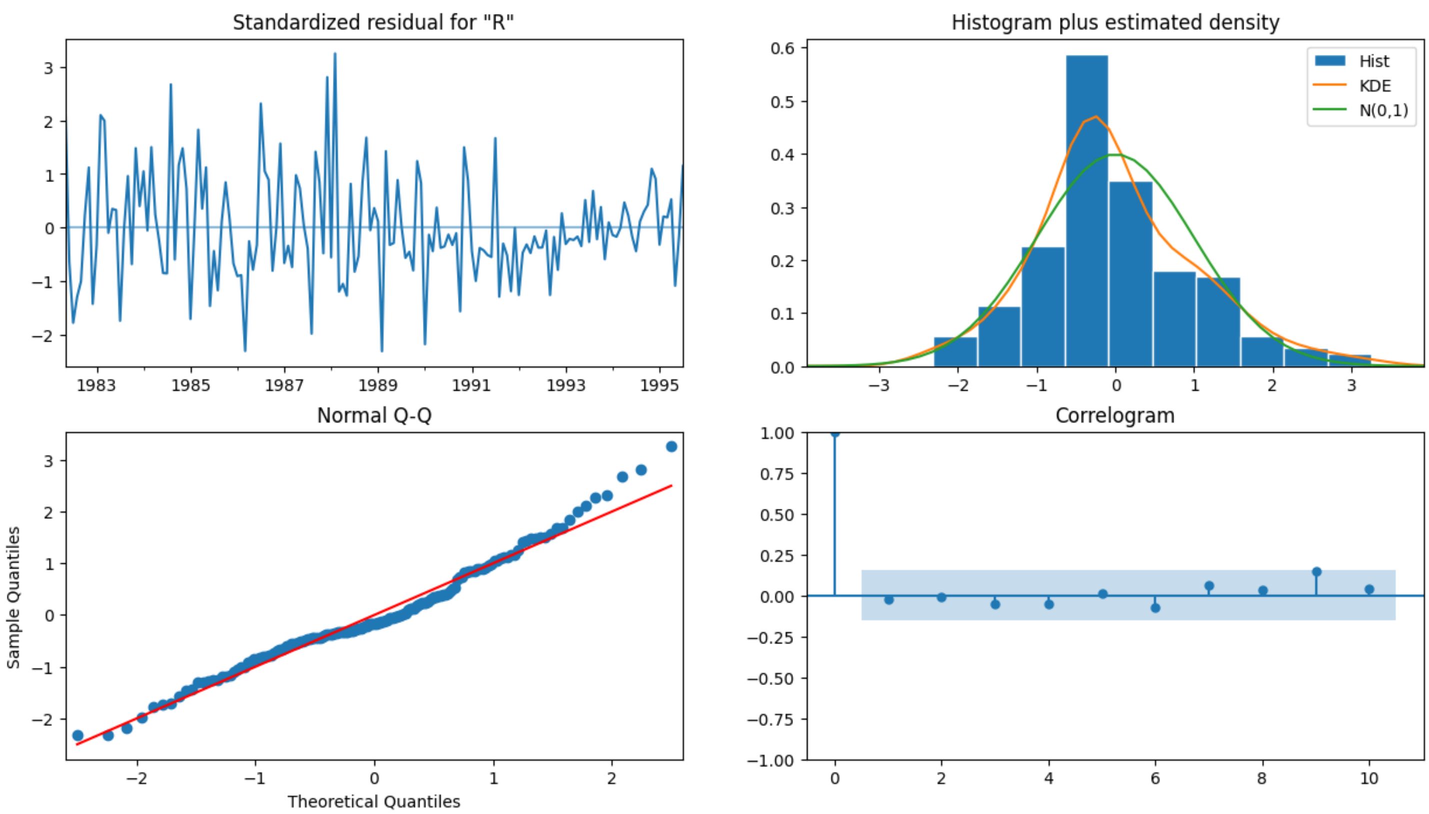

Figure 65: SARIMA Diagnostic Plot on Whole Rose Data

6.7.11.2. Forecast of future 57 months performance using SARIMA Model

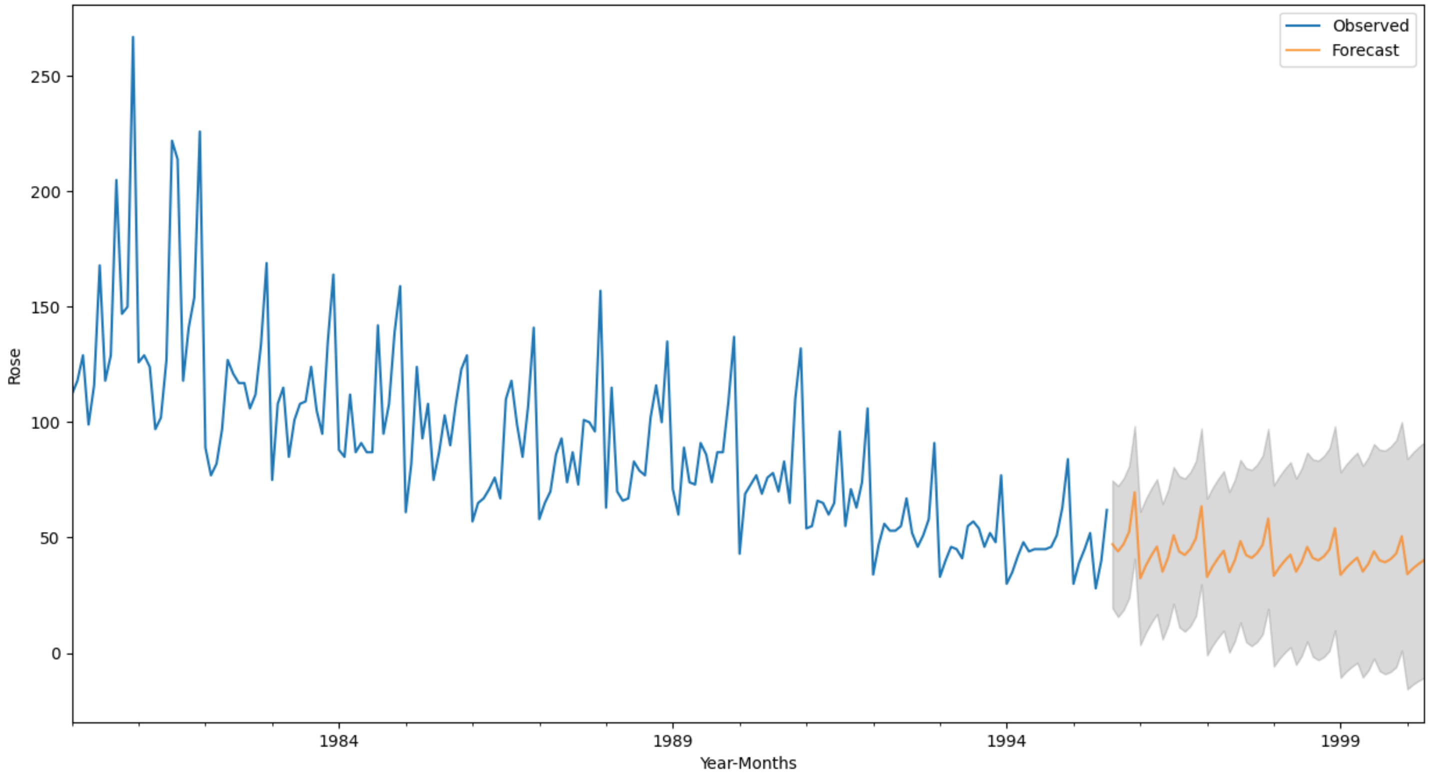

Figure 66: Predict 57 months Rose wine sale into the future using SARIMA Model

6.8. Model Comparison

Model	RMSE Score
Alpha=0.0999, Beta=1.9932826568022853e-06, Gamma=0.0001, Triple Exponential Smoothing	9.337258
2pointTrailing Moving Average	11.801167
4pointTrailing Moving Average	15.370676
6pointTrailing Moving Average	15.867384
9pointTrailing Moving Average	16.345032
Alpha=0.3,Beta=0.4,DoubleExponentialSmoothing	18.337178
Alpha=0.9, Simple Exponential Smoothing	22.513502
SARIMA (0, 1, 2) (2, 0, 2, 12)	25.36382
Alpha=0.995,SimpleExponentialSmoothing	29.243074
ARIMA (1,1,2)	30.487641
ARIMA (1, 0, 2)	38.667285
RegressionOnTime	48.771302
Simple Average	52.431977
Alpha=0.5, Beta=0.3, Gamma=1.0, Triple Exponential Smoothing	62.140816

Table 7: Model Comparison based on RMSE Score

- The best/lowest RMSE Score obtained in TES with Alpha=0.0999, Beta=1.9932826568022853e-06, Gamma=0.0001
- The next best score obtained in 2 point moving average and the highest RMSE score obtained for TES with Alpha=0.5, Beta=0.3, Gamma=1.0

6.9. Forecast with best model

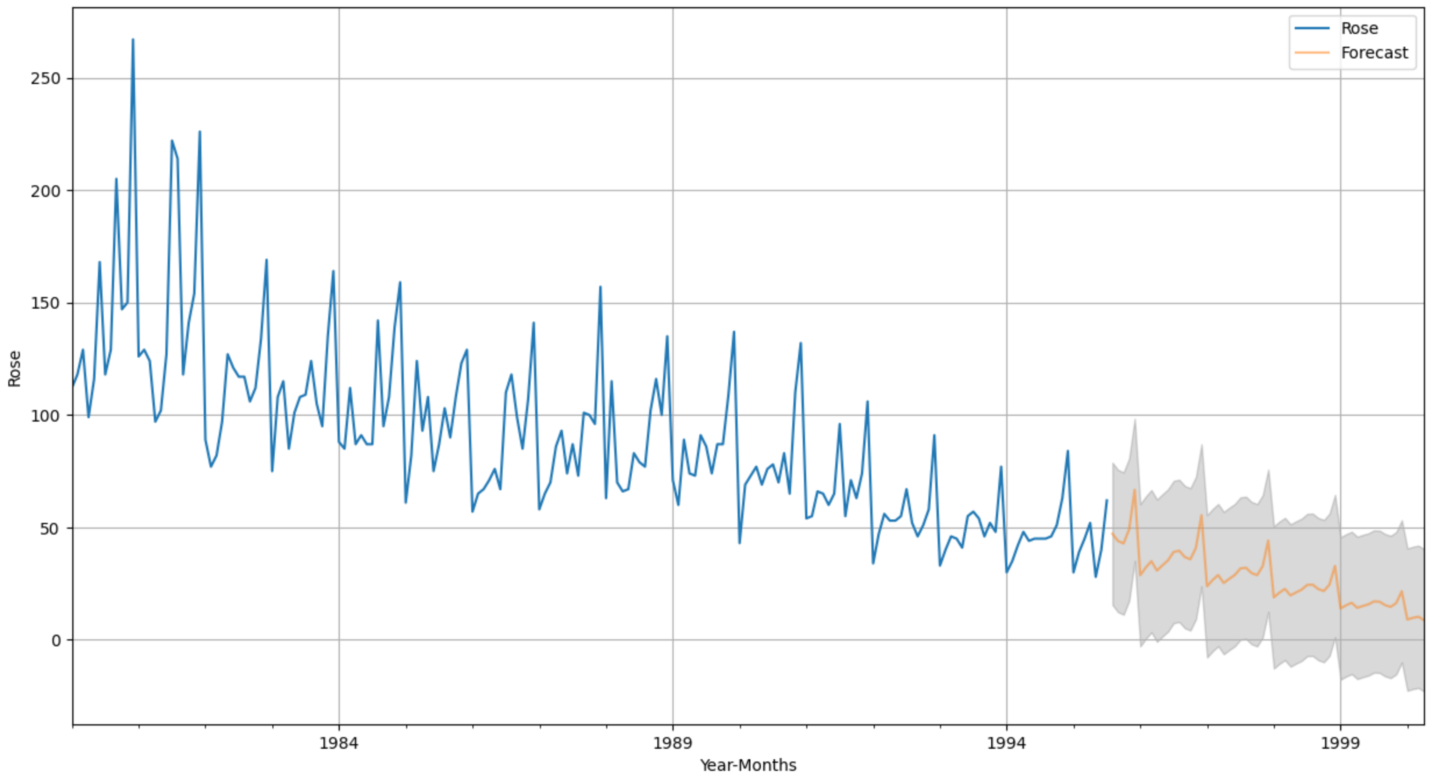

Figure 67: TES Model Forecast on whole ROSE data

- The above trend represents the prediction on Rose wine sale over next 57 months (Model: TES with Alpha=0.0999, Beta=1.9932826568022853e-06, Gamma=0.0001)
- Rose wine sale will decrease.

6.10. Actionable Insights & Recommendations

Sales Trends:

- Declining trend in rose wine sales over the years.
- Seasonal spikes in December, indicating potential holiday-driven demand.

Best Forecasting Model:

- Triple Exponential Smoothing (TES) with parameters (Alpha=0.0999, Beta=0.000002, Gamma=0.0001) achieved the lowest RMSE of 9.34, making it the most reliable for forecasting.

Seasonality Insights:

- Seasonality observed with a periodicity of 12 months.
- Multiplicative decomposition captures seasonal and residual patterns effectively.

Recommendations:

- Revive Demand: Introduce promotional offers or bundle deals during non-peak seasons to stimulate sales.
- Focus on Holiday Season: Align advertising and distribution strategies with the December sales peak.
- New Market Exploration: Investigate new demographics or regions that may favor rose wine to counter the overall decline in sales.